

RELAYSPEC: REUSING FROZEN DRAFTERS THROUGH LINEAR FEATURE INTERFACES

Anonymous authors

Paper under double-blind review

ABSTRACT

Feature-conditioned speculative drafters are trained around the hidden states of a particular language model. RelaySpec reuses these drafters with a new target by learning a linear conditioning interface while keeping the prediction network frozen and removing the source transformer from generation. Across DFlash and EAGLE-3 retargeted from Qwen3-4B to 8B and 14B, the original fits achieve 2.35–5.11 times autoregressive throughput on MATH-500. Frozen-checkpoint tests show that 512 calibration examples retain 97.5% and 98.9% of the throughput obtained with 2,048 examples on 256 GSM8K questions. A richer generated-rollout recipe reaches native DFlash throughput, exceeding it by 2.6% in two matched 128-request NuminaMath runs. Controlled capacity, optimization and adaptation studies support a low-cost dense interface while showing that better feature fitting does not consistently accelerate decoding. Separate Llama and cross-tokenizer Qwen-to-Llama transfers reach 2.52 and 2.75 times matched FP32-target AR throughput, with exact token agreement on all 128 requests per transfer. Expanded calibration improves cross-family throughput by 14.3% but leaves Llama-to-Llama throughput essentially unchanged. These findings establish practical operating points for inherited drafters and characterize the data, capacity and numerical limits of interface-only adaptation.

1 INTRODUCTION

A deployment may inherit a speculative drafter trained for one language model but need to serve another: a larger model, a smaller model, or one from a different family. Feature-conditioned drafters such as DFlash and EAGLE-3 rely on the original model’s internal states (Chen et al., 2026; Li et al., 2025). Keeping that *source* model supplies compatible features but retains another transformer in the generation loop. Training a new drafter incurs a different adaptation cost. We ask how much of the inherited drafter can be reused by changing only its conditioning interface.

RelaySpec learns a linear map from a new *target*’s hidden states to the context consumed by the frozen drafter. During calibration, source and target process shared text to provide paired supervision. During generation, verification already computes the target states needed for the next draft. The map converts them into the drafter’s input, allowing the source transformer to be removed. The target, drafter, inherited embedding and vocabulary projection remain fixed. The target verifier continues to decide which tokens are committed (Leviathan et al., 2023).

The central idea is to adapt the conditioning interface while retaining the drafter’s trained prediction weights. Its value depends on whether proposal quality survives the changed inputs well enough to offset the removed source computation. Matched AR, native-drafter and source-reuse controls measure this tradeoff across both drafter families and expose code workloads where retaining the source is faster.

Contributions and main findings.

1. **Retarget a frozen drafter through its input interface.** RelaySpec replaces source conditioning with mapped target features, removing the source transformer while preserving the drafter’s prediction weights and target-controlled verification. We implement this in DFlash and EAGLE-3 (Section 4).

2. **Establish a useful regime for simple calibration.** On frozen GSM8K tests, 512 examples retain 97.5% and 98.9% of 2,048-example throughput. Dense maps lead the completed 8B capacity comparisons, despite testing MLPs, longer fits and regularization. Better feature fitting need not produce faster decoding (Section 5.5.1).
3. **Show that inherited drafters can reach native throughput.** A richer linear recipe using 16,384 generated rollouts exceeds native DFlash throughput by 2.6% in two matched runs, with 128/128 exact AR sequences. This establishes a higher-effort calibration option (Section 5.6).
4. **Demonstrate reuse across model sizes and tokenizers.** Llama-to-Llama and Qwen-to-Llama transfers reach $2.52\times$ and $2.75\times$ matched FP32-target AR throughput, with 128/128 exact sequences per pair. Expanded calibration helps the cross-family recipe but leaves the Llama comparison unchanged within uncertainty (Section 5.7).

2 RELATED WORK AND POSITIONING

Retargeting an inherited drafter. DFlash and EAGLE-3 condition their proposals on target features (Chen et al., 2026; Li et al., 2025), tying an inherited drafter to its training interface. TriSpec already supports adapter-only finetuning into a fixed EAGLE-family drafter, mapping proxy features to the expected feature space (Jiang et al., 2026). Adapter-only tuning is therefore not our novelty claim. TriSpec uses a proxy to approve some tokens without full-target verification. RelaySpec instead removes the source transformer from generation, retargets its inherited drafter to a new verifier, and retains target-controlled acceptance. Our contribution is this deployment setting and its measured data, capacity and source-removal behavior across two drafting families.

Target-independent drafting and dynamic steering. PARD adapts an autoregressive draft model for parallel token prediction and reuses it across a target family (An et al., 2026). Thus reuse across targets is established prior work. Its target-independent drafter offers zero additional target-specific fitting at deployment. RelaySpec instead makes a small target-specific investment to reuse a feature-conditioned checkpoint and exploit the new target’s internal states. SD^2 is a closer alignment alternative: it computes steering from verifier states and injects it into a pretrained drafter (Berdoz et al., 2026). Our fitted input map replaces source conditioning, whereas SD^2 learns steering within the drafter. The experiments include its public frozen-drafter configuration and PARD’s released checkpoint, with their own matched AR controls (Appendix D). These comparisons expose practical alternatives without attributing differences in runtime or inherited models to the alignment mechanism alone.

Distillation and drafter adaptation. OmniDraft uses online hybrid distillation and an n-gram cache to address cross-vocabulary deployment (Kinattinkara Ramakrishnan et al., 2025). Our primary study uses a shared vocabulary. A separate Qwen-to-Llama extension aligns shared text boundaries during fitting and translates proposals between tokenizers before target verification (Section 5.7). It tests one heterogeneous pair, not a general cross-vocabulary sampling scheme (Timor et al., 2025a). RepSpec changes draft training through structural re-parameterization (Huo et al., 2026). VSD optimizes sequence acceptance, and the Draft-OPD preprint studies on-policy teacher feedback (Zou et al., 2026; Lei et al., 2026). These approaches improve a drafter’s learned prediction behavior. RelaySpec freezes the prediction weights and regresses their input context. Our matched warm-budget connector-CE and rank-32 LoRA comparisons test this alternative locally (Table 50), without ranking untested objectives or full-drafter training.

Frozen interfaces and the meaning of feature alignment. Adapters, low-rank updates and model stitching already adapt or connect frozen networks (Houlsby et al., 2019; Hu et al., 2022; Bansal et al., 2021). EDSD selects two target feature layers using representation metrics while co-designing drafter architecture and training (Cheng et al., 2026). Our native-interface studies instead reduce and recalibrate the input projection of a released drafter whose prediction network stays fixed. Successful stitching need not imply equivalent information (Smith et al., 2025). We therefore evaluate decoded speed and quality alongside feature error, using matched-capacity maps and frozen-checkpoint confirmation.

Proposal structure and execution. Tree policies, prediction heads and execution scheduling improve different parts of speculative decoding (Cai et al., 2024; Li et al., 2024c; Wang et al., 2025a; Miao et al., 2024; Kumar et al., 2026). Self-speculative methods reuse parts of the target (Zhang et al., 2024; Elhoushi et al., 2024). RelaySpec instead changes the conditioning path of an inherited drafter. Its compatibility with other scheduling policies is not established by our experiments.

3 PRELIMINARIES

3.1 DRAFTING AND TARGET VERIFICATION

An autoregressive model generates one token at a time. Blockwise and speculative methods propose several tokens, then check them in a target pass (Stern et al., 2018; Xia et al., 2023; Leviathan et al., 2023). In greedy decoding, the verifier commits the longest matching prefix and a target-selected next token. RelaySpec preserves this decoder and changes the features supplied to its drafter.

3.2 THROUGHPUT AND ACCEPTED PROGRESS

Our primary metric is end-to-end output throughput: total generated tokens divided by total request time, including prompt processing. Let T_A , T_R and T_N denote this throughput for plain AR, RelaySpec and the native target-specific drafter. We report

$$\text{AR speedup} = T_R/T_A, \quad \text{native throughput retained} = 100 T_R/T_N \%. \quad (1)$$

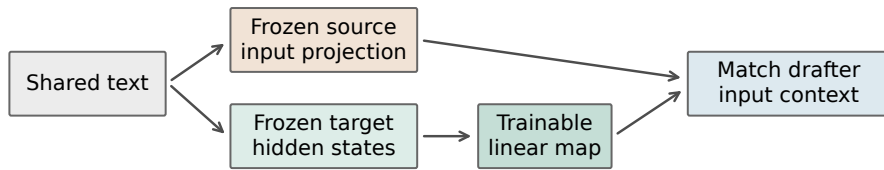
Throughput includes prompt processing. We also report summed AR time relative to summed relay time because output lengths can differ. Accepted progress counts tokens advanced per verification cycle, including the target-supplied token. Removing source computation saves cycle time but can reduce accepted progress. Appendix E specifies this tradeoff and the position-wise acceptance convention.

4 METHOD

4.1 FIT THE INPUT CONSUMED BY THE FROZEN DRAFTER

We run the source and new target on the same text and collect hidden states after five selected transformer blocks. Concatenating these states gives the target vector h_t and source vector s_t at position t . The drafter’s frozen projection F converts source features to its input width. We learn a bias-free matrix W that supplies this input from h_t . Figure 1 separates fitting from generation.

Fit once for the new target



Generate with the new target

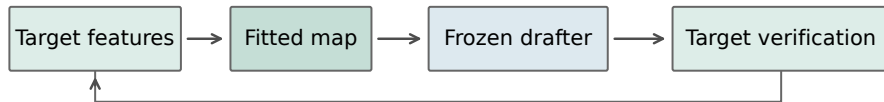


Figure 1: Fit once, then reuse the frozen drafter. The source and target process identical text during fitting. Only the map is updated. During generation, target features from committed positions feed the next draft.

DFlash consumes normalized projected features, while EAGLE-3 consumes the raw projection. Let N_{in} be fixed RMS normalization (Zhang & Sennrich, 2019) and N_{out} the released output

normalization. Teacher and mapped contexts are

$$\begin{aligned} \text{DFlash: } c_t &= N_{\text{out}}(F s_t), \quad \hat{c}_t = N_{\text{out}}(W N_{\text{in}}(h_t)), \\ \text{EAGLE-3: } c_t &= F s_t, \quad \hat{c}_t = W h_t. \end{aligned} \quad (2)$$

We fit W by minimizing the squared context error relative to the teacher’s squared magnitude. For a record with n token positions,

$$\mathcal{L}(W) = \frac{1}{n} \sum_{t=1}^n \frac{\|\hat{c}_t - c_t\|_2^2}{\max(\|c_t\|_2^2, \epsilon)}. \quad (3)$$

We average token losses within records and records across workers, with ϵ the smallest positive normal FP32 value. Teacher-magnitude normalization prevents larger vectors from dominating. The normalized DFlash objective differs from raw linear least squares. The original fits use 4,096 math problem-and-solution conversations, capped at 192 tokens. Supervision is the source context vector. Only W is updated, with 52.43M/65.54M weights for 8B/14B targets. Appendix A gives dimensions, layers and fitting settings.

A richer linear calibration recipe. We also fit separate per-layer maps W_i with a layer-alignment term in addition to context regression:

$$\mathcal{L}_{\text{layer+context}} = \mathcal{L}_{\text{context}} + \frac{1}{5n} \sum_{t,i} \frac{\|W_i h_{t,i} - s_{t,i}\|_2^2}{\|s_{t,i}\|_2^2 + 10^{-6}}. \quad (4)$$

Here $\hat{c}_t = N_{\text{out}}(F \text{concat}_i W_i h_{t,i})$, and the context term uses the same additive 10^{-6} denominator. The two terms have equal weight. These maps take raw inputs and fold into one projection before inference. Thus both recipes retain a linear interface, but their input normalization and supervision differ. The generated-rollout experiment in Section 5.6 tests the complete richer recipe, not an isolated loss ablation.

4.2 DRAFT WITH THE NEW TARGET’S FEATURES

The target caches prompt states, and the map supplies the frozen drafter’s context. The target verifies proposals in parallel, commits the matching prefix plus its next token, and crops the cache to the committed prefix. Mapped states feed the next cycle (Figure 6).

The standard greedy equivalence argument requires block verification to return the same token choices as single-token evaluation of each identical prefix (Leviathan et al., 2023). Finite-precision implementations can change those choices. We therefore measure task accuracy and token agreement separately from throughput. The map itself never supplies the scores used for target verification.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Models and runtime. The primary suite retargets the released Qwen3-4B DFlash and DeepSpec EAGLE-3 drafters to Qwen3-8B and Qwen3-14B (Yang et al., 2025; DeepSeek-AI, 2026). DFlash uses block length 16, EAGLE-3 a seven-position chain. Native controls use released target-specific drafters in the same implementation. Four L40S GPUs each process one sequential request per worker. Historical block and memory studies use RTX 6000 Ada. Within-run methods share hardware. These are single-request measurements, not batched serving (Kwon et al., 2023; Zheng et al., 2024). Table 4 gives the primary protocol. Extensions specify separate runtimes and AR controls.

Workloads and data use. We evaluate math, code and conversation workloads because speculative speed varies across tasks (Xia et al., 2024; Abramovich et al., 2026). MATH-500 supplies 500 questions (Lightman et al., 2024). The breadth suite contains 128 GSM8K questions, 164 HumanEval tasks, 378 MBPP tasks and 80 two-turn MT-Bench conversations (Hendrycks et al., 2021; Cobbe et al., 2021; Chen et al., 2021; Austin et al., 2021; Zheng et al., 2023). The 32-question development set informed interface and loss choices. Historical evaluations retain this exposure history. Fitting and evaluation problem texts have zero normalized exact matches. Appendix H quantifies related templates using token overlap. All completed workload cells in each reported suite are included. Extended calibration uses NuminaMath-CoT (Li et al., 2024a).

Controlled scaling protocol. Original 4,096-record maps remain fixed across the broad evaluation. Additional sweeps vary records, capacity and updates using shared frozen features and 1,024 validation records. The longer-output 8B comparisons use 128 exposed MATH questions with a 2,048-token cap. Initial 14B and adaptation screens use sixteen questions at 256 tokens, followed by 32 GSM8K questions at 2,048 tokens for 14B. These development studies remain separate from the broad evaluation.

Quality and pairing. We score the primary MATH-500 outputs used for the AR timing comparison with the pinned Qwen math scorer. Historical code comparisons use EvalPlus base and expanded tests (Liu et al., 2023). MT-Bench measures two-turn generation time, with each method conditioning its second turn on its own first answer. Its bootstrap unit is the conversation. Other confidence intervals resample matched requests and condition on the fitted checkpoint. Model loading and external task scoring sit outside the request timer.

5.2 ACCELERATION OVER PLAIN AUTOREGRESSIVE DECODING

RelaySpec accelerates all four primary MATH-500 configurations (Figure 2a). DFlash reaches 186.49 and 116.88 tokens/s at 8B and 14B, while EAGLE-3 reaches 84.92 and 59.18. AR-relative ratios span 2.35–5.11, with all paired intervals above one.

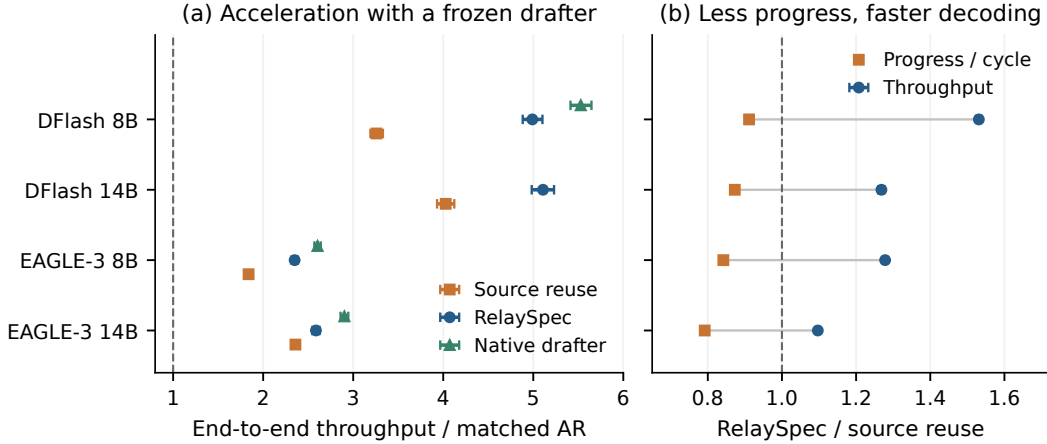


Figure 2: MATH-500 reuse with matched controls. (a) End-to-end throughput relative to each run’s AR. No native DFlash-14B drafter was evaluated. (b) RelaySpec advances fewer tokens per cycle than source reuse but achieves higher throughput. Whiskers are paired 95% request intervals for throughput, conditional on fitted maps. Progress ratios are descriptive points.

DFlash uses parallel proposals and the EAGLE-3 path drafts sequentially. These released-model comparisons also use different library versions, so comparisons between drafter families do not isolate architecture.

5.3 NATIVE RETENTION AND SOURCE REMOVAL

RelaySpec retains 90.3% of native DFlash-8B throughput and 90.2%/89.1% of native EAGLE-3 at 8B/14B (Table 7). These compare released drafters, not matched original training costs. Against source reuse, throughput ratios are 1.531/1.268 for DFlash and 1.278/1.097 for EAGLE-3 at 8B/14B. Holding the inherited drafter fixed, removing source computation outweighs the lower progress per cycle (Figure 2b).

5.4 MATH ACCURACY AND OUTPUT AGREEMENT

Table 1 scores the outputs timed in Figure 2. Observed AR-to-relay accuracy differences range from -0.2 to $+0.8$ percentage points, and each paired interval includes zero. These measured outcomes support the accuracy comparison without treating token equality as a substitute for task quality.

Table 1: MATH-500 accuracy in percent. Difference is RelaySpec minus AR in percentage points. Token matches count identical full output sequences.

Drafter	Target	AR score	Relay score	Difference [95% CI]	Token matches
DFlash	8B	75.8	76.6	+0.8 [-1.4, +3.0]	117/500
DFlash	14B	79.2	79.2	+0.0 [-2.2, +2.0]	113/500
EAGLE-3	8B	75.8	75.6	-0.2 [-2.4, +2.0]	135/500
EAGLE-3	14B	79.2	79.4	+0.2 [-1.8, +2.2]	106/500

BF16 AR and block verification can select different tokens on an identical prefix because their matrix shapes and cache histories differ (Yuan et al., 2025). The historical comparisons in Table 1 therefore do not establish exact AR reproduction. Their accuracy intervals also do not prove a one-point noninferiority margin. We retain these results and test numerical agreement separately using batch-invariant BF16 execution (Section 5.6) and FP32 target execution (Section 5.7). All 128 requests match AR in each of those separate comparisons. Neither precision setting is a guarantee for untested inputs (Appendix K).

Speed across workloads. RelaySpec is 1.85–3.69 times faster than AR across all eight DFlash breadth cells (Appendix Figure 8). At 14B, code remains faster than AR even where source reuse beats the relay. AR measures acceleration, while source reuse measures the benefit of removing its source.

Fitting work and deployment memory. The 52.43–65.54M-weight maps fit in 85.6–118.5 seconds on four RTX 6000 Ada GPUs, including feature computation and optimization after loading. Source removal saves 7.80/7.63 GiB in separate EAGLE-3 8B/14B processes. Required source embeddings and vocabulary projections remain resident. Appendix I reports absolute memory and separate setup costs.

5.5 CALIBRATION COST AND INTERFACE CAPACITY

5.5.1 HOW MUCH DATA AND MAPPER CAPACITY ARE NEEDED?

Holding 8,192 batch-four updates fixed separates distinct records from optimization work. Throughput rises from 96.9 tokens/s with 16 records to 192.8 with 512, then ranges from 196.9 to 198.4 at 2,048–32,768 (Figure 3a). Thus 16 records give $2.58\times$ AR, and 512 retain 97.2% of the observed best. All eleven maps score 105/128. These single-seed development results do not establish a general data threshold. Appendix C retains the historical MATH-only study.

The capacity study crosses 512/2,048 records with dense, factored linear and GELU MLP maps over seven widths at 8,192 updates. Factored maps and MLPs have matched parameter counts. Validation gains flatten with capacity (Appendix Figure 13). On 128 questions at a 2,048-token cap, dense $N = 512$ retains 97.78% [97.09, 98.47]% of $N = 2048$ throughput. A 23.59M-parameter factored linear map retains 95.00% [94.14, 95.85]% at $N = 512$, but its interval crosses the 95% boundary.

Confirmation with frozen checkpoints. Before generating confirmation outputs, we fix the dense checkpoints and analysis criteria, then evaluate 256 GSM8K questions excluded from the recorded development history (Table 2). Both families exceed the predeclared 95% speed-retention threshold. The two data sizes have identical per-question correctness within each family. Relative to AR, conservative accuracy-difference intervals are $[-3.07, 3.07]$ points for DFlash and $[-1.63, 3.14]$ for EAGLE-3. Neither establishes the one-point noninferiority margin.

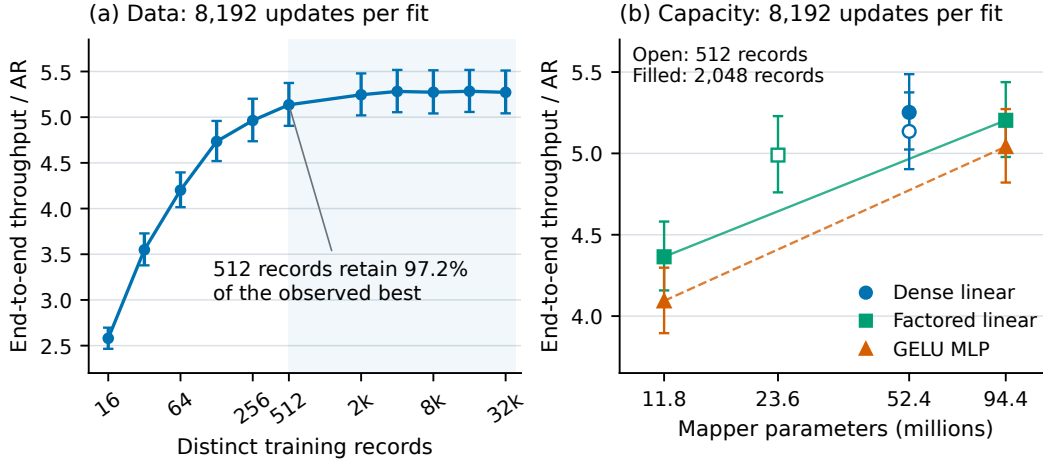


Figure 3: DFlash-8B calibration on 128 exposed MATH questions, with 8,192 updates per fit and a 2,048-output-token cap. (a) Eleven record counts share controls. (b) Seven candidates from a separate capacity campaign, using its own AR. Open symbols use 512 records, filled symbols 2,048. Paired 95% request intervals exclude fitting-seed variation and selection uncertainty.

Table 2: Frozen small-data confirmation on 256 GSM8K questions. Retention is a percentage relative to the 2,048-example mapper in the same family, with paired 95% intervals. Outputs are capped at 2,048 tokens.

Family	Examples	Tokens/s	Retention (95% CI)	Correct	At cap
DFlash	512	157.8	97.5 [96.9,98.1]	236/256	0
DFlash	2,048	161.9	100 (reference)	236/256	0
EAGLE-3	512	100.0	98.9 [98.1,99.7]	238/256	0
EAGLE-3	2,048	101.1	100 (reference)	238/256	0

Better feature fitting need not mean faster decoding. The 94.37M-parameter MLP-4096 at $N = 2048$ reaches 190.15 tokens/s, below dense (Figure 3b). Extending the width-512 MLP from 8,192 to 32,768 updates on the same 2,048 records raises throughput only from 154.39 to 157.68. Regularization gives no matched endpoint benefit from L2 and only small weight-decay gains in feature validation. Dense mapping leads both completed 8B long-output capacity comparisons, including matched-parameter nonlinear alternatives. This is a ranking of the tested recipes, not an optimized function-class comparison. Appendix Figure 14 relates both fitting losses to decoding.

What the interface reveals. The 14B full-answer development screen repeats the dense/MLP ordering, while lower-rate EAGLE-3 follow-ups show sensitivity to optimization (Appendix C.6). Selecting two feature layers cuts projection weights by 60% and retains 97.6% retargeted DFlash throughput. Separate native DFlash and EAGLE-3 projections retain 97.1% and 98.8% throughput on frozen GSM8K tests. Code can require a third layer (Appendix C). Mixed math/instruction fitting improves code and chat speed but slightly lowers math speed at two seeds (Appendix F). These findings make interface design a useful control over reuse, while limiting claims of a single best recipe.

5.5.2 FEATURE FITTING WINS THE MATCHED ADAPTATION COMPARISONS

Feature regression achieves the highest throughput at every tested warm budget (Figure 4). The comparison fixes 512 training records, charges CE and LoRA for their initial mapper, and gives each an equal three-rate development screen. Across all three budgets, CE retains 73.2–76.6% and the two LoRA fits retain 79.4–82.7% of feature throughput. Every paired 95% interval lies below parity. This short development screen does not establish full-answer quality or the ranking of other objectives. Cache preparation and export are separate costs (Appendix C.7).

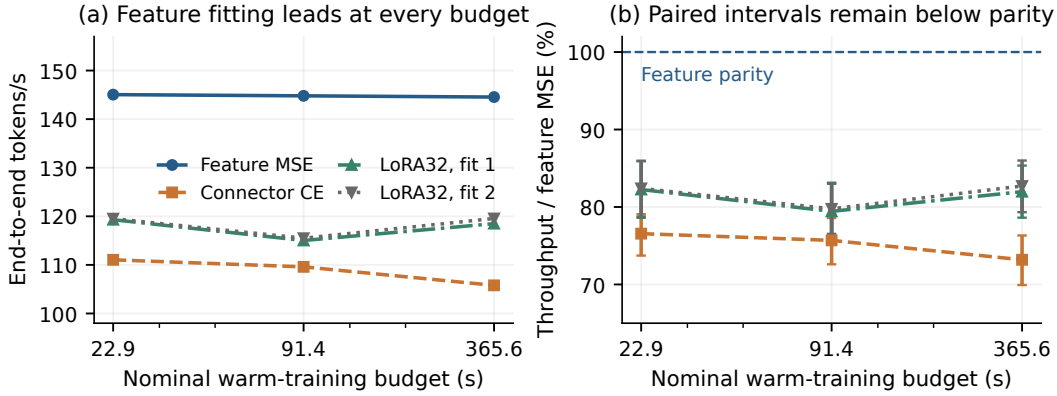


Figure 4: Matched warm-training budgets on 512 records in one DFlash-8B runtime. (a) Feature regression leads all twelve fits. (b) CE and two LoRA fits remain below feature parity, with paired 95% request intervals. This is a sixteen-question development screen with a 256-token cap. Table 51 gives actual timing and updates.

Public-runtime baselines. Table 3 places the completed public baselines beside RelaySpec. The relay has the highest measured throughput, but the separate runtime controls prevent an algorithm-only ranking. Table 53 distinguishes their adaptation mechanisms.

Table 3: Public-runtime comparison on 128 MATH development questions, 2,048-token cap. Correct counts are out of 128, with each system’s AR in parentheses. Precision, attention and libraries differ, so speedups use own-runtime AR. Settings and paired quality uncertainty appear in Appendix D.

System	Tok/s	Own AR tok/s	Speedup [95% CI]	Correct (AR)
RelaySpec ($N = 512$)	193.68	37.72	5.14 [4.90, 5.37]	105 (104)
PARD	105.63	31.67	3.34 [3.22, 3.45]	105 (103)
SD ² (frozen drafter)	19.92	12.48	1.60 [1.56, 1.63]	103 (103)

5.6 REACHING NATIVE THROUGHPUT WITH RICHER CALIBRATION

We train Equation 4 on 16,384 generated NuminaMath rollouts (Li et al., 2024a) for three epochs, then evaluate 128 requests from a separate NuminaMath split in a batch-invariant BF16 runtime. RelaySpec reaches 167.83 and 167.97 tokens/s in two runs, versus 163.57 and 163.67 for a native Qwen3-8B DFlash drafter. Reversing GPU assignments preserves the 2.6% advantage (Figure 5b). AR, native and relay outputs match on all 128 requests in both runs. AR is measured once and reused. Thus reusing a 4B drafter need not impose a native-throughput deficit in this fitted recipe and runtime.

This recipe costs more: rollout generation takes 96.4 minutes on two GPUs, feature capture and validation 52.0 minutes on two, and mapper epoch computation 14.4 minutes on one. The model and data preparation costs are not part of the original 86–119-second fitting claim. Data, loss, optimization and runtime all change, so this result does not identify which change caused the gain (Appendix J).

5.7 BEYOND QWEN SCALING: NEW FAMILIES AND TOKENIZERS

We reuse a Llama-3.1-8B DFlash drafter with Llama-3.2-3B, and a Qwen3-4B DFlash drafter with Llama-3.1-8B (Grattafiori et al., 2024; Meta, 2024). For different tokenizers, fitting pairs hidden states at shared text-prefix endpoints. Generation converts source proposals into target token IDs, which the target then verifies. Only proposed input and conditioning change, not commit authority.

We freeze checkpoints and block lengths after development, then evaluate 128 additional MATH questions at a 2,048-token cap. Targets and maps use FP32 with TF32 disabled, while frozen drafters use BF16. Both old and selected maps match their matched FP32 AR on all 128 requests in each pair (Figure 5a). The selected cross-family map raises end-to-end throughput by 14.3% [11.5, 17.2]%,

reaching $2.75\times$ AR. The larger Llama map changes throughput by only 0.45% $[-0.63, 1.54]\%$. More calibration work helps the tested cross-family recipe, but the within-family result favors retaining its cheaper original map.

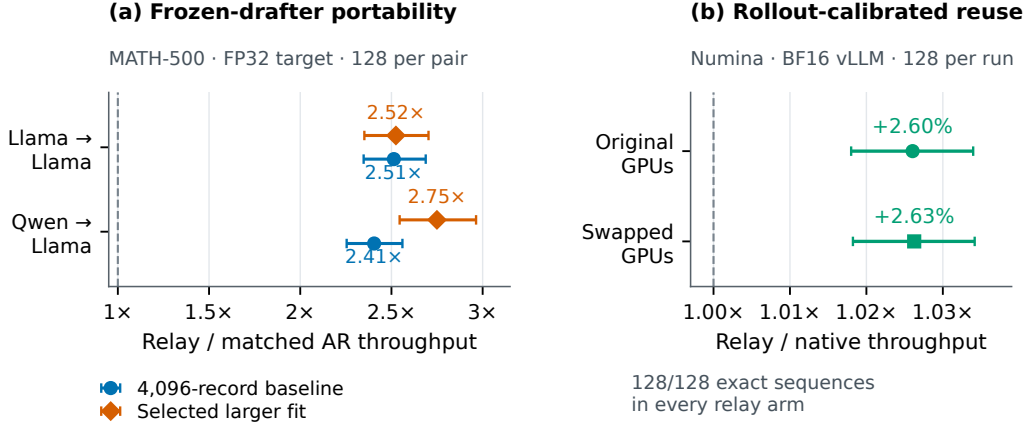


Figure 5: Two separate transfer evaluations, each with 128 requests and a 2,048-output-token cap. (a) MATH, FP32 target: selected maps reach $2.52\times$ and $2.75\times$ matched AR. (b) NuminaMath, batch-invariant BF16: richer calibration reaches $1.026\times$ native throughput under both GPU assignments. All mapper outputs match their own AR reference. Paired 95% request intervals exclude fitting-seed variation.

Cross-family verification calls fall from 28,000 to 24,487 for identical outputs, while Llama calls remain nearly unchanged (Appendix Figure 25). The 128 questions exclude recent family development but belong to the broader project’s MATH pool. These speeds are not pooled with the BF16 Qwen suite (Appendix K).

6 DISCUSSION AND LIMITATIONS

Main insights. A drafter’s learned prediction function can remain useful after its conditioning model changes. A small linear interface often suffices, but its feature error alone does not determine speed. Richer fitting reaches native throughput in one setting and helps cross-family reuse, while the larger Llama fit adds no resolved gain. Interface selection therefore needs both feature and decoding measurements.

Scope and limitations. Mechanism comparisons hold drafter and runtime fixed, while public baselines differ in runtime and training history. Historical BF16 and later exact matches require their respective numerical references. Frozen small-data tests support speed retention, not tight accuracy noninferiority. Coverage remains limited to the evaluated pairs, one cross-tokenizer transfer and one request per worker. Development exposure, related templates, few fitting seeds and capped outputs limit generalization. The rollout recipe changes data, supervision and optimization together.

7 CONCLUSION

RelaySpec makes a frozen drafter’s conditioning interface a practical unit of adaptation. Calibrated reuse removes the source transformer and supports both small-data efficiency and native-level throughput in a tested richer recipe.

AI USE STATEMENT

Generative AI assisted with implementation, code review, literature search, analysis and manuscript editing. Numerical tables and plots are generated from recorded artifacts with checked method,

request and scorer identity. Scientific interpretation and submission decisions remain the authors' responsibility.

REPRODUCIBILITY STATEMENT

The appendix specifies model identifiers, feature layers, fitting settings, measurement boundaries, complete workload results and data checks. The accompanying implementation includes configurations, request records, scorer provenance and scripts that regenerate the reported assets.

REFERENCES

- Talor Abramovich, Maor Ashkenazi, Izzy Putterman, Benjamin Chislett, Tiyasa Mitra, Bitu Darvish Rouhani, Ran Zilberstein, and Yonatan Geifman. SPEED-Bench: A unified and diverse benchmark for speculative decoding. In *43rd International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2604.09557>. Accepted paper.
- Zihao An, Huajun Bai, Ziqiong Liu, Dong Li, and Emad Barsoum. PARD: Accelerating LLM inference with low-cost PARallel draft model adaptation. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=XbOyv7iVGL>.
- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021. URL <https://arxiv.org/abs/2108.07732>.
- Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems 34*, 2021. URL <https://papers.nips.cc/paper/2021/hash/01ded4259d101feb739b06c399e9cd9c-Abstract.html>.
- Frédéric Berdoz, Peer Rheinboldt, and Roger Wattenhofer. Steering pretrained drafters during speculative decoding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 30067–30075, 2026. doi: 10.1609/aaai.v40i36.40255. URL <https://ojs.aaai.org/index.php/AAAI/article/view/40255>.
- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/cai24b.html>.
- Jian Chen, Yesheng Liang, and Zhijian Liu. DFlash: Block diffusion for flash speculative decoding. In *43rd International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2602.06036>. Accepted paper.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021. URL <https://arxiv.org/abs/2107.03374>.
- Longkai Cheng, Ximing Wang, Jiangcai Zhu, Kailai Shao, Chao Chen, and Haixiang Hu. EDS: Entropy-driven design for faster speculative decoding. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 46243–46260. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.2145. URL <https://aclanthology.org/2026.acl-long.2145/>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reichihiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.

- DeepSeek-AI. DeepSpec: A full-stack codebase for training and evaluating speculative decoding draft models. Official open-source software, 2026. URL <https://github.com/deepseek-ai/DeepSpec>.
- Mostafa Elhoushi, Akshat Shrivastava, Diana Liskovich, Basil Hosmer, Bram Wasti, Liangzhen Lai, Anas Mahmoud, Bilge Acun, Saurabh Agarwal, Ahmed Roman, Ahmed Aly, Beidi Chen, and Carole-Jean Wu. LayerSkip: Enabling Early Exit Inference and Self-Speculative Decoding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. URL <https://aclanthology.org/2024.acl-long.681/>.
- Yichao Fu, Peter Bailis, Ion Stoica, and Hao Zhang. Break the Sequential Dependency of LLM Inference Using Lookahead Decoding. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/fu24a.html>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems 34, Datasets and Benchmarks Track*, 2021. URL <https://openreview.net/forum?id=7Bywt2mQsCe>.
- Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning*, 2019. URL <https://proceedings.mlr.press/v97/houlsby19a.html>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*, 2022. URL <https://www.microsoft.com/en-us/research/publication/lora-low-rank-adaptation-of-large-language-models/>.
- Shijing Hu, Jingyang Li, Xingyu Xie, Zhihui Lu, Kim-Chuan Toh, and Pan Zhou. GRIFFIN: Effective token alignment for faster speculative decoding. In *Advances in Neural Information Processing Systems*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/b6e67ae290635d0874c4cb43ba2a2c9b-Abstract-Conference.html.
- Feiye Huo, Jianchao Tan, Jiahao Liu, Zixu Jiang, Jiacheng Li, Jingang Wang, Xunliang Cai, and Shengli Sun. RepSpec: Structural re-parameterized draft model training for speculative decoding. In *International Conference on Learning Representations*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/d70ea003729b440b89a2f958a5554c1f-Abstract-Conference.html.
- Haoyun Jiang, Junqi He, Feng Hong, Xinlong Yang, Jianwei Zhang, Zheng Li, Zhengyang Zhuge, Zhiyong Chen, Bo Han, Junyang Lin, and Jiangchao Yao. TriSpec: Ternary speculative decoding via lightweight proxy verification. *arXiv preprint arXiv:2601.23180*, 2026. doi: 10.48550/arXiv.2601.23180. URL <https://arxiv.org/abs/2601.23180>.
- Ramchalam Kinattinkara Ramakrishnan, Zhaocong Yuan, Jay Zhuo, Chen Feng, Yicheng Lin, Chenzheng Su, and Xiaopeng Zhang. OmniDraft: A cross-vocabulary, online adaptive drafter for on-device speculative decoding. In *Advances in Neural Information Processing Systems*, volume 38, pp. 42319–42348, 2025. doi: 10.52202/085713-1409. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/3c2fe1417eed1c6ff9acf169617981ea-Abstract-Conference.html.
- Tanishq Kumar, Tri Dao, and Avner May. Speculative speculative decoding. In *International Conference on Learning Representations*, 2026. URL

- https://proceedings.iclr.cc/paper_files/paper/2026/hash/1b96f01343ff10150e6719eb163e1536-Abstract-Conference.html.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023. URL <https://doi.org/10.1145/3600006.3613165>.
- Haodi Lei, Yafu Li, Haoran Zhang, Shunkai Zhang, Qianjia Cheng, Xiaoye Qu, Ganqu Cui, Bowen Zhou, Ning Ding, Yun Luo, and Yu Cheng. Draft-OPD: On-policy distillation for speculative draft models. *arXiv preprint arXiv:2605.29343*, 2026. URL <https://arxiv.org/abs/2605.29343>.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 19274–19286, 2023. URL <https://proceedings.mlr.press/v202/leviathan23a.html>.
- Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Costa Huang, Kashif Rasul, Longhui Yu, Albert Jiang, Ziju Shen, Zihan Qin, Bin Dong, Li Zhou, Yann Fleureau, Guillaume Lample, and Stanislas Polu. NuminaMath. Numina dataset repository, 2024a. URL <https://huggingface.co/datasets/AI-MO/NuminaMath-CoT>. Revision 9d8d210c9f6a36c8f3cd84045668c9b7800ef517.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty. In *Proceedings of the 41st International Conference on Machine Learning*, 2024b. URL <https://proceedings.mlr.press/v235/li24bt.html>.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024c. URL <https://aclanthology.org/2024.emnlp-main.422/>.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE-3: Scaling up inference acceleration of large language models via training-time test. In *Advances in Neural Information Processing Systems 38*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/c7b5a35ea98b62512a869c19ea7b03cb-Abstract-Conference.html.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/aca97732e30bcf1303bc22ac3924fd16-Abstract-Conference.html.
- Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by ChatGPT really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems 36*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/43e9d647ccd3e4b7b5baab53f0368686-Abstract-Conference.html.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1711.05101>.
- Meta. Llama 3.2 model card. Official model documentation, 2024. URL <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>.
- Xupeng Miao, Gabriele Oliaro, Zhihao Zhang, Xinhao Cheng, Zeyu Wang, Zhengxin Zhang, Rae Ying Yee Wong, Alan Zhu, Lijie Yang, Xiaoxiang Shi, Chunan Shi, Zhuoming Chen, Daiyaan Arfeen, Reyna Abhyankar, and Zhihao Jia. SpecInfer: Accelerating Large Language Model Serving with Tree-based Speculative Inference and Verification. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3*, 2024. URL <https://arxiv.org/abs/2305.09781>.

- Damian Smith, Harvey Mannering, and Antonia Marcu. Functional alignment can mislead: Examining model stitching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 55972–55998, 2025. URL <https://proceedings.mlr.press/v267/smith25a.html>.
- Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise Parallel Decoding for Deep Autoregressive Models. In *Advances in Neural Information Processing Systems 31*, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/c4127b9194fe8562c64dc0f5bf2c93bc-Abstract.html>.
- Nadav Timor, Jonathan Mamou, Daniel Korat, Moshe Berchansky, Gaurav Jain, Oren Pereg, Moshe Wasserblat, and David Harel. Accelerating LLM inference with lossless speculative decoding algorithms for heterogeneous vocabularies. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025a. URL <https://proceedings.mlr.press/v267/timor25a.html>.
- Nadav Timor, Jonathan Mamou, Daniel Korat, Moshe Berchansky, Oren Pereg, Moshe Wasserblat, Tomer Galanti, Michal Gordon-Kiwkowitz, and David Harel. Distributed speculative inference: Speculation parallelism for provably faster lossless language model inference. In *International Conference on Learning Representations*, 2025b. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/b36554b97da741b1c48c9de05c73993e-Abstract-Conference.html.
- Jikai Wang, Yi Su, Juntao Li, Qingrong Xia, Zi Ye, Xinyu Duan, Zhefeng Wang, and Min Zhang. OPT-Tree: Speculative Decoding with Adaptive Draft Tree Structure. *Transactions of the Association for Computational Linguistics*, 2025a. URL <https://aclanthology.org/2025.tacl-1.8/>.
- Xin Wang, Yu Zheng, Zhongwei Wan, and Mi Zhang. SVD-LLM: Truncation-aware singular value decomposition for large language model compression. In *International Conference on Learning Representations*, 2025b. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/3104e1ab39875cf54fe1eb4473e7c5a1-Abstract-Conference.html.
- Heming Xia, Tao Ge, Peiyi Wang, Si-Qing Chen, Furu Wei, and Zhifang Sui. Speculative Decoding: Exploiting Speculative Execution for Accelerating Seq2seq Generation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL <https://aclanthology.org/2023.findings-emnlp.257/>.
- Heming Xia, Zhe Yang, Qingxiu Dong, Peiyi Wang, Yongqi Li, Tao Ge, Tianyu Liu, Wenjie Li, and Zhifang Sui. Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024. URL <https://aclanthology.org/2024.findings-acl.456/>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. Understanding and mitigating numerical sources of nondeterminism in LLM inference. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-5653. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/f80094a824ba5912d4a2de169c404a40-Abstract-Conference.html.
- Zhihang Yuan, Yuzhang Shang, Yue Song, Dawei Yang, Qiang Wu, Yan Yan, and Guangyu Sun. ASVD: Activation-aware singular value decomposition for compressing large language models. *arXiv preprint arXiv:2312.05821*, 2023. URL <https://arxiv.org/abs/2312.05821>.
- Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. In *Advances in Neural Information Processing Systems 32*, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/1e8a19426224ca89e83cef47f1e7f53b-Abstract.html>.

- Jun Zhang, Jue Wang, Huan Li, Lidan Shou, Ke Chen, Gang Chen, and Sharad Mehrotra. Draft & Verify: Lossless Large Language Model Acceleration via Self-Speculative Decoding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. URL <https://aclanthology.org/2024.acl-long.607/>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient Execution of Structured Language Model Programs. In *Advances in Neural Information Processing Systems 37*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/724be4472168f31balc9ac630f15dec8-Abstract-Conference.html.
- Xiandong Zou, Jianshu Li, Jing Huang, and Pan Zhou. Variational speculative decoding: Rethinking draft training from token likelihood to sequence acceptance. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://ink.library.smu.edu.sg/sis_research/11190/.

APPENDIX

The appendix gives implementation details (Appendix A), complete workload results (Appendix B), fitting studies (Appendix C), acceptance measurements (Appendix E), data checks (Appendix H) and fitting-time and memory measurements (Appendix I).

A IMPLEMENTATION AND MEASUREMENT DETAILS

Table 4: Primary Qwen-suite protocol. Its original fitting choices stay fixed across workloads. The generated-rollout and Llama extensions use the separate protocols in Appendices J and K.

Component	Setting
Primary baseline	Plain greedy AR with the same target, prompt and runtime
Other controls	Native target-specific drafter and optimized source reuse
Generation	BF16, SDPA attention, thinking disabled, maximum 2,048 new tokens
Relay fitting	4,096 records, 192-token cap, 1,024 AdamW updates (Loshchilov & Hutter, 2019), four records/update
Optimizer	Learning rate 6×10^{-4} , zero weight decay, gradient clipping 1.0
Timing	Synchronized request timer, warmup excluded, rotated method order
Uncertainty	10,000 paired resamples, whole conversations for two-turn chat

Primary-suite released models. The source is Qwen/Qwen3-4B. Targets are Qwen/Qwen3-8B and Qwen/Qwen3-14B. The source DFlash is z-lab/Qwen3-4B-DFlash-b16, and its native 8B control is z-lab/Qwen3-8B-DFlash-b16. EAGLE-3 uses DeepSpec’s eagle3_qwen3_4b_ttt7, eagle3_qwen3_8b_ttt7 and eagle3_qwen3_14b_ttt7 releases. All model and upstream revisions are fixed in the accompanying configurations.

Features and dimensions. Layer indices count transformer blocks from zero. Source and Qwen3-8B states come from blocks [1, 9, 17, 25, 33]. Qwen3-14B states come from [1, 10, 19, 28, 37]. Joining five 4,096-wide states gives a 20,480-wide 8B input. Joining five 5,120-wide states gives a 25,600-wide 14B input. Both drafter interfaces have width 2,560. The matrix dimensions are therefore $2,560 \times 20,480$ and $2,560 \times 25,600$. The DFlash input normalization has no trainable gain or bias. The released output normalization remains frozen.

Training. Four workers average record losses at each update. Seed 1729 fixes the selected fit. AdamW uses learning rate 0.0006, zero weight decay and gradient clipping at 1.0. Computation uses BF16, with the feature loss computed in FP32. Each of 1,024 updates processes four records, each capped at 192 rendered tokens. The same map remains fixed across tasks.

Runtime and timing. The main AR runs use Python 3.12.13 and PyTorch 2.9.1 with CUDA 12.8. DFlash uses Transformers 5.3.0. EAGLE-3 uses the pinned 5.10.2 overlay. Each worker owns a full target replica on one L40S GPU. Four workers process independent requests rather than splitting one model over GPUs. One warmup per method precedes measured requests. Method order rotates across requests. The synchronized timer includes prompt processing, drafting, verification, feature updates and runtime bookkeeping. The historical block and memory experiments use RTX 6000 Ada GPUs.

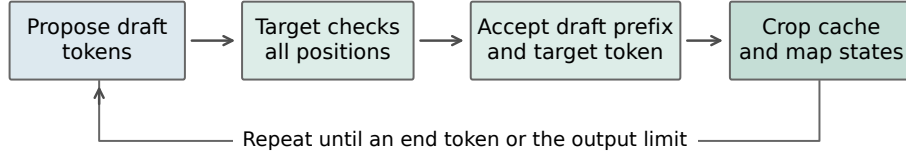


Figure 6: One generation cycle. RelaySpec changes the drafter’s context. The target retains control over committed tokens and the shared decoder retains responsibility for positions, stopping and cache updates.

Stopping and conversation history. Generation stops at the first end token or the output cap. Capped requests remain in both timing and quality aggregates. Each MT-Bench method uses its own first-turn output as second-turn context. Two turns form one resampling unit. The breadth run has 750 initial records and 830 timed requests after expanding 80 conversations to two turns.

Aggregation. Each method’s throughput is the sum of its output-token counts divided by the sum of its request times. Each of 10,000 resamples recomputes this ratio from matched requests, using bootstrap seed 1729. Confidence bounds are the central 95% of resampled estimates. The request-time ratio uses summed times alone. Table 5 reports both measurements and output lengths.

Table 5: Additional MATH-500 measurements. A is AR, S is source reuse and R is RelaySpec. Time ratio divides summed AR time by summed relay time.

Drafter	Target	Mean tokens AR / R	Time ratio AR / R	Throughput R / S	Progress S / R
DFlash	8B	783.22 / 778.81	5.018	1.531	7.66 / 6.98
DFlash	14B	777.66 / 772.33	5.143	1.268	7.44 / 6.49
EAGLE-3	8B	783.22 / 767.79	2.397	1.278	5.15 / 4.33
EAGLE-3	14B	777.66 / 766.20	2.625	1.097	5.07 / 4.01

Table 6: MATH-500 end-to-end tokens/s, including prompt processing. Progress R is mean recorded relay progress per cycle. Ratios and intervals use the AR arm of that same run. Native-drafter comparisons are in Table 7.

Drafter	Target	AR tok/s	Source tok/s	Relay tok/s	Relay / AR [95% CI]	Progress R
DFlash	8B	37.37	121.81	186.49	4.99 [4.88, 5.10]	6.98
DFlash	14B	22.88	92.15	116.88	5.11 [4.98, 5.23]	6.49
EAGLE-3	8B	36.15	66.44	84.92	2.35 [2.32, 2.38]	4.33
EAGLE-3	14B	22.88	53.96	59.18	2.59 [2.55, 2.63]	4.01

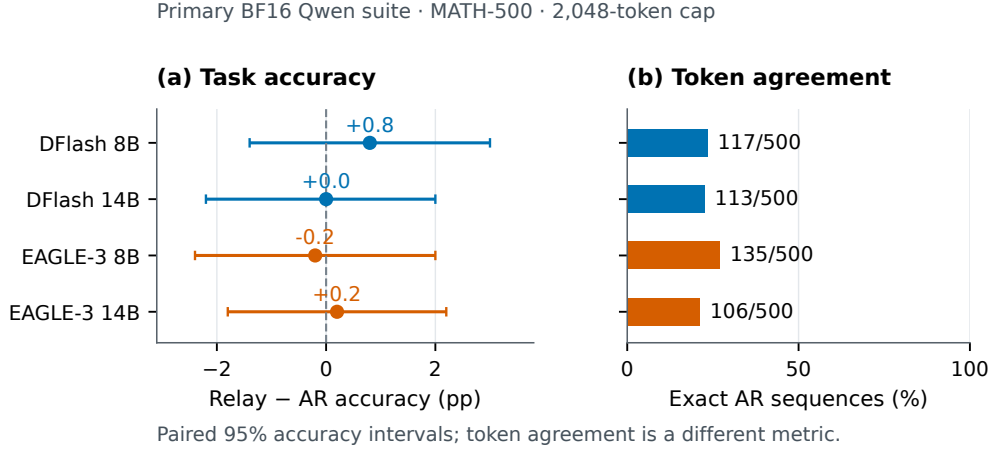


Figure 7: Quality and token agreement are separate outcomes in the primary BF16 Qwen suite. Left: RelaySpec minus matched AR MATH-500 accuracy, with paired 95% request intervals. Right: identical complete output sequences among the same 500 requests, at a 2,048-token cap. These historical measurements are distinct from the later exact batch-invariant BF16 and FP32-target extensions.

B COMPLETE WORKLOAD RESULTS

Table 7: Native throughput retained and mean recorded progress per cycle. N and R denote native drafting and RelaySpec. Progress includes the implementation’s target-supplied token.

Drafter	Target	Native tok/s	Relay tok/s	Retained [95% CI]	Progress N	Progress R
DFlash	8B	206.58	186.49	90.3% [89.8, 90.8]	7.89	6.98
EAGLE-3	8B	94.12	84.92	90.2% [89.8, 90.7]	5.40	4.33
EAGLE-3	14B	66.39	59.18	89.1% [88.6, 89.7]	5.28	4.01

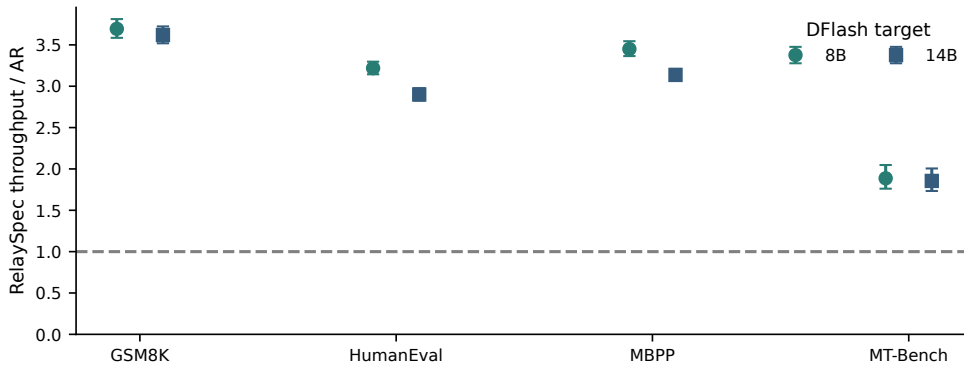


Figure 8: DFlash RelaySpec throughput relative to AR across workloads. Whiskers are paired 95% intervals. Chat intervals resample whole two-turn conversations. Full absolute values are in Table 8.

Table 8: Complete DFlash breadth comparisons with paired AR. Ratios use end-to-end throughput. Every planned task and target cell is included.

Target	Task	Requests	AR tok/s	Relay tok/s	Relay / AR [95% CI]	Relay / source
8B	GSM8K	128	37.60	138.89	3.69 [3.59, 3.81]	1.456
8B	HumanEval	164	37.55	120.87	3.22 [3.14, 3.30]	1.244
8B	MBPP	378	37.49	129.35	3.45 [3.36, 3.54]	1.320
8B	MT-Bench	160	37.47	70.69	1.89 [1.76, 2.05]	1.387
14B	GSM8K	128	26.80	96.98	3.62 [3.52, 3.72]	1.104
14B	HumanEval	164	26.73	77.54	2.90 [2.83, 2.98]	0.890
14B	MBPP	378	26.72	83.82	3.14 [3.07, 3.21]	0.956
14B	MT-Bench	160	26.55	49.26	1.85 [1.73, 2.01]	1.086

The earlier fixed-map source comparisons in Table 9 provide all four workloads for both drafter families on RTX 6000 Ada. They are a separate paired experiment with their own denominator. The EAGLE-3 14B code cells show the reduced margin when lower accepted progress outweighs source-removal savings. This operating range complements the AR acceleration measured in the main comparison.

Table 9: Historical paired source-reuse breadth measurements on RTX 6000 Ada. Ratios compare relay with source throughput within these runs. They are not normalized by an AR arm from another experiment.

Drafter	Target	Task	Requests	Source tok/s	Relay tok/s	Ratio [95% interval]
DFlash	8B	GSM8K	128	116.87	166.39	1.4237 [1.4040, 1.4445]
DFlash	8B	HumanEval	164	118.84	145.00	1.2201 [1.2017, 1.2383]
DFlash	8B	MBPP	378	117.77	151.55	1.2868 [1.2732, 1.3001]
DFlash	8B	MT-Bench	160	60.85	81.62	1.3412 [1.3183, 1.3639]
DFlash	14B	GSM8K	128	88.04	97.07	1.1025 [1.0836, 1.1214]
DFlash	14B	HumanEval	164	87.34	77.61	0.8886 [0.8741, 0.9038]
DFlash	14B	MBPP	378	88.02	83.98	0.9541 [0.9416, 0.9663]
DFlash	14B	MT-Bench	160	45.32	49.30	1.0877 [1.0653, 1.1083]
EAGLE-3	8B	GSM8K	128	85.99	108.45	1.2613 [1.2437, 1.2789]
EAGLE-3	8B	HumanEval	164	71.75	81.10	1.1304 [1.1172, 1.1434]
EAGLE-3	8B	MBPP	378	69.85	82.10	1.1753 [1.1656, 1.1850]
EAGLE-3	8B	MT-Bench	160	42.27	49.45	1.1698 [1.1501, 1.1906]
EAGLE-3	14B	GSM8K	128	67.81	73.41	1.0826 [1.0675, 1.0979]
EAGLE-3	14B	HumanEval	164	55.38	50.58	0.9132 [0.9011, 0.9255]
EAGLE-3	14B	MBPP	378	54.55	50.49	0.9256 [0.9151, 0.9361]
EAGLE-3	14B	MT-Bench	160	33.83	34.57	1.0221 [0.9999, 1.0449]

Table 10: Task scores for the historical fixed-map breadth runs. Source reuse and RelaySpec share each reported score. Code columns are single-program pass percentages under base and expanded EvalPlus tests.

Drafter	Target	GSM8K	HumanEval	HumanEval+	MBPP	MBPP+
DFlash	8B	93.75	85.98	80.49	84.13	73.02
DFlash	14B	94.53	89.02	83.54	88.36	75.13
EAGLE-3	8B	92.97	87.20	82.32	83.07	71.96
EAGLE-3	14B	94.53	90.24	85.98	88.89	74.87

Across these historical math/code comparisons, all 4,680 paired task outcomes agree between source reuse and RelaySpec. This count is four model/drafter pairs times $500 + 128 + 164 + 378 = 1,170$ problems. Expanded code tests evaluate the same programs and do not add independent examples. For the current AR-paired DFlash GSM8K runs, the scorer gives 120/128 correct for all arms at 8B and 121/128 at 14B. Code scores above refer specifically to the historical outputs, preserving scorer/run identity.

AR-paired EAGLE-3 code quality. We additionally score the saved programs from the EAGLE-3 AR-paired breadth runs with official EvalPlus base and expanded tests. Table 11 compares one program per task against AR from the same run. At 8B, RelaySpec passes four fewer MBPP tasks under both suites. At 14B, it passes two more. HumanEval expanded-test totals match AR at both target sizes, although individual task outcomes differ. RelaySpec and source reuse have identical per-task outcomes in all eight cells. The paired intervals cover zero. This does not establish equality or noninferiority. These retrospective comparisons use one fitted map per setting and do not include fitting-seed uncertainty. Conversation quality remains unmeasured.

Table 11: EAGLE-3 AR-paired code quality. Counts are passed tasks. The intervals resample paired tasks. Expanded tests reuse the same programs and do not add independent samples.

Target/task	Suite	AR	Relay	Δ (pp)	Paired 95% interval
8B HumanEval	base	143/164	141/164	-1.22	$[-3.66, +1.22]$
8B HumanEval	plus	133/164	133/164	+0.00	$[-3.05, +3.05]$
8B MBPP	base	318/378	314/378	-1.06	$[-2.91, +0.79]$
8B MBPP	plus	276/378	272/378	-1.06	$[-2.65, +0.53]$
14B HumanEval	base	143/164	143/164	+0.00	$[-2.44, +2.44]$
14B HumanEval	plus	134/164	134/164	+0.00	$[-2.44, +2.44]$
14B MBPP	base	338/378	340/378	+0.53	$[-1.32, +2.38]$
14B MBPP	plus	283/378	285/378	+0.53	$[-1.06, +2.12]$

C FITTING AND INTERFACE STUDIES

Table 12: DFlash-8B block-size ablation on 64 questions, a 512-token cap and RTX 6000 Ada hardware. AR is measured in each run.

Block	AR tok/s	Native draft tok/s	Relay tok/s	Relay / AR	Progress / cycle
8	44.68	175.12	164.90	3.691	4.98
16	44.67	229.87	210.15	4.704	6.48
32	44.76	207.70	131.36	2.935	4.14

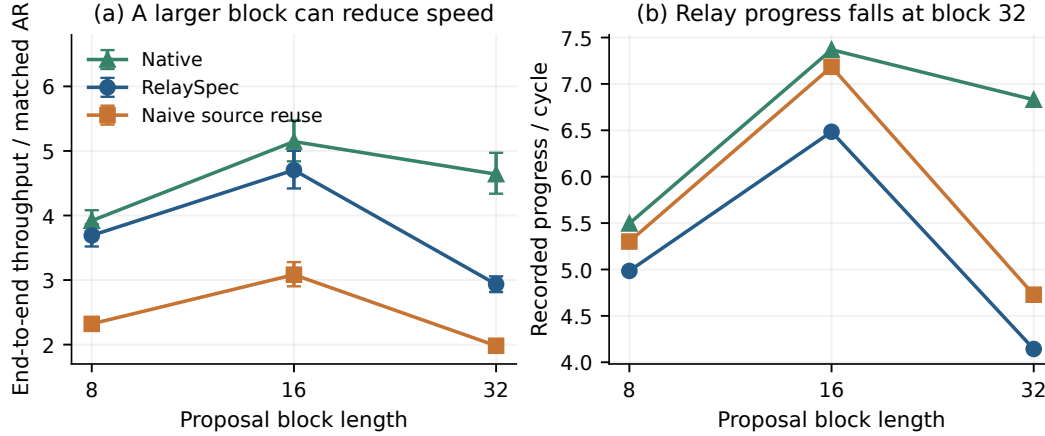


Figure 9: DFlash-8B block-size screen on 64 development questions, 512-output-token cap, RTX 6000 Ada. Larger blocks do not guarantee more accepted progress or higher speed. (a) End-to-end throughput relative to each block run’s AR with paired 95% request intervals. (b) Recorded target-token progress per cycle. The source baseline here is naive source reuse.

Distinct records at fixed fitting work. A separate controlled DFlash-8B study holds fitting at 1,024 updates and four records per update while varying distinct records from 64 to 4,096. Smaller

sets repeat in the same ordered schedule. Table 13 reports end-to-end throughput on the same 128 development-exposed MATH questions. Of the measured settings, 512 is the smallest within 5% of the best observed mapper throughput. The 256-record setting retains 91.4% of that observed best. This is a descriptive, single-seed boundary, not a minimum across settings or a confirmed optimum. The 512–4,096 settings form an unresolved plateau. Selecting the reference from the same evaluations does not provide independent confirmation. Every relay scores 105/128, versus AR’s 104/128.

Table 13: Distinct-data study at 4,096 presentations for every row. Native retention uses each run’s paired native-drafter control. Best retention uses the observed best relay across these settings. All results condition on one fitting seed.

Records	Tokens/s	$\times$ AR	Native retained	Best retained	Correct
64	134.39	3.55	65.7%	72.8%	105/128
128	154.56	4.10	75.5%	83.7%	105/128
256	168.76	4.51	82.6%	91.4%	105/128
512	181.91	4.84	88.9%	98.5%	105/128
1,024	184.55	4.92	90.3%	99.9%	105/128
2,048	183.81	4.88	89.8%	99.5%	105/128
4,096	184.64	4.90	90.2%	100.0%	105/128

Continuous optimization. We also save checkpoints from one uninterrupted 4,096-record fit at 128, 256, 512, 1,024 and 2,048 updates. They have seen respectively 512, 1,024, 2,048, 4,096 and 4,096 distinct records. The final point therefore measures a second pass, not 8,192 distinct examples. End-to-end AR-relative throughput increases from 3.65 to 5.02 across this trajectory. All five checkpoints score 105/128. Unlike the earlier separately fitted budget configurations, these checkpoints share one optimization trajectory. Figure 10 separates this experiment from the fixed-work data study.

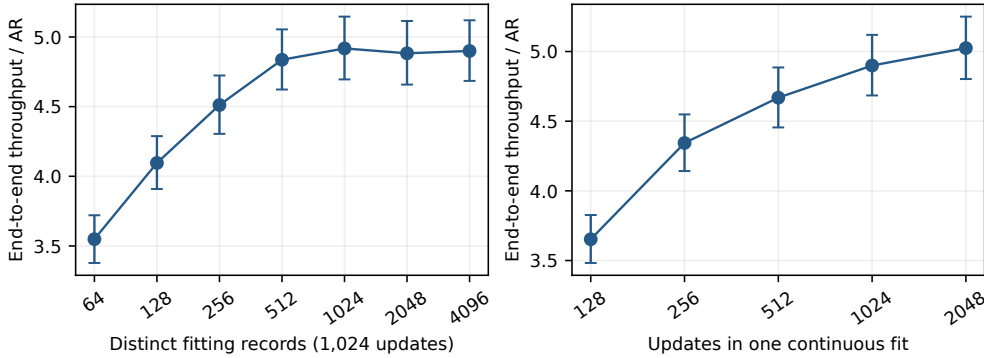


Figure 10: Historical MATH-only fitting study. Left: distinct-data scaling at 1,024 updates. Right: one continuous optimization trajectory. This uses a different training dataset and budget from Figure 3.

Compression should account for the feature distribution. At matched rank 1,024, approximating the mapper’s weight matrix alone can be much weaker than preserving its observed projection outputs (Table 14). For the latter, we sample up to eight feature positions from each of 512 fitting records, approximate the dominant uncentered output subspace U_k of $XW^\top$, and deploy $U_k(U_k^\top W)$, using normalized inputs in X where the mapper requires them. This is an application of data-aware compression, with established precedents in ASVD and SVD-LLM (Yuan et al., 2023; Wang et al., 2025b), not a new general factorization method. EAGLE-3 illustrates the distinction: weight-only SVD retains 54.1% on GSM8K, output-subspace compression 93.9%, and direct teacher fitting 96.1%. On MATH the latter two are nearly tied. Direct fitting remains useful alongside posthoc compression.

Drafter / verifier	Task	Weight SVD	Output-subspace	Direct fitting
EAGLE-3 8B	GSM8K	54.1%	93.9%	96.1%
EAGLE-3 8B	MATH	59.4%	96.7%	96.5%
DFlash 14B	MATH	92.7%	96.6%	97.3%

Table 14: Rank-1,024 throughput retained relative to the paired full map on 32 exposed questions with a 128-token cap. Posthoc arms inherit a trained dense map, while direct fitting starts from the original teacher pairs. The deployment rank is matched, not total adaptation compute.

Retargeting toward a smaller verifier. As a boundary experiment, we reuse the Qwen3-4B DFlash drafter with Qwen3-0.6B, taking target layers [1,7,13,19,25]. The source and target tokenizer files are byte-identical at the pinned revisions. We extract 512 fitting and 128 validation records and fit for 8,192 updates. Two dense seeds use 13.11M mapper parameters. Factored linear and GELU MLP maps at width 1,024 each use 7.86M. Per-worker fitting takes 54–61 seconds after cache extraction, which takes 32.8 seconds on four GPUs including model loading.

On 16 development GSM8K requests, dense RelaySpec reaches 168.0 tokens/s, or 3.26 times AR [3.00,3.50], compared with 102.6 tokens/s for source reuse (Table 18). The second seed repeats the result. AR time divided by RelaySpec time is 3.32 [3.07,3.55]. We report both ratios because outputs differ from AR on nine questions in this BF16 run. All methods finish below the 512-token cap. Correctness is 10/16 for speculative methods and 9/16 for AR, with a conservative paired accuracy-difference interval of $[-23.9, 33.8]$ percentage points for dense RelaySpec. This does not establish accuracy superiority or noninferiority.

Source verification occupies 46.8% of source-reuse request time. RelaySpec removes it while reducing progress per cycle from 5.45 to 4.61. This supports the source-removal mechanism even when the verifier is smaller than the drafter’s source model. It remains a small, single-workload development result rather than a confirmed deployment recommendation for small verifiers.

To investigate the output disagreement, we repeat the same sixteen questions with the same fitted checkpoint in BF16 and full FP32. All source, draft, target and mapper computations change precision together, and FP32 disables TF32. Within-precision AR agreement is 7/16 in BF16 and 16/16 in FP32. The BF16 and FP32 AR outputs themselves match on only 5/16 questions. This intervention supports a numerical explanation in this sample, but also changes kernel dispatch and does not isolate target precision alone. It is not a proof of agreement on other prompts. The four diagnostic workers complete in 61–78 seconds. Numerical sensitivity and FP32 mitigation have prior study, including LayerCast (Yuan et al., 2025). Our contribution here is a controlled diagnostic of the inherited-drafter interface, not discovery of this numerical phenomenon or a new precision algorithm.

The pattern extends to the tested code and dialogue screens (Table 15). Code agreement rises from 5/8 to 8/8, and dialogue agreement from 8/16 to 16/16 timed turns across eight conversations. Each precision has its own AR baseline. Code outputs are not execution-scored, and dialogue responses have no quality scores. The results support applicability beyond math in this small verifier, while distinguishing token agreement from task performance.

Workload	Precision	Units	AR matches	Throughput / AR (95% CI)
GSM8K	BF16	16	7/16	3.29 [3.02, 3.54]
GSM8K	FP32	16	16/16	3.37 [3.12, 3.63]
Code	BF16	8	5/8	2.52 [2.25, 2.79]
Code	FP32	8	8/8	2.79 [2.50, 3.10]
Dialogue	BF16	8	8/16	1.94 [1.58, 2.71]
Dialogue	FP32	8	16/16	1.95 [1.57, 2.78]

Table 15: Qwen3-0.6B precision diagnostics with the frozen dense mapper. Units are questions or whole conversations. AR matches count timed turns. Paired intervals resample the units within each precision. GSM8K and code use 512-token caps, dialogue uses 256 tokens per turn.

We also run the same sixteen GSM8K questions with the main Qwen3-8B verifier and its frozen dense mapper. Exact AR agreement rises from 12/16 in BF16 to 16/16 in full FP32. No answer reaches the

512-token cap. The BF16 workers finish in 87–104 seconds and FP32 workers in 152–179 seconds. Runtime records confirm the requested parameter precision and disabled TF32 for FP32. Together with the smaller-verifier screens, this supports the numerical diagnosis across the two tested verifier sizes, without extending the agreement claim beyond these requests and execution settings.

Is output-head precision sufficient? We next promote only the verifier’s output projection to FP32, leaving the transformer, input embeddings and mapper in BF16. We copy the output head before promotion to preserve input-embedding precision if the weights are tied. On the same sixteen GSM8K questions, agreement with each runtime’s own AR rises from 7/16 to 9/16 for Qwen3-0.6B and from 12/16 to 16/16 for Qwen3-8B (Table 16). Head-only FP32 does not reproduce full-FP32 AR: those matches are 7/16 and 13/16, respectively. Thus the output projection is a useful intervention in the tested 8B setting, but does not explain all precision sensitivity or provide full-FP32 equivalence. These sixteen requests were already exposed during diagnosis.

Target	Head	Own AR matches	FP32 AR matches	Tokens/s	Throughput / AR
0.6B	BF16	7/16	5/16	168.1	3.24
0.6B	FP32	9/16	7/16	161.5	3.23
8B	BF16	12/16	9/16	164.6	4.30
8B	FP32	16/16	13/16	154.8	4.28

Table 16: Output-head precision with a BF16 transformer. Own AR uses the same runtime as RelaySpec. FP32 AR runs the entire model in FP32 and is a distinct reference. Rates concern the same questions but can time different outputs across precision settings.

The initial 8B agreement does not generalize to a broader development screen (Table 17). On 32 additional GSM8K requests, FP32-head agreement with own-runtime AR rises from 15/32 to 20/32. On sixteen MATH requests it rises from 3/16 to 5/16. Released native DFlash and RelaySpec have identical output strings on every request within each precision setting, including their disagreements with AR. This points to a shared execution issue in these samples, rather than an effect unique to retargeting. Head promotion is only a partial mitigation. Native DFlash also uses the target output head for proposal logits, so its intervention changes that projection too. The MATH screen is heavily truncated at 512 tokens and does not measure full-answer accuracy.

Workload	Head	Units	Native matches	Relay matches	AR capped
GSM8K	BF16	32	15/32	15/32	1/32
GSM8K	FP32	32	20/32	20/32	1/32
MATH	BF16	16	3/16	3/16	11/16
MATH	FP32	16	5/16	5/16	11/16

Table 17: Broader 8B head-precision diagnostic. Matches compare complete recorded token sequences with each runtime’s own AR, including capped sequences. MATH AR reaches the cap on 11/16 requests in both settings. Native and RelaySpec reach it on 11/16 in BF16 and 10/16 with the FP32 head. One GSM8K request is capped for every method in both settings.

Method	Tokens/s	Throughput / AR	AR time / time	Correct	AR matches
AR	51.6	1.00	1.00	9/16	16/16
Source reuse	102.6	1.99	2.03	10/16	7/16
Dense, seed 1729	168.0	3.26	3.32	10/16	7/16
Dense, duplicate	167.0	3.24	3.30	10/16	7/16
Dense, seed 1730	168.2	3.26	3.32	10/16	7/16
Factored linear-1024	161.2	3.13	3.18	10/16	7/16
MLP-1024	147.9	2.87	2.92	10/16	7/16

Table 18: Downward retargeting to Qwen3-0.6B on 16 GSM8K requests. Timing includes prompt processing. Correctness and exact AR-output matches are separate. The duplicate row repeats the same checkpoint and passes the trajectory-equivalence check. All models run in BF16.

Selecting target features. The interface also permits a controlled test of target-layer sufficiency. We refit DFlash’s linear mapper using only target layers 25 and 33, keeping the released drafter frozen, the same 512 fitting records and 8,192 batch-four updates. This reduces the mapper from 52.43M to 20.97M parameters. It does not remove target-model layers or reduce whole-model parameters by 60%. Initial development screens selected this two-layer interface before confirmation.

A separate protocol then fixed both checkpoint hashes, the first 64 lexicographically ordered IDs from the previously unused GSM8K reserve, a 2,048-token cap, and a 95% throughput-retention criterion. An updated exposure audit found no reserve IDs in 964 collected rank files across both worktrees. The original fitting-data exact/lexical exclusions remain inherited. Four disjoint 16-question shards each compare the same two methods on one L40S. The paired bootstrap resamples all 64 requests 10,000 times with seed 1729. The lower 95% endpoint exceeds the fixed 95% retention threshold (Table 19). Both maps produce identical token sequences on all 64 questions and score 62/64. None reaches the cap. A conservative paired accuracy-difference interval is $[-6.62, 6.62]$ percentage points, so this sample does not establish one-point quality noninferiority.

Interface	Parameters	Tokens/s	Retention (95% CI)	Correct
Five target layers	52.43M	160.67	100.00 [100.00, 100.00]%	62/64
Two target layers	20.97M	156.86	97.63 [96.23, 99.03]%	62/64

Table 19: Fixed DFlash-8B interface comparison on 64 fresh GSM8K requests. Parameter counts concern the mapper. Throughput includes request overhead. Intervals resample requests and do not include fitting-seed variation.

The recorded profiles explain why reducing mapper size need not improve end-to-end speed. In this fixed comparison, full-map computation occupies 0.74% of total request time, including its prefill application, while full-target verification occupies 81.9%. The two-layer map reduces its own share to 0.58%, but modestly lower progress per verification cycle raises total verification time. Preserving acceptance therefore matters more than mapper arithmetic in this setting. These are recorded region attributions, not isolated kernel or memory measurements.

The result has a family-dependent boundary. On 16 exposed, 128-token MATH requests, two-layer refitting retains 99.3% of full-map throughput with a 14B DFlash verifier, versus 93.7% with EAGLE-3. A second 8B fitting seed retains 97.6% for DFlash and 94.8% for EAGLE-3 on 16 exposed GSM8K requests. These are development screens, not additional confirmation tests. They motivate studying which features a frozen interface requires rather than asserting that two target layers always suffice.

At equal two-layer mapper size, depth matters (Figure 11). Refitting from early layers 1 and 9 retains 56.3% of full-map throughput for DFlash and 57.0% for EAGLE-3. Spaced layers 9 and 25 retain 91.5% and 91.4%, while late layers 25 and 33 retain 98.7% and 96.2% on the same 16 development questions. All fits use 512 records, 8,192 updates and seed 1729. This controls mapper capacity, supporting an information-depth explanation within these settings. It does not establish that this particular layer pair is optimal across tasks or models.

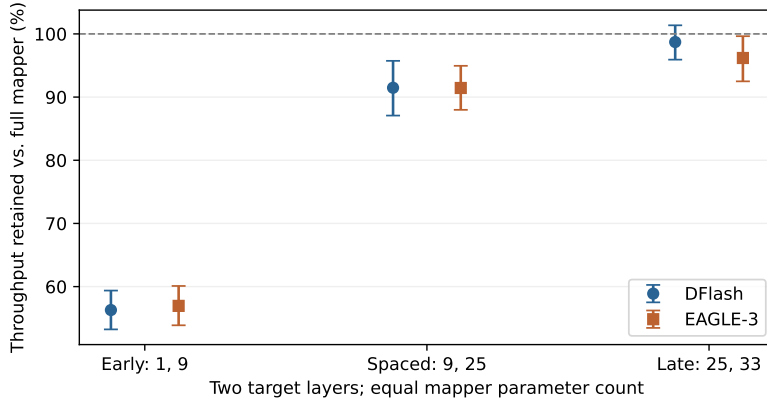


Figure 11: Equal-capacity target-layer comparison on 16 exposed GSM8K questions with a 128-token cap. Each arm is paired with its own full-map baseline. Late-arm rows are the matching subset of the earlier 32-request screen. Paired 95% request intervals exclude fitting-seed variation.

Can nonlinearity compensate for early-layer inputs? We fit an early-layer GELU MLP and a factorized linear map at width 1,950, giving exactly 20,966,400 parameters in each case. Both use layers [1,9], 512 records and 8,192 updates at seed 1729. Each decoding worker also evaluates the existing early- and late-layer dense maps. The MLP lowers training error in both drafter families, but does not close the gap to late-layer inputs (Table 20). It worsens DFlash validation error. For EAGLE-3 it slightly improves validation error, yet throughput remains far below the late-layer control. Late-layer dense maps retain 97.4–98.6% in the DFlash workers and 92.6–92.7% in EAGLE-3, compared with 50.0–57.9% for the fitted early-layer maps. This limits the tested nonlinear fitting approach, not the recoverability of later-layer information in principle.

Family	Early-layer map	Train error	Val. error	Progress/cycle	Retention
DFlash	Factored linear	0.333	0.409	2.61	56.0%
DFlash	GELU MLP	0.271	0.439	2.33	50.0%
EAGLE-3	Factored linear	0.333	0.411	2.57	57.9%
EAGLE-3	GELU MLP	0.272	0.399	2.56	57.4%

Table 20: Early-layer reconstruction at matched parameter count. Retention is relative to each worker’s paired full map on eight exposed GSM8K requests with a 128-token cap. Progress includes the committed target token. Errors use the same fitting and validation diagnostics within each family. One fitting seed is tested.

Cross-domain limits of layer sufficiency. Without changing the Numina-fitted maps, we repeat the depth screen on code and dialogue (Table 21). DFlash’s late-layer map retains 99.7% on code and 97.8% on dialogue. EAGLE-3 retains 85.4% and 91.9%, and its spaced-layer map is slightly faster on dialogue. Thus late-layer sufficiency transfers well in the tested DFlash screens, but is not a family-independent rule. These small exposed screens do not establish code test-pass rates, dialogue quality, or a generally optimal layer choice.

Drafter	Workload	Early	Spaced	Late
DFlash	Code	40.5%	84.8%	99.7%
DFlash	Dialogue	65.8%	91.3%	97.8%
EAGLE-3	Code	52.0%	84.9%	85.4%
EAGLE-3	Dialogue	69.6%	93.0%	91.9%

Table 21: Throughput retained relative to each arm’s paired full mapper. Code uses eight MBPP requests with a 512-token cap. Dialogue uses eight two-turn MT-Bench conversations with a 256-token per-turn cap. Early, spaced and late denote layers [1, 9], [9, 25] and [25, 33]. All compact maps have equal parameter count and the same fitting data and budget.

Native interface compression as an application. The same interface abstraction supports a secondary experiment without retargeting: compress the released Qwen3-8B DFlash input projection by rank-1,536 SVD, leaving the rest of the native drafter unchanged. The rank was fixed from development screens before generating any of the 64 confirmation answers above. We retain both the released native path and a full-rank projection repacked into the relay path as controls. The compressed projection has 37.75M rather than 83.89M parameters (55% fewer). This is not a whole-drafter parameter reduction.

Table 22 uses the same 64 requests and output cap. Relative to the released native path, compression retains 98.66% [97.58, 99.75]% end-to-end throughput. All three methods score 62/64 with identical outputs and no cap-length outputs. Repacking alone retains 99.16%, so the table distinguishes packaging overhead from compression. This is a preselected secondary comparison, not another primary significance test. It demonstrates an application of inspecting and replacing the feature interface. SVD itself is not a novel method, and activation-aware LLM compression already has established precedents (Yuan et al., 2023; Wang et al., 2025b).

Interface	Parameters	Tokens/s	Retention (95% CI)	Correct
Released native	83.89M	178.65	100.00 [100.00, 100.00]%	62/64
Repacked native	83.89M	177.16	99.16 [99.00, 99.28]%	62/64
Rank 1,536	37.75M	176.25	98.66 [97.58, 99.75]%	62/64

Table 22: Preselected native DFlash projection compression on 64 GSM8K requests. Retention is relative to the released native implementation. Only projection parameters are counted.

Distilling a native interface from fewer layers. The layer result also suggests an alternative to weight decomposition: fit a smaller interface directly to the released native projection’s outputs. Using cached target features from 512 fitting records, we apply the frozen native projection and its output normalization to construct supervision, then fit a one-layer or two-layer input map for 8,192 updates. The native drafter remains frozen. Teacher hashes, target identities and layer lists are checked, and the fitting path passes its batched versus individual-example gradient check.

The two-layer native map has 33.55M parameters (60% fewer than the 83.89M native projection), while one layer has 16.78M (80% fewer). Table 23 shows the tradeoff on development workloads: two layers retain 97.1–97.9% throughput relative to released native, while one layer loses substantially more, especially on code. Rank-1,536 SVD retains slightly more throughput with slightly more parameters (37.75M). Thus the student provides another compact-interface operating point, not a uniform improvement over decomposition. These students were selected after the fixed native-SVD comparison and are not evaluated as new candidates on its confirmation questions.

Workload	Units	One layer	Two layers	SVD rank 1,536
GSM8K	32	84.7%	97.7%	98.8%
MATH	32	88.8%	97.1%	98.1%
Code	8	71.6%	97.2%	98.6%
Dialogue	8	87.0%	97.9%	99.2%

Table 23: Native interface students versus released-native throughput. GSM8K/MATH use 32 questions and a 128-token cap, code eight MBPP requests and a 512-token cap, and dialogue eight two-turn conversations with a 256-token per-turn cap. All are exposed development workloads.

Executable-code development check. A separate fixed set of 32 MBPP tasks compares all five frozen native variants at a 1,024-token cap. These tasks exclude the earlier autoresearch MBPP IDs but appear in historical evaluations, so they are development examples. Every variant produces the same tokens and extracted code, passes 28/32 published base-test suites and has no cap-length outputs. The four failures are shared. Tests execute in an isolated Python process using the first Python code block, without EvalPlus extended tests. Two layers retain 95.4% [93.5,97.3]% of released-native throughput, versus 71.6% for one layer and 97.7% for rank-1,536 SVD. This supports observed functional agreement alongside a workload-dependent efficiency penalty. The conservative paired accuracy interval is $[-12.80, +12.80]$ percentage points, not tight quality noninferiority.

Calibration data depends on the interface role. We fit native and retargeted two-layer maps on the same 16- or 128-record prefixes, at 8,192 batch-four updates and two fitting seeds. Each map is compared with its role’s original 512-record two-layer reference on the same eight exposed GSM8K questions, with a 512-token cap. At 128 records, native maps retain 97.2–97.9% throughput across seeds, versus 87.7–88.9% for retargeting. At 16 records these ranges fall to 66.7–67.0% and 46.3–47.1%, respectively. The 16-record fits closely match training features but have much larger validation errors. These standard recipes retain different teacher outputs, output widths and normalization, so the contrast does not isolate an intrinsic difference in sample complexity. The two seeds share questions and reference maps. This is a calibration-sensitivity screen, not a minimum-data guarantee or independent quality confirmation.

Inherited weights change the calibration requirement. A native interface permits an additional control: retain the released projection columns for layers 25 and 33, discarding the other columns. This untrained crop already preserves much of native throughput (Table 24). Initializing the same compact map from these columns instead of random weights improves sparse-data fitting. At 128 updates on 16 records, validation error falls from 0.347 to 0.038 and GSM8K throughput retention rises from 51.7% to 99.3%, relative to the original 512-record compact map. At 128 records the corresponding retentions are 72.4% and 101.0%. All non-initialization fitting settings are matched. The two 128-update inherited fits take 1.55–1.58 seconds of optimizer-update time and 8.8–10.4 seconds within the fitting worker, excluding frozen-feature extraction and original drafter training. Longer inherited fits at 1,024 and 8,192 updates give no clear GSM8K throughput advantage in this screen. Table 24 pairs the short and long fit candidates with untrained cropping across four workloads. This separates inherited performance from the additional effect of calibration. All paired outputs match, but MATH and dialogue include capped outputs, and code and dialogue quality are unscored. Each workload has eight exposed requests, with two turns per dialogue conversation. The fixed record order and copied initialization produce identical weight storage when the nominal fitting seed changes, so those repeats are not independent fitting-seed evidence. Four additional disjoint 16-record training windows, verified against the frozen cache index, retain 99.1–100.0% of the 512-record compact reference after 128 updates on the same GSM8K screen. This checks calibration-record sensitivity without adding independent test questions. This is an initialization-dependent calibration finding, not a universal minimum-data claim or a new general pruning algorithm.

Task	Updates	Crop	Rnd16	Inh16	Rnd128	Inh128	Ref512
GSM8K	8,192	95.7	66.6	99.8	96.7	100.6	99.8
MATH	8,192	90.8	69.0	93.5	95.5	96.2	97.4
Code	8,192	90.6	52.6	92.6	92.6	94.2	98.0
Dialogue	8,192	98.4	72.7	98.4	96.2	98.1	98.8
GSM8K	128	95.9	51.6	98.6	72.4	100.6	99.9
MATH	128	90.8	52.0	94.5	76.2	95.2	97.4
Code	128	90.8	36.8	94.4	56.2	95.2	98.2
Dialogue	128	98.4	63.1	97.7	75.0	99.2	98.8

Table 24: Native initialization controls. Values are throughput percentages relative to each worker’s full repacked-native projection. Crop uses no fitting. Rnd and Inh denote random and inherited-column initialization, with 16 or 128 records. Ref512 is the original 512-record compact map. All compact maps retain two layers and 33.55M projection weights instead of 83.89M. Each row pairs all seven methods. Short and long budgets are separate campaigns on the same exposed requests, with a 512-token cap. Full paired request intervals are retained in the accompanying artifacts.

A scalar-reweighting control. We freeze the cropped native columns and learn only one gain per retained layer, folding both gains into the exported projection. All four settings (16/128 records, 128 updates, learning rates 0.003/0.03) are paired with untrained cropping and full-matrix fits on the same four development workloads. Scalar/crop throughput ratios range from 99.8% to 100.8%. All 16 paired 95% intervals include one. On MATH, scalar fits retain 95.7–96.6% of full-matrix fitting throughput at the same record count, with all four paired intervals below one. These exposed, capped screens suggest that the fitted feature transformation adds more than layer reweighting in this setting. They do not establish superiority over all scalar optimizers: full fits use learning rate 0.0006, and intervals exclude selection and multiple-comparison uncertainty.

A low-rank correction between gains and full fitting. Freezing the inherited columns and fitting an additive low-rank correction provides another calibration control, using the standard low-rank adaptation construction (Hu et al., 2022). Rank 128 trains 1.57M parameters instead of 33.55M, a 95.3% reduction in trainable weights. Folding the correction retains the same dense deployed projection, so this is not an additional inference-memory saving. At 128 records and 128 updates, the higher of two tested rates (0.006) retains 97.3–100.5% of full-matrix fitting throughput across three fitting seeds and four workloads. Validation error remains 0.038–0.042, versus 0.029 for full fitting. The individual paired intervals and every rank/rate arm are retained in the artifacts. These are the same eight exposed requests per workload, with capped MATH and dialogue outputs. The range describes twelve seed/workload point estimates, not a confidence bound or a new independent benchmark sample. The result supports a parameter-efficient approximation to full calibration, with seed and workload dependence, rather than exact equivalence or a novel low-rank algorithm.

A separately frozen short-fit test. Before decoding, we fix the 16-record, 128-update inherited map as the primary candidate on 64 further GSM8K questions (sorted reserve positions 128–191). An exact-ID audit finds no matches in 2,726 prior local result files. This is not a global or pretraining exposure guarantee. The predeclared criterion requires the lower paired 95% throughput-retention bound against released native to reach 95%. The primary candidate retains 95.09% [94.03,96.13]% and *fails* this criterion (Table 25). The 128-record control retains 96.60% [95.48,97.70]%, but remains descriptive rather than replacing the primary candidate. All five methods produce identical outputs, score 62/64, and avoid the 2,048-token cap. The conservative paired accuracy interval remains [−6.62, +6.62] percentage points. No optional extension or refitting uses these outcomes. Thus the short fits support efficient calibration, while the fresh test limits a 16-record sufficiency claim inferred from the smaller development screen.

Interface	Parameters	Tokens/s	Retention (95% CI)	Correct
Released native	83.89M	158.99	100.00 [100.00, 100.00]%	62/64
Repacked native	83.89M	157.66	99.16 [99.08, 99.24]%	62/64
Untrained crop	33.55M	147.50	92.77 [91.62, 93.93]%	62/64
16 records, 128 updates	33.55M	151.19	95.09 [94.03, 96.13]%	62/64
128 records, 128 updates	33.55M	153.59	96.60 [95.48, 97.70]%	62/64

Table 25: Frozen short-fit comparison on 64 additional GSM8K questions. Only the 16-record map is primary. All fitted compact maps inherit native columns and use 128 updates. Intervals use 10,000 paired question bootstrap replicates, seed 1729, and exclude fitting-data and fitting-seed variation.

Frozen native-student confirmation. We freeze the two-layer student’s checkpoint and a new 64-question set before decoding. These are sorted reserve positions 64–127, disjoint from the preceding interface/SVD confirmation. A scan of 932 prior local result files finds no matching question IDs. This does not establish global or pretraining non-exposure. The primary criterion requires the lower endpoint of the paired 95% throughput interval against released native to exceed 95%. We retain the one-layer student, rank-1,536 SVD and full repacked projection as descriptive controls, with no optional sample extension or candidate refitting.

The two-layer student retains 97.05% [95.93, 98.21]% of released-native throughput and passes the fixed criterion (Table 26). One layer retains only 84.55% [83.11, 86.10]%, while SVD retains 98.08% [97.13, 99.06]% with slightly more parameters than two layers. Every method produces identical recorded outputs, scoring 60/64, and none reaches the 2,048-token cap. The conservative paired accuracy interval for two layers versus released native is still [−6.62, +6.62] percentage points: identical observed answers do not establish tight accuracy noninferiority. Together with retargeting, this supports a compact late-layer conditioning interface in two uses, without claiming that two layers suffice for every drafter or verifier.

Interface	Parameters	Tokens/s	Retention (95% CI)	Correct
Released native	83.89M	170.22	100.00 [100.00, 100.00]%	60/64
Repacked native	83.89M	168.91	99.23 [99.18, 99.28]%	60/64
Two-layer student	33.55M	165.21	97.05 [95.93, 98.21]%	60/64
One-layer student	16.78M	143.93	84.55 [83.11, 86.10]%	60/64
Rank 1,536	37.75M	166.96	98.08 [97.13, 99.06]%	60/64

Table 26: Fixed native-student comparison on a separate set of 64 GSM8K questions. Retention uses the released-native reference and 10,000 paired question bootstrap replicates, seed 1729. Only the two-layer comparison is primary. Counts refer to the input projection, not the whole drafter. Fitting-seed variation is outside these intervals.

Native conditioning transfers across drafter families. We repeat the native-interface intervention in EAGLE-3, retaining layers 25 and 33 and learning its raw post-fusion vector without adding an external normalization layer. Repacking the released full projection first reproduces tokens, acceptance and verifier-call trajectories in four operational pilots. At 128 updates, inherited columns reduce validation error from 0.369 to 0.046 with 16 fitting records and from 0.281 to 0.043 with 128 records, compared with random initialization. With 512 records, extending inherited fitting from 128 to 8,192 updates improves validation error from 0.034 to 0.029. The four-workload screen nevertheless shows a tradeoff: GSM8K retention rises from 96.4% to 100.3% and MATH from 95.3% to 99.6%, while code falls from 95.3% to 93.5%. These are eight exposed requests per workload, with two turns per dialogue and a 256-token cap. They do not measure full-answer quality or support uniformly better decoding from a lower fitting loss.

Task	Crop	Rnd128	Inh128	Rnd8192	Inh8192	SVD
GSM8K	81.6	65.5	96.4	98.3	100.3	88.5
MATH	78.8	70.0	95.3	95.7	99.6	92.9
Code	82.9	45.8	95.3	92.5	93.5	93.5
Dialogue	82.7	57.4	93.7	95.2	95.7	87.8

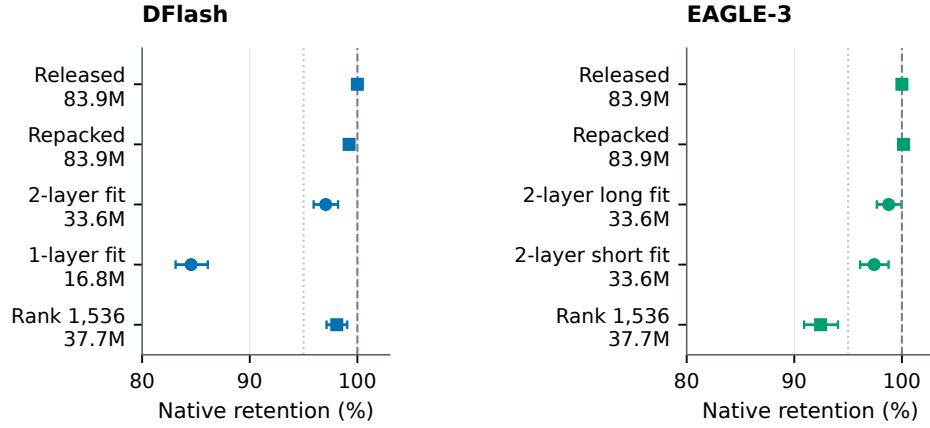
Table 27: Native EAGLE-3 throughput retention (%) against full repacked native on exposed development requests. Rnd/Inh denote random/inherited initialization. All fitted maps use 512 records and the indicated update count. Crop is untrained and SVD has rank 1,536. Each task has eight requests, with two turns per dialogue and a 256-token cap.

Before confirmation decoding, we freeze the inherited 512-record, 8,192-update map as primary and the final 64 reserve questions (sorted positions 192–255). An exact-ID scan of 2,838 prior local result files finds no matches, without guaranteeing global or pretraining non-exposure. The primary retains 98.78% [97.67,99.96]% of released-native throughput, passing the predeclared 95% lower-bound criterion (Table 28). All five methods score 59/64, produce identical outputs and avoid the 2,048-token cap. The conservative paired accuracy interval is still $[-6.62, +6.62]$ percentage points. The shorter inherited fit and weight-SVD remain descriptive controls. Weight-SVD retains 92.44% with 37.75M parameters, versus 33.55M for the calibrated map, but requires no calibration data. Activation-aware compression is evaluated separately on exposed development tasks below. The result supports compact native conditioning in both tested drafter families, without claiming a generally optimal compression method.

Interface	Parameters	Tokens/s	Retention (95% CI)	Correct
Released native	83.89M	101.60	100.00 [100.00, 100.00]%	59/64
Repacked native	83.89M	101.75	100.15 [99.99, 100.36]%	59/64
128-update inherited	33.55M	98.98	97.42 [96.11, 98.76]%	59/64
8,192-update inherited	33.55M	100.35	98.78 [97.67, 99.96]%	59/64
Rank 1,536	37.75M	93.91	92.44 [90.91, 94.06]%	59/64

Table 28: Separately frozen native EAGLE-3 comparison on 64 GSM8K questions. Only the 8,192-update inherited map is primary. Retention uses released native and 10,000 paired question bootstrap draws, seed 1729. Counts concern projection weights, excluding the rest of the drafter. Intervals exclude fitting-data and fitting-seed variation.

Native Qwen3-8B projection · separate frozen 64-question GSM8K sets



95% paired request intervals · 2,048-token cap · counts cover only the projection.
Dotted line: 95% retention criterion. Layers and optimization differ across controls.

Figure 12: Native Qwen3-8B projection compression on separate frozen 64-question GSM8K sets. Throughput is relative to each released native drafter, with paired 95% request intervals and a 2,048-token cap. Labels count projection weights only. All five arms per family are included. The dotted line marks 95% retention. EAGLE-3 inherited two-layer fits use 128 or 8,192 updates. Families differ in question sets and fitting budgets.

Code benefits from an additional late layer. The native EAGLE-3 code deficit motivates a fixed-budget layer intervention. We retain layers 25 and 33 and separately add layer 1, 9 or 17, using inherited columns, 512 math records and 128 updates for every fitted map. An eight-question screen motivates a separate 32-question MBPP development comparison, disjoint from those eight. All three additions remain in the comparison, without selecting a confirmatory primary. The late triple [17,25,33] retains 98.67% [97.35,99.90]% of released-native throughput with 40% fewer projection weights, versus 92.94% [91.46,94.46]% for two layers (Table 29). The other triples retain 94.58% and 96.61%. All six methods pass 25/32 published base-test suites with no 1,024-token cap hits. Three-layer maps match all native token sequences, while two layers match 31/32. Tests use the same isolated base-assertion harness as the DFlash code study, without extended tests. Equal observed pass counts do not establish tight quality noninferiority. These results support workload-dependent interface capacity, not a general two-layer requirement or a globally optimal layer selection.

Interface	Weights	Tokens/s	Native retained (95% CI)	Tests passed
Released native	83.89M	81.91	100.00 [100.00,100.00]%	25/32
Repacked full	83.89M	82.04	100.16 [99.95,100.41]%	25/32
Layers 25,33	33.55M	76.13	92.94 [91.46,94.46]%	25/32
Layers 17,25,33	50.33M	80.82	98.67 [97.35,99.90]%	25/32
Layers 9,25,33	50.33M	79.14	96.61 [95.17,98.12]%	25/32
Layers 1,25,33	50.33M	77.47	94.58 [93.13,96.14]%	25/32

Table 29: Native EAGLE-3 on 32 MBPP development tasks, all frozen layer choices retained. Paired 95% question-bootstrap intervals use 10,000 draws, seed 1729, without multiplicity adjustment. Fitted maps share 512 math records, 128 updates and inherited initialization.

A subsequent control separates calibration from retained native weights. For a fixed first 16 tasks of this cohort, we compare each fitted map with its untrained column crop in the same worker (Table 30). The late-three-layer crop already retains 94.66% of native throughput. Fitting improves it by 2.62% [0.21,5.02]%, while the two-layer crop gains 13.80% [11.16,16.55]%. All four fitting/crop

intervals exceed parity in this subset, without adjustment for multiple comparisons. Every arm passes 12/16 base-test suites and avoids the cap. This is a paired calibration control on reused development questions, not another independent sample. It shows that inherited columns explain much of the compact map’s performance, while learned correction provides a measurable addition.

Layers	Crop / native	Fitted / native	Fitted / crop (95% CI)
25,33	81.30%	92.51%	1.138 [1.112,1.165]
17,25,33	94.66%	97.14%	1.026 [1.002,1.050]
9,25,33	92.31%	95.90%	1.039 [1.017,1.060]
1,25,33	86.88%	93.62%	1.078 [1.046,1.108]

Table 30: Untrained cropping versus calibrated native EAGLE projections on the same fixed 16-task subset. Every layer choice is retained. Fitted/crop ratios are paired within worker, with 95% question-bootstrap intervals. This subset does not add 64 independent questions.

Native compression with calibration activations. A stronger low-rank control projects the native weight matrix onto the leading output subspace of mapped calibration activations, using the same construction described above. It uses 512 math records, at most eight sampled positions per record, and rank 1,536. Weight-only SVD has the same 37.75M projection parameters but uses no calibration data. The original two-layer maps have 33.55M parameters and inherit 512-record, 8,192-update fitting. These comparisons match deployment rank for the two SVD methods, not total adaptation work.

After an eight-question screen, we retain all five frozen methods on 32 exposed GSM8K questions and 32 MBPP tasks in both families (Table 31). The eight-question screen is a subset, not an additional independent sample. Activation-aware SVD retains 99.5% and 98.0% of native DFlash throughput on math and code, versus 98.3% and 98.6% for weight-only SVD. Their paired differences remain unresolved. In native EAGLE-3, activation-aware SVD retains 94.1% on math and 86.1% on code, versus 95.3% and 90.0% for weight-only SVD. On code its activation/weight ratio is 0.957 [0.938,0.977], whereas the two-layer map retains 93.0% of native throughput with fewer weights. Thus accounting for calibration activations does not consistently improve native decoding, and the relative merits of layer reduction and low-rank compression depend on drafter family and workload.

All methods share correctness counts within each cell: 30/32 on GSM8K, 28/32 on DFlash code, and 27/32 on EAGLE-3 code. Every EAGLE-3 code arm has one cap-length output. All other cells avoid the cap. Equal scores on these reused questions do not establish quality noninferiority.

Family	Task	Interface	Tokens/s	Native retained (95% CI)	Correct
DFlash	GSM8K	Released	167.69	100.00 [100.00,100.00]%	30/32
DFlash	GSM8K	Repacked full	166.35	99.20 [99.06,99.34]%	30/32
DFlash	GSM8K	Two layers	163.04	97.22 [95.55,99.00]%	30/32
DFlash	GSM8K	Weight SVD	164.88	98.32 [96.88,99.62]%	30/32
DFlash	GSM8K	Activation SVD	166.88	99.52 [98.49,100.68]%	30/32
DFlash	MBPP	Released	175.27	100.00 [100.00,100.00]%	28/32
DFlash	MBPP	Repacked full	175.07	99.88 [99.21,100.84]%	28/32
DFlash	MBPP	Two layers	168.22	95.97 [93.98,98.03]%	28/32
DFlash	MBPP	Weight SVD	172.76	98.57 [97.34,99.91]%	28/32
DFlash	MBPP	Activation SVD	171.76	97.99 [96.35,99.59]%	28/32
EAGLE-3	GSM8K	Released	101.22	100.00 [100.00,100.00]%	30/32
EAGLE-3	GSM8K	Repacked full	101.76	100.53 [100.02,101.23]%	30/32
EAGLE-3	GSM8K	Two layers	97.72	96.54 [94.59,98.53]%	30/32
EAGLE-3	GSM8K	Weight SVD	96.42	95.26 [92.96,97.48]%	30/32
EAGLE-3	GSM8K	Activation SVD	95.25	94.10 [91.96,96.24]%	30/32
EAGLE-3	MBPP	Released	85.55	100.00 [100.00,100.00]%	27/32
EAGLE-3	MBPP	Repacked full	85.60	100.06 [99.86,100.30]%	27/32
EAGLE-3	MBPP	Two layers	79.54	92.98 [91.45,94.55]%	27/32
EAGLE-3	MBPP	Weight SVD	77.01	90.01 [88.07,91.97]%	27/32
EAGLE-3	MBPP	Activation SVD	73.68	86.13 [83.82,88.39]%	27/32

Table 31: Native-interface compression on 32 development questions per workload and family. Intervals use 10,000 paired question bootstrap draws. Both SVD methods use rank 1,536. The two-layer maps retain layers 25 and 33. Caps are 2,048 tokens on GSM8K and 1,024 on MBPP. Every EAGLE-3 MBPP arm reaches the cap once, with no cap hits elsewhere. Code scores use published base-test suites. No multiplicity correction or new confirmation sample is claimed.

A smaller-verifier boundary. Repeating the depth screen with the 0.6B verifier gives a weaker compact interface (Table 32). At matched 5.24M parameters, early, spaced and late two-layer maps retain 52.5%, 79.9% and 82.2% of the full five-layer mapper’s throughput. One layer retains 60.7% with 2.62M parameters. Late features remain more useful in this screen, but the near-full retention observed at 8B does not carry over to this smaller verifier. This is an eight-question development result with one fitting seed, not a general lower bound on the required layers.

Verifier	Target layers	Train error	Val. error	Progress/cycle	Retention
Qwen3-0.6B	Early [1,7]	0.432	0.459	2.34	52.5%
Qwen3-0.6B	Spaced [7,19]	0.355	0.374	3.70	79.9%
Qwen3-0.6B	Late [19,25]	0.332	0.351	3.76	82.2%
Qwen3-0.6B	Last [25]	0.417	0.430	2.75	60.7%

Table 32: Downward-retargeting depth screen, 4B-trained DFlash to a 0.6B verifier. All fits use 512 records and 8,192 updates. Decoding uses eight exposed GSM8K requests with a 512-token cap. Retention is relative to each worker’s paired full mapper. Errors use the common fitting and validation sets.

A matched-capacity nonlinear follow-up does not recover that deficit (Table 33). Width-1,137 MLP and factorized maps each have 5,239,296 parameters, versus 5,242,880 for late dense. MLPs lower training error from about 0.332 to 0.303–0.304 and validation error from 0.351 to 0.345, yet retain only 81.1–83.1% throughput at two fitting seeds. Factorized maps retain 83.1–84.2%. Their paired late-dense controls retain 82.8–84.3%. The same eight questions are reused across seeds. This closes the tested fitting intervention, without establishing that later features are unrecoverable in principle.

Seed	Architecture	Train error	Val. error	Progress/cycle	Retention
1729	GELU MLP	0.304	0.345	3.69	81.1%
1729	Factorized linear	0.336	0.353	3.76	84.2%
1730	GELU MLP	0.303	0.345	3.78	83.1%
1730	Factorized linear	0.336	0.353	3.77	83.1%

Table 33: Matched-size nonlinear follow-up for 0.6B late-layer mapping. Both architectures use layers [19,25], 512 records and 8,192 updates. Retention is relative to each worker’s full mapper on the same eight exposed questions. Fitting seeds do not add independent evaluation samples. Better feature regression does not close the decoding deficit.

Expanded NuminaMath data curve. Table 34 reports all eleven completed sizes at 8,192 batch-four updates, spanning 2,048 epochs at 16 records to one epoch at 32,768. The same 1,024 feature-validation records are used throughout. Larger data marginally improves feature loss but does not improve measured decoding beyond the plateau. The 512-record mapper retains 97.2% of the observed best throughput, while the 256-record mapper retains 93.9%. These percentages are descriptive and conditional on the evaluated fitting seed and development questions.

Table 34: NuminaMath fitting at fixed 8,192 updates on 128 paired MATH questions with a 2,048-token output cap. All eleven mapper outputs score 105/128, versus AR’s 104/128.

Records	Tokens/s	Speedup / AR	Correct / 128
16	96.87	2.58×	105
32	133.23	3.55×	105
64	157.74	4.20×	105
128	177.75	4.73×	105
256	186.36	4.96×	105
512	192.81	5.14×	105
2,048	196.93	5.25×	105
4,096	198.30	5.28×	105
8,192	197.98	5.27×	105
16,384	198.37	5.28×	105
32,768	197.94	5.27×	105

Table 35: Completed DFlash-8B fitting studies on the same 128-question set. Records count distinct fitting examples. Two passes over 4,096 records give 8,192 presentations. Source reuse is paired within each run.

Fitting choice	Records	Updates	Relay tok/s	Relay / source [95% CI]	Score
512 records	512	128	136.78	1.150 [1.118, 1.181]	82.03
1,024 records	1,024	256	162.04	1.372 [1.347, 1.397]	82.03
2,048 records	2,048	512	173.72	1.477 [1.458, 1.495]	82.03
4,096 records, two passes	4,096	2,048	187.85	1.573 [1.558, 1.588]	82.03
Ridge surrogate	4,096	<i>solve</i>	115.18	0.955 [0.927, 0.982]	82.03
Nonlinear width 512	4,096	1,024	104.49	0.872 [0.841, 0.903]	82.03

The data-budget configurations use the same normalized linear interface and relative-error objective. The nonlinear map uses an intermediate width of 512 and about 11.8M parameters, compared with 52.4M for the dense 8B map. Its lower throughput is consistent with a less effective fitted interface, but the comparison also changes capacity. It does not isolate a disadvantage of nonlinear functions. The ridge configuration solves a raw linear surrogate with ridge coefficient 1.0. Since the deployed DFlash interface includes output normalization, this is a different objective from Equation 3.

Matched capacity on a separate fitting source. We additionally fit 30 DFlash-8B configurations on nested sets of 512 and 2,048 NuminaMath-CoT records,¹ using a fixed 1,024-record fitting

¹AI-MO/NuminaMath-CoT, revision 9d8d210c9f6a, training shard 0 of 5.

validation split. The source-stratified sets include `cn_k12`, `orca_math` and `synthetic_math` examples. Their lexical filter excludes exact normalized problem/solution matches and near matches at five-shingle Jaccard similarity at least 0.6 for texts with at least ten unique shingles. This filter does not establish semantic or template independence. These results use a separate fitting source from the historical MATH study above.

Each configuration receives 8,192 updates with four records per update: 64 passes at $N = 512$ and 16 at $N = 2048$. Frozen features are cached once. Factored linear maps and bias-free two-layer GELU MLPs use widths 64, 128, 256, 512, 1,024, 2,048 and 4,096, giving matched parameter counts $h(20,480 + 2,560)$. The dense reference has 52.43M parameters. The rank of a linear function remains bounded by 2,560 even when its factorization width is larger. All cells share the data order, relative-interface objective, learning rate 6×10^{-4} and seed 1729.

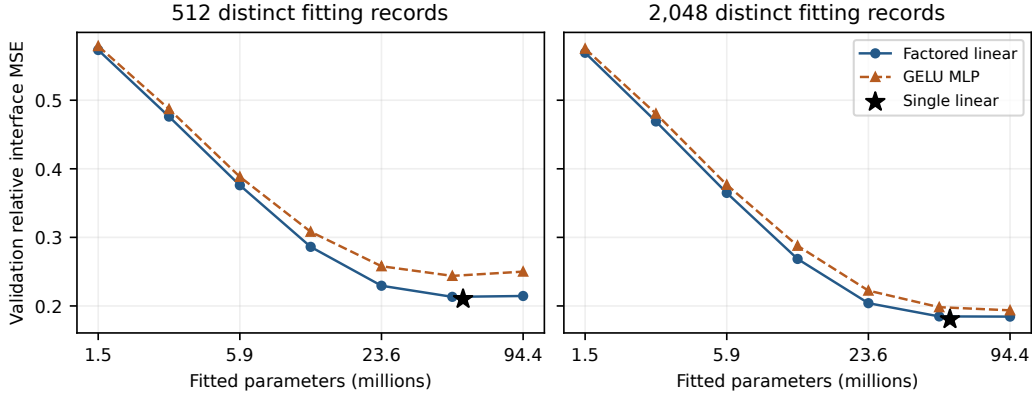


Figure 13: Complete primary capacity study at 8,192 updates per cell. Markers compare validation interface error, with one fitting seed. The curves do not establish decoding speed or answer quality.

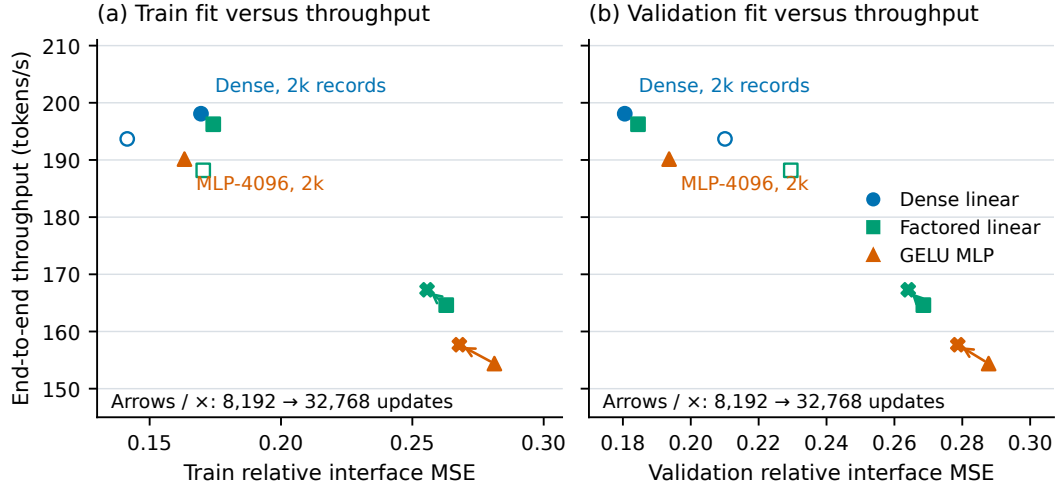


Figure 14: Feature fitting and throughput on 128 development questions, with a 2,048-token cap. Open/filled markers use 512/2,048 records. Dense-2k is faster than MLP-4096-2k despite its higher training loss. Arrows connect 8,192 to 32,768 updates for width-512 factored and MLP maps. Lower validation loss generally accompanies higher throughput here. These descriptive comparisons do not isolate architecture, data or optimization.

Table 36: Validation relative interface MSE for the capacity study. Linear and MLP entries at the same width have equal parameter counts. The dense row supplies the single-linear-layer reference.

Width	Params (M)	Linear, $N=512$	MLP, $N=512$	Linear, $N=2048$	MLP, $N=2048$
64	1.47	0.5732	0.5790	0.5692	0.5752
128	2.95	0.4761	0.4871	0.4690	0.4804
256	5.90	0.3760	0.3883	0.3648	0.3767
512	11.80	0.2862	0.3082	0.2685	0.2878
1024	23.59	0.2295	0.2579	0.2040	0.2224
2048	47.19	0.2133	0.2438	0.1847	0.1982
4096	94.37	0.2145	0.2501	0.1845	0.1936
Dense	52.43	0.2101	—	0.1805	—

At this optimization budget, wider factorizations approach the dense reference, while MLPs have higher validation error at every matched width. Increasing MLP width from 2,048 to 4,096 at $N = 512$ lowers diagnostic training error from 0.1586 to 0.1278 but raises validation error from 0.2438 to 0.2501. At $N = 2048$, validation error instead decreases from 0.1982 to 0.1936. This interaction motivates checking regularization and convergence before attributing the ranking to function class.

More epochs on fixed smaller datasets. We continue the width-512 linear and MLP fits to 32,768 total updates, preserving Adam moments, random states and cyclic data position. A separate bounded pilot reproduced uninterrupted fitting bit for bit. Figure 15 shows late-training diagnostics on the same first 256 training records and all 1,024 validation records. The $N = 2048$ MLP improves from 0.2878 to 0.2787 validation error between 8,192 and 32,768 updates. At $N = 512$, linear training error falls from 0.2443 to 0.2350 while validation error rises from 0.2862 to 0.2873. The latter is consistent with mild overfitting on this diagnostic split. These single-seed fitting trajectories require separate decoding and quality evaluation, including replication across targets and workloads.

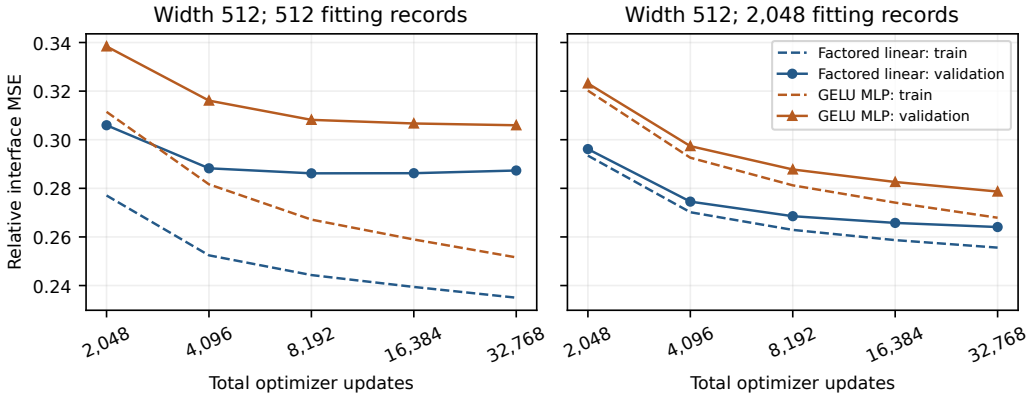


Figure 15: Late-training trajectories, starting at 2,048 updates. Dashed lines use the fixed training diagnostic subset. Solid lines use fitting validation. The horizontal axis counts updates, while the distinct datasets remain fixed at 512 or 2,048 records.

Explicit regularization on fixed smaller datasets. We complete 42 nonzero-penalty fits for the dense, width-512 factored linear and width-512 GELU MLP connectors at $N \in \{512, 2048\}$. Each fit uses 8,192 updates, the same 1,024 validation records and seed 1729. Explicit L2 coefficients are $10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}$. Separate AdamW arms use decay $10^{-4}, 10^{-3}, 10^{-2}$. No tested L2 coefficient improves its matched zero-penalty endpoint. The largest AdamW improvement is a 0.17% relative decrease in validation error for the width-512 linear connector at $N = 512$. Neither MLP setting improves with either penalty family. Table 37 reports the best validation value within each penalty family, so these minima are development selections. Figure 16 retains all tested coefficients.

This grid does not establish a decoding gain or a penalty effect across seeds, and it does not resolve regularization for the wider MLPs.

Table 37: Matched regularization endpoints. Lower relative interface MSE is better. Each nonzero minimum is selected from four L2 or three AdamW coefficients using the same fitting validation split.

N	Mapper	No penalty	Best tested L2	Best tested AdamW
512	Dense	0.21008	0.21138	0.21011
512	Factored linear 512	0.28617	0.28738	0.28570
512	GELU MLP 512	0.30818	0.30827	0.30822
2048	Dense	0.18055	0.18424	0.18055
2048	Factored linear 512	0.26853	0.26970	0.26852
2048	GELU MLP 512	0.28776	0.28864	0.28779

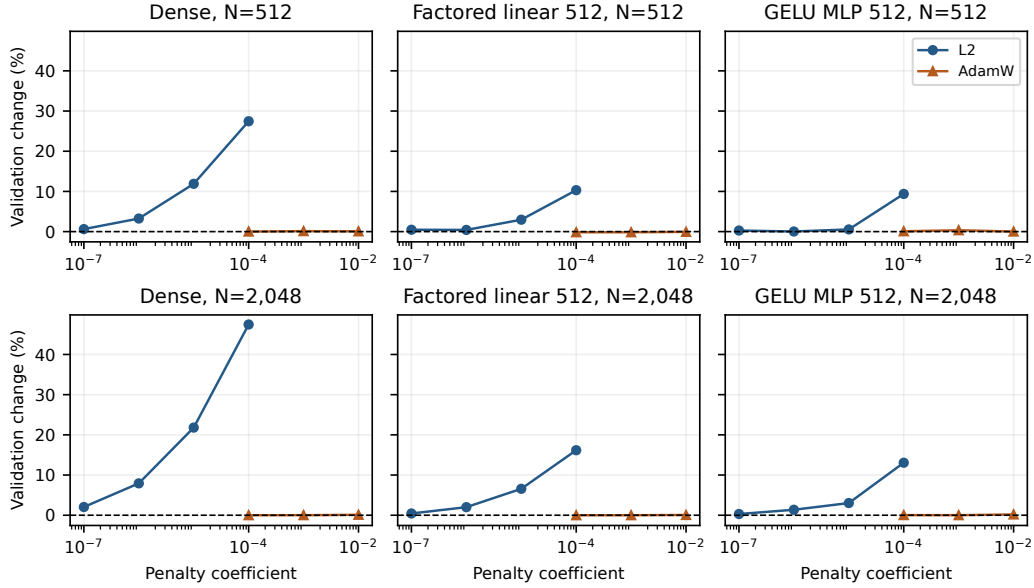


Figure 16: All 42 penalty endpoints relative to their matched zero-penalty validation error. Positive values indicate worse fitting validation. The two curves are separate objective-L2 and AdamW experiments.

Table 38: Matched EAGLE-3 8B normalization comparison on 32 development questions after 128 updates. Throughput ratios use source reuse.

Input to linear map	Relative error	Throughput ratio [95% interval]	Progress retained
Normalized features	0.413	1.035 [1.003, 1.068]	69.0%
Raw features	0.266	1.237 [1.213, 1.263]	82.6%

Table 39: DFlash objective comparison on 32 development questions. Candidates share fitting data, update count, normalization and optimizer. Each ratio uses the source-reuse arm of that candidate’s paired run.

Target	Objective	Throughput ratio [95% interval]	Token matches
8B	Relative error	1.523 [1.488, 1.558]	32/32
8B	Squared error + cosine	1.509 [1.476, 1.545]	32/32
14B	Relative error	1.265 [1.243, 1.289]	32/32
14B	Squared error + cosine	1.251 [1.229, 1.276]	32/32

Relative error weights a record’s feature mismatch by teacher magnitude. Squared error plus 0.1 times cosine distance combines coordinate error with directional agreement. Direct candidate-to-candidate throughput ratios are 1.0085 [1.0005, 1.0174] at 8B and 1.0114 [1.0046, 1.0177] at 14B. These are modest gains with one fewer mixing coefficient.

C.1 EAGLE-3 FITTING REPLICATION

We repeat the fixed smaller-data fitting study with the EAGLE-3 source drafter and the 8B target. The grid contains dense, factored linear and GELU MLP connectors, with widths 128, 512 and 2048 for the latter two. Each primary fit uses 8,192 updates at $N \in \{512, 2048\}$ and the same 1,024 fitting-validation records. Two additional seeds repeat the dense $N = 2048$ setting. This experiment uses the EAGLE input convention without the DFlash input normalization. Error values are interpreted within each drafter family’s conditioning interface.

Table 40: EAGLE-3 fitting validation. End denotes the 8,192-update endpoint. Best denotes the minimum saved fitting-validation error, with its update count. These are development selections for seed 1729, not decoding results. M weights counts millions of trainable mapper parameters.

Mapper	M weights	$N = 512$			$N = 2048$		
		End	Best	Update	End	Best	Update
Dense	52.43	0.24787	0.21666	1024	0.21203	0.21061	2048
Linear 128	2.95	0.41980	0.41980	8192	0.41505	0.41505	8192
Linear 512	11.80	0.28632	0.28505	2048	0.27188	0.27188	8192
Linear 2048	47.19	0.23486	0.22877	2048	0.21132	0.21132	8192
MLP 128	2.95	0.46554	0.46554	8192	0.45645	0.45645	8192
MLP 512	11.80	0.31749	0.31749	8192	0.29577	0.29577	8192
MLP 2048	47.19	0.28154	0.27124	2048	0.23196	0.22683	4096

The dense $N = 512$ mapper reaches 0.21666 at 1,024 updates, then rises to 0.24787 at the final update. Its endpoint is worse than the width-2048 linear mapper, but its best saved validation value is better than that linear mapper’s 0.22877. At $N = 2048$, the dense and width-2048 linear minima are 0.21061 and 0.21132. Thus endpoint rankings alone can reflect different convergence trajectories. At every matched width in this grid, the MLP’s best saved validation error exceeds the linear mapper’s minimum. The three dense $N = 2048$ endpoints range from 0.212034 to 0.212043. This narrow seed spread concerns one fitting setting. It does not establish seed stability for the other architectures or for decoding performance.

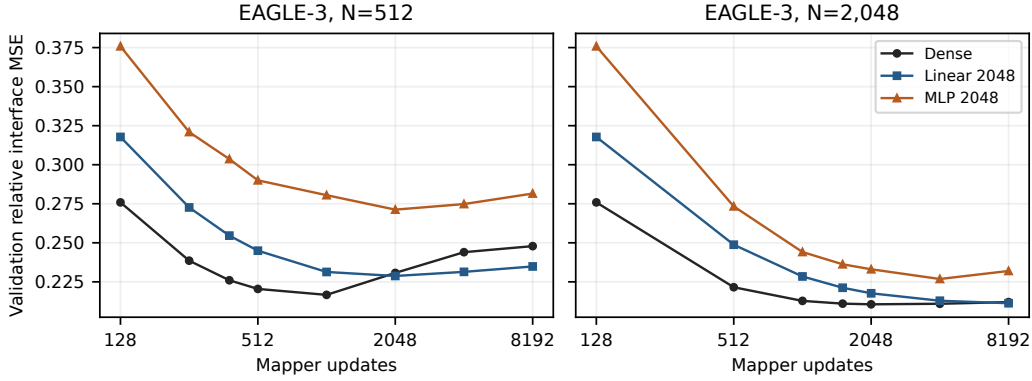


Figure 17: EAGLE-3 fitting-validation trajectories at fixed distinct-data counts. Additional updates can worsen validation, most visibly for the dense mapper at $N = 512$. The horizontal axis counts updates rather than new examples. The subsequent checkpoint comparison measures actual decoding.

C.2 FEATURE-VALIDATION MINIMA AND DECODING

We compare each distinct saved feature-validation minimum with its 8,192-update endpoint in two declared decoding campaigns. Each uses 16 paired development questions, greedy decoding and a 256-token cap. Checkpoint selection precedes decoding. Table 41 includes every distinct early-versus-endpoint pair in these campaigns.

Table 41: Decoding throughput at the minimum feature-validation checkpoint relative to the same fit’s endpoint. Intervals resample paired requests 10,000 times. They are descriptive, without multiple-comparison adjustment, and do not measure variation across fitting seeds.

Family	Mapper	N	Learning rate	Selected update	Early / end [95% CI]
DFlash	Linear 4096	512	0.0006	4096	1.009 [0.981, 1.040]
DFlash	MLP 4096	512	0.0006	4096	0.978 [0.961, 0.998]
DFlash	Linear 4096	512	0.0002	4096	1.008 [0.985, 1.029]
DFlash	Linear 4096	512	0.0018	4096	1.003 [0.978, 1.029]
DFlash	MLP 4096	512	0.0002	2048	1.011 [0.986, 1.039]
DFlash	MLP 4096	512	0.0018	4096	0.961 [0.939, 0.983]
EAGLE-3	Dense	512	0.0006	1024	0.950 [0.924, 0.976]
EAGLE-3	Linear 512	512	0.0006	2048	0.969 [0.948, 0.988]
EAGLE-3	Linear 2048	512	0.0006	2048	0.964 [0.922, 1.003]
EAGLE-3	MLP 2048	512	0.0006	2048	0.968 [0.942, 0.996]
EAGLE-3	Dense	2048	0.0006	2048	0.980 [0.969, 0.992]
EAGLE-3	MLP 2048	2048	0.0006	4096	1.019 [0.995, 1.046]

Lower feature-validation loss does not consistently predict faster decoding. For EAGLE-3’s dense $N = 512$ fit, the selected 1,024-update checkpoint retains 94.99% of endpoint throughput, with interval [92.39, 97.59]%. For the DFlash width-4096 MLP at $N = 512$ and learning rate 0.0006, the selected 4,096-update checkpoint retains 97.77%, with interval [96.09, 99.76]%. Other intervals cross equal throughput. These results support treating feature regression as a surrogate and measuring decoding when selecting a connector. They do not establish full-answer quality or justify selection on untouched evaluation data.

Table 42: Observed throughput ranges across three fitting seeds in a separate 16-question, 256-token development campaign. Selected uses each fit’s minimum saved feature-validation checkpoint. These ranges describe the three measured fits and runtime variation, not confidence intervals.

Mapper	N	Learning rate	Endpoint tok/s range	Selected tok/s range
Dense	512	0.0006	184.34–184.68	183.36–184.58
Dense	2048	0.0006	186.11–189.22	186.11–189.22
Linear 1024	512	0.0006	179.47–181.23	179.47–181.23
Linear 4096	2048	0.0002	185.36–187.81	185.36–187.81
MLP 4096	2048	0.0002	183.92–184.78	183.92–184.78
MLP 4096	512	0.0002	174.53–177.30	175.57–177.56

All six settings in Table 42 use the same cached features and seeds 1729, 1730 and 1731. The dense $N = 512$ endpoints span 184.34–184.68 tokens/s. The width-4096 $N = 2048$ MLP improves on its $N = 512$ counterpart, while the dense connector remains competitive. These short development runs do not establish the minimum sufficient data count or distinguish small architecture gaps from runtime variation.

C.3 LONGER DEVELOPMENT OUTPUTS FOR SMALL-DATA MAPPERS

We evaluate nine saved DFlash connectors on 128 paired MATH development questions with a 2,048-token cap, greedy decoding and the same source, target and native-drafter references. These exposed questions provide a larger development comparison than the 16-question, 256-token screen. They are separate from any untouched confirmation set. The continued width-512 fits reuse the same 2,048 distinct training examples.

Table 43: Capped development outcomes for small-data DFlash fitting. Throughput retention uses the dense $N = 2048$, 8,192-update connector. Intervals use 10,000 paired request bootstrap samples. Correct counts scored answers. Cap counts outputs reaching 2,048 tokens, which can include EOS exactly at the limit and are not exact truncation counts.

Method	N	Updates	Tok/s	Dense retained (%) [95% CI]	Correct	Cap
AR	–	–	37.72	19.04 [18.23, 19.90]	104/128	11
Native drafter	–	–	204.57	103.27 [102.35, 104.17]	105/128	9
Source reuse	–	–	120.96	61.06 [60.53, 61.59]	105/128	9
Dense	512	8192	193.68	97.78 [97.09, 98.47]	105/128	9
Dense	2048	8192	198.09	100.00 [100.00, 100.00]	105/128	9
Linear 1024	512	8192	188.19	95.00 [94.14, 95.85]	105/128	9
Linear 4096	2048	8192	196.25	99.07 [98.50, 99.60]	105/128	9
MLP 4096	2048	8192	190.15	95.99 [95.30, 96.70]	105/128	9
Linear 512	2048	8192	164.61	83.10 [81.81, 84.38]	105/128	9
Linear 512	2048	32768	167.29	84.46 [83.15, 85.71]	105/128	9
MLP 512	2048	8192	154.39	77.94 [76.50, 79.35]	105/128	9
MLP 512	2048	32768	157.68	79.60 [78.07, 81.06]	105/128	9

Dense fitting on 512 examples retains 97.78% of the dense $N = 2048$ throughput, with interval [97.09, 98.47]%. This establishes a lower-data performance point in this setting, but does not establish that 512 is minimal. The width-1024 linear mapper at $N = 512$ retains 95.00%, with interval [94.14, 95.85]%, leaving its 95% boundary unresolved. The width-4096 linear mapper is slightly below the dense connector in this larger sample, so the earlier short-screen point ranking should not be interpreted as an established architecture advantage.

All nine connectors score 105 of 128 answers, compared with 104 for AR. Their paired accuracy-difference intervals are $[-0.03125, 0.046875]$. These intervals do not establish a one-percentage-point noninferiority margin. Nine connector outputs and eleven AR outputs reach the token cap. The observed scores therefore concern the configured capped evaluation, not uncapped answer quality. Longer width-512 fitting gives modest throughput increases while remaining below the wider or dense connectors. This comparison does not treat additional updates as additional data.

C.4 EAGLE-3 LONGER-OUTPUT CAPACITY COMPARISON

Two disjoint 64-question shards evaluate eight saved EAGLE-3 maps and three shared controls on the same 128 exposed MATH development questions. All 1,408 expected method/request pairs are present. The 2,048-token cap, fitting checkpoints, source revisions and pinned scorer remain fixed across shards. An eight-request pilot precedes this evaluation.

Table 44: EAGLE-3 with an 8B target on 128 exposed development questions. Dense retention uses the $N = 2048$ endpoint. Intervals are paired 95% request-bootstrap intervals conditional on the fits. Cap counts all outputs reaching 2,048 tokens. It is not an exact truncation count.

Method	N	Tokens/s	Dense retained (%) [95% CI]	Correct	Cap
AR	–	36.32	38.19 [37.14, 39.25]	104/128	11
Native drafter	–	94.89	99.78 [98.86, 100.80]	105/128	7
Source reuse	–	66.80	70.24 [69.69, 70.84]	105/128	7
Dense	512	93.47	98.28 [97.59, 99.01]	105/128	7
Dense	2048	95.10	100.00 [100.00, 100.00]	105/128	7
Linear 2048	512	93.85	98.69 [98.08, 99.31]	105/128	7
Linear 2048	2048	93.59	98.41 [97.76, 99.12]	105/128	7
MLP 2048	512	84.67	89.03 [88.24, 89.87]	105/128	7
MLP 2048	2048	89.25	93.85 [93.27, 94.46]	105/128	7
Linear 512	2048	84.91	89.29 [88.54, 90.09]	105/128	7
MLP 512	2048	79.76	83.87 [82.94, 84.84]	105/128	7

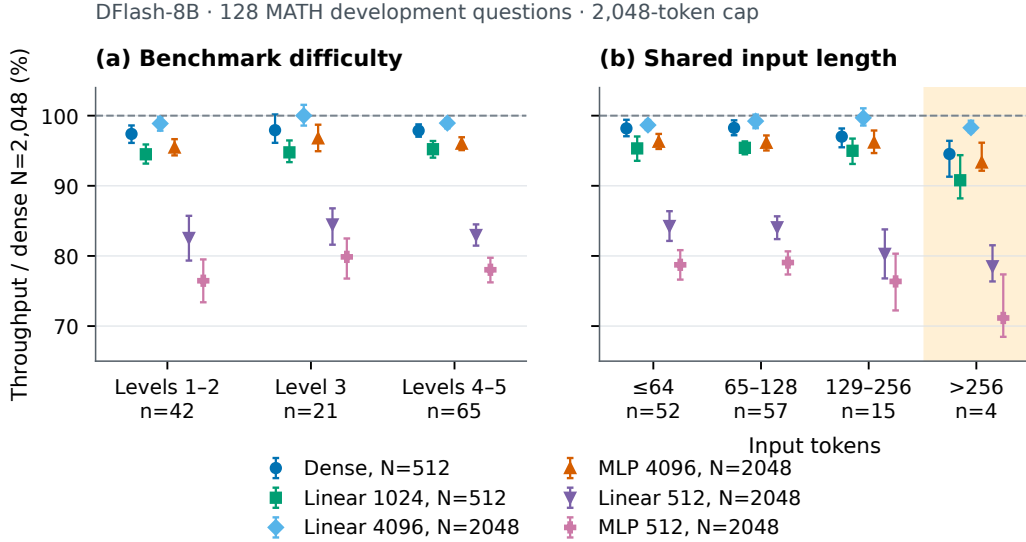
Dense $N = 512$ retains 98.28% of dense $N = 2048$ throughput. Linear-2048 at $N = 512$ retains 98.69%, while the two MLP-2048 fits retain 89.03% and 93.85% at $N = 512$ and $N = 2048$. The width-512 endpoints are slower. Every speculative method scores 105/128 with seven cap-length outputs. AR scores 104/128 with eleven. All speculative methods have identical per-question correctness in this sample, but the conservative paired 95% difference interval is still $[-3.37, 3.37]$ percentage points. Against AR, dense $N = 2048$ has four beneficial and three adverse correctness discordances. Equal or close scores therefore do not establish tight noninferiority or uncapped answer quality. The observed small-data retention replicates across the two drafter families under these development conditions, not across independent benchmark samples.

C.5 INPUT-ONLY TASK COMPLEXITY

We stratify the same 128 development requests using the benchmark’s original difficulty levels and subjects, together with shared tokenized input length. Group definitions are fixed before computing subgroup outcomes. They do not use candidate success, generated output length or latency. This is a retrospective analysis of exposed development data, not a new confirmatory test. All twelve methods and all requests remain in the analysis.

Table 45: Throughput retained from dense $N = 2048$ within each MATH difficulty group, in percent with descriptive paired 95% intervals. All fits use 8,192 updates except the continued width-512 fits, which use 32,768 updates on the same 2,048 examples. For fitted maps, unspecified N is 2,048. Intervals do not adjust for multiple subgroup comparisons or model selection.

Setting	Levels 1–2 ($n = 42$)	Level 3 ($n = 21$)	Levels 4–5 ($n = 65$)
AR	20.1 [18.6, 21.5]	19.0 [16.5, 22.1]	18.7 [17.8, 19.7]
Native drafter	101.4 [99.1, 104.1]	104.2 [102.0, 106.6]	103.7 [102.8, 104.6]
Source reuse	60.6 [59.2, 61.8]	60.7 [59.2, 62.1]	61.3 [60.6, 61.9]
Dense, $N = 512$	97.4 [96.1, 98.6]	97.9 [96.1, 100.2]	97.9 [97.0, 98.8]
Dense, $N = 2048$	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Linear 1024, $N = 512$	94.5 [93.2, 95.9]	94.8 [93.4, 96.5]	95.2 [94.0, 96.4]
Linear 4096, $N = 2048$	98.9 [97.9, 99.8]	100.0 [98.6, 101.5]	98.9 [98.2, 99.6]
MLP 4096, $N = 2048$	95.5 [94.3, 96.6]	96.8 [94.9, 98.7]	96.0 [95.1, 96.9]
Linear 512	82.6 [79.3, 85.7]	84.5 [81.6, 86.8]	83.0 [81.5, 84.5]
Linear 512, continued	84.1 [80.6, 87.5]	86.2 [84.0, 88.2]	84.2 [82.7, 85.7]
MLP 512	76.5 [73.4, 79.5]	79.9 [76.8, 82.5]	78.0 [76.2, 79.7]
MLP 512, continued	78.4 [74.6, 82.0]	79.8 [77.0, 82.0]	80.0 [78.1, 81.7]



95% paired request intervals; shaded $n=4$ group is exploratory.

Figure 18: Input-only subgroup analysis of 128 exposed MATH questions. Six original 8,192-update fits illustrate small-data and capacity contrasts, relative to dense $N = 2048$ within each group. Labels give group counts, with the four-request input group shaded. Whiskers are descriptive paired 95% request intervals without multiplicity or selection adjustment. The full twelve-method and subject analyses remain in the evidence.

Dense $N = 512$ retains 97.4–97.9% of dense $N = 2048$ throughput across the three difficulty groups. The width-1024 linear map stays near the 95% boundary, while MLP-4096 retains 95.4–96.8%. The smaller width-512 maps remain slower in every group. These results describe the tested capacities and workloads. They do not establish an optimal size across settings.

For levels 1–2, level 3 and levels 4–5, respectively, AR answers 38/42, 19/21 and 47/65 questions correctly. Every mapper answers 38/42, 18/21 and 49/65 correctly. All eleven AR cap hits occur in the highest difficulty group, as do eight of the nine mapper cap hits. Difficulty therefore coincides with changes in capped answer quality and output length. It is not a controlled intervention on decoding cost.

Input-length bins of at most 64, 65–128, 129–256 and more than 256 tokens contain 52, 57, 15 and 4 requests. In the final four-request group, dense $N = 512$ retains 94.5% [91.3, 96.4]% and linear-1024 retains 90.8% [88.2, 94.4]% of dense $N = 2048$ throughput. The small sample and multiple comparisons make this a signal for replication, not evidence of a causal length effect or a new tuning rule. All seven original subject strata, input assignments, method outcomes and conservative paired accuracy intervals are retained in the accompanying artifacts. No additional generation or fitting is used for this analysis.

C.6 CAPACITY REPLICATION WITH A 14B TARGET

We repeat the fixed-data capacity comparison with Qwen3-14B for both DFlash and EAGLE-3. Each family has fourteen primary fits: dense, factored linear and GELU MLP widths 512, 1,024 and 4,096 at each of $N = 512$ and $N = 2048$. Two additional dense fits use seed 1730. All receive 8,192 batch-four updates at learning rate 6×10^{-4} , with no L2 penalty or weight decay. Validation uses the same 1,024 records within each family. Training diagnostics use a 256-record prefix. The five target taps give 25,600 input dimensions and the output has 2,560 dimensions. Dense therefore has 65.54M parameters, versus 14.42M, 28.84M and 115.34M at the three matched factorized/MLP widths. Width 4,096 does not increase the maximum rank of a linear function beyond the output dimension.

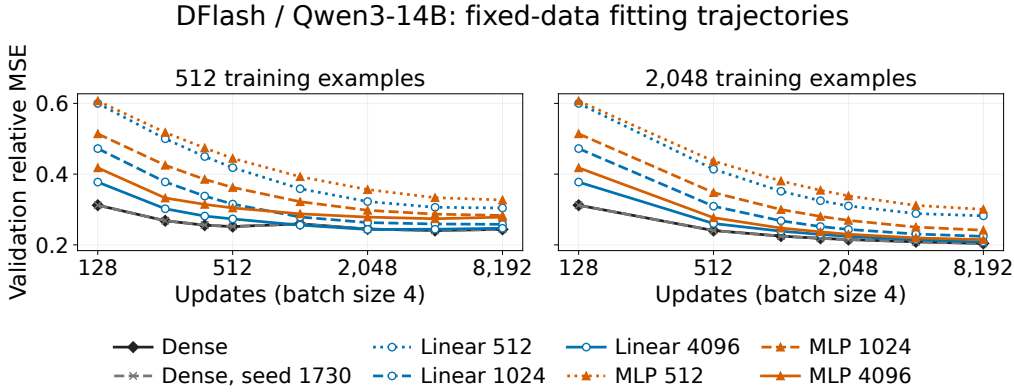


Figure 19: DFlash with Qwen3-14B. Validation error across the saved training trajectory, with all capacities and both dense seeds retained. The decoding endpoint remains fixed at 8,192 updates.

Table 46: DFlash-14B fixed endpoints. Throughput retention uses dense $N = 2048$, seed 1729. Intervals are paired 95% request-bootstrap intervals on sixteen exposed questions capped at 256 output tokens. They exclude fitting-seed and selection uncertainty.

Mapper	N	Params (M)	Train / val.	Tokens/s	Retained (%)
Dense	512	65.54	0.153 / 0.244	123.38	96.0 [92.3, 99.3]
Dense (seed 1730)	512	65.54	0.153 / 0.244	122.08	94.9 [91.5, 98.1]
Linear 512	512	14.42	0.251 / 0.304	99.16	77.1 [72.6, 81.8]
Linear 1024	512	28.84	0.180 / 0.258	117.39	91.3 [87.7, 95.1]
Linear 4096	512	115.34	0.155 / 0.247	121.65	94.6 [91.4, 97.9]
MLP 512	512	14.42	0.277 / 0.327	92.11	71.6 [68.1, 75.8]
MLP 1024	512	28.84	0.208 / 0.283	109.02	84.8 [79.7, 90.3]
MLP 4096	512	115.34	0.140 / 0.278	112.85	87.8 [83.5, 92.0]
Dense	2048	65.54	0.188 / 0.204	128.58	100.0 [100.0, 100.0]
Dense (seed 1730)	2048	65.54	0.188 / 0.204	129.55	100.8 [99.5, 102.1]
Linear 512	2048	14.42	0.274 / 0.282	105.53	82.1 [76.9, 87.4]
Linear 1024	2048	28.84	0.212 / 0.224	127.67	99.3 [96.5, 102.3]
Linear 4096	2048	115.34	0.193 / 0.208	126.96	98.7 [96.5, 101.2]
MLP 512	2048	14.42	0.292 / 0.300	96.46	75.0 [70.9, 79.2]
MLP 1024	2048	28.84	0.227 / 0.241	121.21	94.3 [90.9, 98.1]
MLP 4096	2048	115.34	0.174 / 0.215	122.32	95.1 [92.6, 98.2]

With DFlash, the wide MLP at $N = 2048$ has lower training error than dense, but higher validation error and lower measured throughput. The width-1024 linear map nearly matches dense throughput with fewer parameters. These comparisons use fixed endpoints, not checkpoints chosen after inspecting decoding. Every method reaches the output cap on these sixteen questions, so the short scores do not establish full-answer quality. The runtime includes AR and source-reuse controls. No native 14B DFlash checkpoint is substituted from another target.

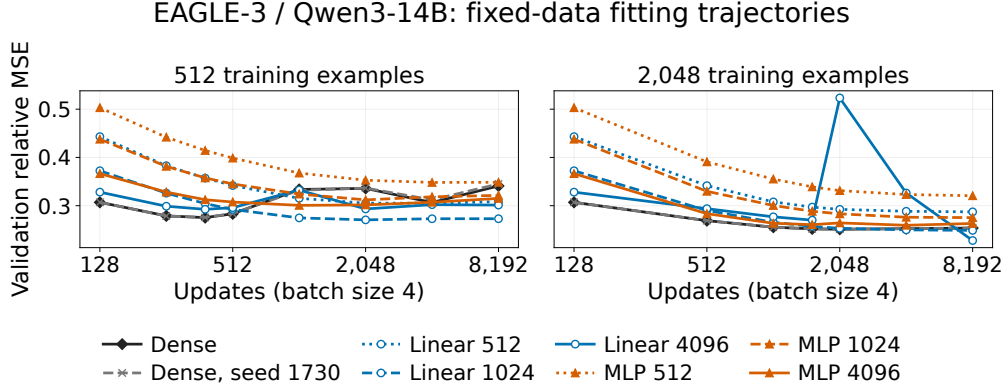


Figure 20: EAGLE-3 with Qwen3-14B under the same data-count and update budgets. The width-4096 linear trajectory retains its intermediate validation spike. Curves show optimization behavior, not a ranking of decoding performance or a selected stopping rule.

Table 47: EAGLE-3-14B fixed endpoints under its own runtime. Retention and intervals use the same within-family protocol as Table 46. Absolute throughput across the two tables is not an algorithm-only comparison.

Mapper	N	Params (M)	Train / val.	Tokens/s	Retained (%)
Dense	512	65.54	0.234 / 0.341	62.24	94.4 [91.7, 97.2]
Dense (seed 1730)	512	65.54	0.238 / 0.345	63.32	96.1 [92.7, 99.5]
Linear 512	512	14.42	0.256 / 0.308	56.92	86.4 [83.5, 89.4]
Linear 1024	512	28.84	0.197 / 0.273	62.50	94.8 [91.9, 98.0]
Linear 4096	512	115.34	0.226 / 0.301	60.90	92.4 [89.8, 95.1]
MLP 512	512	14.42	0.302 / 0.349	53.15	80.6 [77.4, 84.1]
MLP 1024	512	28.84	0.253 / 0.321	59.45	90.2 [87.3, 92.8]
MLP 4096	512	115.34	0.224 / 0.315	52.64	79.9 [76.6, 83.5]
Dense	2048	65.54	0.231 / 0.254	65.91	100.0 [100.0, 100.0]
Dense (seed 1730)	2048	65.54	0.231 / 0.254	66.45	100.8 [99.3, 102.4]
Linear 512	2048	14.42	0.279 / 0.287	58.92	89.4 [86.0, 92.7]
Linear 1024	2048	28.84	0.236 / 0.249	63.41	96.2 [94.6, 97.8]
Linear 4096	2048	115.34	0.216 / 0.228	65.61	99.5 [97.5, 101.9]
MLP 512	2048	14.42	0.311 / 0.321	56.54	85.8 [82.2, 89.3]
MLP 1024	2048	28.84	0.259 / 0.275	61.76	93.7 [90.5, 97.0]
MLP 4096	2048	115.34	0.227 / 0.263	64.02	97.1 [94.9, 99.4]

EAGLE-3 differs from DFlash in its fitting dynamics. At $N = 512$, both dense seeds have their lowest saved validation error at 384 updates, followed by substantial increases. At $N = 2048$, the wide linear endpoint has lower validation error than dense, but nearly equal decoding throughput. Its intermediate spike also shows that the shared learning rate does not produce uniformly smooth training. We complete all four previously declared lower-rate checks at 8,192 updates, preserving the original records, feature cache and seed. Lower rates improve every endpoint’s feature-validation error and avoid the wide-linear spike at the saved checkpoints (Table 48). Paired decoding compares each new map with its original-rate counterpart. The best-fitting dense map is not the fastest: rate 0.00006 gives validation error 0.226, versus 0.250 at 0.0002, but the latter has higher throughput. Thus optimization affects the capacity comparison, while feature error remains an imperfect decoding surrogate.

Mapper	Records	LR	Original val.	New val.	Ratio (95% CI)
Dense	512	0.0002	0.341	0.250	1.069 [1.042,1.098]
Dense	512	6e-05	0.341	0.226	1.037 [1.011,1.064]
Linear-4096	2,048	0.0002	0.228	0.213	1.017 [0.993,1.040]
MLP-4096	2,048	0.0002	0.263	0.214	1.044 [1.014,1.070]

Table 48: Targeted EAGLE-3/14B learning-rate checks, one fitting seed. Original learning rate is 0.0006. Ratios compare the lower-rate map with the original map at matched architecture, records and 8,192 updates on eight exposed MATH questions, with paired 95% request intervals. All four checks are retained. Four of eight outputs reach the 512-token cap for every method, so these are development throughput results, not full-answer quality confirmation.

A fixed full-answer follow-up at 14B. We freeze the original-rate dense $N \in \{512, 2048\}$ checkpoints and the matched-size linear-4096 and MLP-4096 maps at $N = 2048$, all at 8,192 updates. A separate development campaign evaluates them with AR on 32 fixed GSM8K questions in each drafter family, using a 2,048-token cap. The questions were previously exposed and are shared with the sampled 8B study, so this is not independent confirmation. Every declared output is retained. All ten family/method cells score 31/32 and avoid the cap (Table 49). Dense $N = 512$ retains 97.53% [95.77,99.48]% of dense $N = 2048$ throughput for DFlash and 97.17% [95.36,99.25]% for EAGLE-3. The wider MLP retains 92.14% and 95.73%, respectively, while matched-size linear maps retain 98.80% and 100.28%. This extends the data-efficiency and capacity comparison to completed answers at 14B under the original fitting recipe. It does not compare the best optimizer for each architecture or establish tight quality noninferiority. Broader tasks and fitting-seed uncertainty remain outside this 32-question follow-up.

Family	Method	Weights	Tokens/s	Dense retained (95% CI)	Correct	Cap
DFlash	AR	–	22.98	22.72 [21.43,23.98]%	31/32	0
DFlash	Dense,512	65.54M	98.67	97.53 [95.77,99.48]%	31/32	0
DFlash	Dense,2048	65.54M	101.17	100.00 [100.00,100.00]%	31/32	0
DFlash	Linear4096,2048	115.34M	99.96	98.80 [97.81,99.95]%	31/32	0
DFlash	MLP4096,2048	115.34M	93.21	92.14 [90.37,94.10]%	31/32	0
EAGLE-3	AR	–	22.97	32.54 [31.17,33.90]%	31/32	0
EAGLE-3	Dense,512	65.54M	68.60	97.17 [95.36,99.25]%	31/32	0
EAGLE-3	Dense,2048	65.54M	70.59	100.00 [100.00,100.00]%	31/32	0
EAGLE-3	Linear4096,2048	115.34M	70.79	100.28 [98.42,102.40]%	31/32	0
EAGLE-3	MLP4096,2048	115.34M	67.58	95.73 [92.32,99.08]%	31/32	0

Table 49: Full-answer 14B development comparison, 32 shared GSM8K questions per family. All fitted checkpoints use learning rate 0.0006 and 8,192 updates. Weight counts concern the mapper. Paired 95% intervals use 10,000 question-bootstrap draws, seed 1729, conditioning on the fixed checkpoints. Correctness does not establish one-point noninferiority.

C.7 MATCHED WARM-TRAINING BUDGETS ON 512 EXAMPLES

Table 50: End-to-end tokens/s after matched warm-training budgets, using one common DFlash-8B runtime and the same 512 records. Columns are nominal training seconds. Bold marks the best tested adaptation in each column. All twelve fits and paired intervals appear in Table 51.

Adaptation	22.9 s	91.4 s	365.6 s
Feature MSE	145.05	144.80	144.55
Connector CE	111.04	109.60	105.79
LoRA32 (1729)	119.30	115.02	118.49
LoRA32 (1730)	119.52	115.49	119.55

We compare feature regression, connector cross-entropy (CE) and rank-32 drafter LoRA on the same 512-example pool. CE updates the connector while LoRA holds the initial connector fixed. Both

use target-greedy teacher labels on the last 15 proposal positions. An equal three-rate development screen selects learning rate 2×10^{-5} for each adaptation family. Feature regression uses its declared rate 6×10^{-4} . The two LoRA seeds vary the update initialization and share the same initial mapper. They do not measure variation across initial mapper fits.

We measure 91.413 seconds of optimizer-update time for the 8,192-update feature baseline and declare budgets of 0.25, 1 and 4 times that cost. CE and LoRA begin with a 128-update mapper, whose measured 1.546 seconds of training counts toward each budget. Each worker stops after the first completed update reaching its remaining budget. Recorded overshoot is at most one update, ranging from 0.002 to 0.201 seconds across the twelve fits. The three four-GPU fitting jobs finish in 2m33s, 2m35s and 7m12s.

Table 51: Measured training costs and subsequent decoding in one common development campaign. Worker time includes model/cache setup, validation and export where performed, plus the measured initial mapper worker cost for CE and LoRA. It excludes Python startup and shared cache construction. Ratios compare against feature regression at the same warm budget.

Budget	Objective	Updates	Warm s	Worker s	Tok/s	Feature ratio [95% CI]
0.25×	Feature MSE	2026	22.86	52.04	145.05	1.000 [1.000, 1.000]
0.25×	Connector CE	106	22.94	91.89	111.04	0.766 [0.737, 0.791]
0.25×	LoRA32 (1729)	95	23.00	94.66	119.30	0.823 [0.786, 0.859]
0.25×	LoRA32 (1730)	94	22.86	95.20	119.52	0.824 [0.788, 0.860]
1×	Feature MSE	8144	91.42	119.71	144.80	1.000 [1.000, 1.000]
1×	Connector CE	474	91.58	161.04	109.60	0.757 [0.726, 0.786]
1×	LoRA32 (1729)	402	91.45	165.73	115.02	0.794 [0.757, 0.831]
1×	LoRA32 (1730)	407	91.50	164.61	115.49	0.798 [0.765, 0.830]
4×	Feature MSE	32446	365.65	390.11	144.55	1.000 [1.000, 1.000]
4×	Connector CE	1899	365.72	435.56	105.79	0.732 [0.699, 0.763]
4×	LoRA32 (1729)	1651	365.85	438.51	118.49	0.820 [0.786, 0.853]
4×	LoRA32 (1730)	1646	365.76	437.99	119.55	0.827 [0.793, 0.860]

Table 52: References measured within the same budget-decoding campaign. The initial mapper has 128 feature-regression updates. The 8,192-update reference is an existing saved checkpoint.

Reference	Tok/s	Initial mapper ratio [95% CI]
AR	28.61	0.240 [0.207, 0.277]
Native drafter	151.71	1.275 [1.205, 1.348]
Source reuse	84.71	0.712 [0.684, 0.737]
Initial mapper	119.01	1.000 [1.000, 1.000]
Dense, 8192 updates	145.82	1.225 [1.154, 1.302]

Feature regression reaches 145.05, 144.80 and 144.55 tokens/s at the three budgets. Relative to the quarter-budget fit, the 1-times and 4-times fits retain 99.83% [97.18, 102.53]% and 99.66% [97.35, 102.11]%, respectively. These intervals do not establish a benefit from longer fitting. Connector CE retains 73.2–76.6% of same-budget feature throughput, and the two LoRA seeds retain 79.4–82.7%. This result concerns the declared teacher-forced objective, initialization, rank and development-selected learning rates. It does not establish that other adaptation objectives or full drafter training fail.

Warm training is a marginal cost with cached features available. The historical full-cache extraction took 712.58 seconds and produced 292.8 GB. We reuse its first 512 training records and report the original construction cost separately. Dividing that cost by the record count would not measure a fresh 512-example extraction. The timed fits do not add training examples or repeat large-data scaling.

All seventeen methods decode the same sixteen exposed development questions with a 256-token cap. Tables 51 and 52 report descriptive paired request bootstrap intervals. They do not provide full-answer quality, untouched confirmation or corrections for selecting among multiple comparisons. Abso-

lute throughput is interpreted within this common campaign and its deterministic training/runtime configuration.

C.8 DRAFT LENGTH AND INTERFACE CHOICES

The block-size experiment directly compares AR, native DFlash and the relay at lengths 8, 16 and 32 on 64 fixed requests. Block 16 gives the highest relay throughput, 210.15 tokens/s (Table 12). Block 8 retains more source progress proportionally but advances fewer tokens per cycle. Block 32 adds verification width while reducing accepted progress for the released block-16 drafter. This supports keeping its trained block length instead of maximizing the relative gain over source reuse.

The EAGLE-3 normalization comparison holds parameter count, data and 128-update budget fixed. Raw target features reduce relative context error from 0.413 to 0.266 and increase the source-relative throughput ratio from 1.035 to 1.237. The raw interface consumes feature magnitude, which input normalization removes. This gives a concrete reason for preserving scale in the selected map.

Appendix C reports the DFlash objective, ridge and normalization comparisons with their distinct fitting recipes.

D EXTERNAL BASELINES IN THEIR PUBLIC RUNTIMES

Table 53: Mechanism and evaluation scope of the closest alternatives. All rows inherit pretrained models. “No new fit” for PARD refers to retargeting its released checkpoint, not its original training cost. DFlash and EAGLE-3 are the inherited drafter families rather than independent competitors to RelaySpec’s interface replacement.

Method	New-target adaptation	Use of target features	Evidence here
RelaySpec	Fit input map, freeze drafter	Replace source conditioning	Qwen and Llama targets, two drafter families
PARD (An et al., 2026)	No new fit for released drafter	Target-independent proposals	Public runtime, 128-question capped quality
SD ² (Berdoz et al., 2026)	Fit steering in tested frozen-drafter option	Inject steering within drafter	Rate and epoch sweeps, 128-question capped quality
Connector CE	Fit input map with token CE	Same replacement interface	Common-runtime matched warm budgets
Drafter LoRA	Fit initial map, then LoRA with map frozen	Adapt feature-conditioned drafter	Common-runtime matched warm budgets, two seeds

We evaluate SD²’s public frozen-drafter option and the released PARD parallel drafter on sixteen exposed MATH development requests with a 256-token cap. Each system has a correct cached AR control using its own target runtime and identical pretokenized inputs. These are short development comparisons. Below we separately evaluate capped answer quality. No untouched confirmation is included in these measurements.

SD² trains 402,665,472 steering parameters with merged guidance from target layers 3, 16 and 33 into all drafter layers. Inherited target and drafter weights remain frozen. Each of six objective/rate settings sees 512 distinct records once, using 128 batch-four updates and 95,835 non-padding supervised positions. Records are capped at 192 tokens. Training takes 47.0–48.6 seconds per fit. The initial TVD and KL settings also have exact fitting replicas, whose costs remain in the experiment record. This supervision differs from the feature-regression and last-15-position adaptation objectives in the preceding study.

Table 54: One-epoch SD² development comparison. Intervals resample paired requests 10,000 times and do not adjust for rate selection. Exact AR counts identical capped token sequences.

Setting	Tok/s	AR ratio [95% CI]	Independent ratio [95% CI]	Exact AR
Matched AR	12.64	1.000 [1.000, 1.000]	0.671 [0.622, 0.719]	16/16
Independent drafter	18.83	1.490 [1.391, 1.607]	1.000 [1.000, 1.000]	6/16
Zero guidance	18.56	1.469 [1.371, 1.584]	0.985 [0.983, 0.988]	6/16
TVD, 10^{-5}	18.38	1.454 [1.360, 1.567]	0.976 [0.955, 0.997]	4/16
KL, 10^{-5}	18.17	1.438 [1.341, 1.552]	0.965 [0.937, 0.993]	9/16
TVD, 4×10^{-6}	18.52	1.466 [1.370, 1.574]	0.984 [0.961, 1.006]	6/16
TVD, 2×10^{-5}	18.47	1.462 [1.365, 1.570]	0.981 [0.953, 1.011]	6/16
KL, 4×10^{-6}	18.66	1.477 [1.384, 1.588]	0.991 [0.962, 1.016]	5/16
KL, 2×10^{-5}	18.42	1.458 [1.363, 1.569]	0.978 [0.955, 1.003]	10/16

The public SD² evaluation configuration uses FP16 target storage, BF16 drafter and steering storage, and BF16 autocast under Transformers 4.52.4. We verify original FP32 training weights before conversion and separately fingerprint the BF16 inference tensors. The best fitted point, KL at 4×10^{-6} , retains 99.11% [96.19, 101.62]% of independent-drafter throughput. This screen does not establish a steering gain. We freeze this rate before checking additional epochs on the same small pool.

Table 55: Released PARD with zero new-target fitting, Transformers 4.51.3, BF16 weights, eager attention and static caches. The released checkpoint carries inherited proposer training.

Setting	Tok/s	AR ratio [95% CI]	Exact AR
Matched eager AR	31.27	1.000 [1.000, 1.000]	16/16
Released PARD	98.60	3.153 [2.831, 3.558]	4/16

Every SD² committed token matches its actual verifier decision, and immediate versus deferred observation preserves tokens and acceptance. Each timed PARD request has a separate verifier-observed replica with identical raw output and acceptance counts. PARD’s measured generation omits that added observer. Both timers include prefills and cache setup, preserve raw EOS/cap overshoot and exclude detokenization. Earlier failed exact-AR gates remain failed. Controlled cache, query and attention replays expose changes in rounded target scores, while future-query perturbations preserve earlier logits in the tested calls. The sequence differences in these tables require separate task-quality evidence. Absolute speeds across these public runtimes do not isolate algorithm effects, and shared model residency does not measure isolated serving memory.

D.1 ADDITIONAL EPOCHS AND CAPPED ANSWER QUALITY

After the six-rate screen, we fix KL and the selected learning rate and fit for 1, 2, 4 and 8 epochs on the same 512 examples. These are four fresh fits, each with its own learning-rate decay horizon, rather than checkpoints from one training trajectory. The one-epoch fit exactly reproduces the selected earlier training fingerprint. All four endpoints are evaluated together on the same sixteen development requests, with the public SD² runtime and matched controls.

Table 56: SD² epoch budget at fixed 512 distinct examples. Training KL is the mean over updates in each final epoch. It is not held-out loss. Independent-drafter throughput is 18.75 tokens/s. Intervals resample paired requests and do not adjust for model selection.

Epochs	Fit seconds	Final-epoch training KL	Tok/s	Independent ratio [95% CI]
1	48.44	0.3923	18.56	0.9900 [0.9607, 1.0149]
2	97.01	0.2787	18.45	0.9841 [0.9639, 1.0033]
4	193.81	0.2027	18.39	0.9808 [0.9555, 1.0039]
8	388.08	0.1254	18.26	0.9738 [0.9488, 0.9997]

Training KL decreases from 0.3923 to 0.1254, while every fitted endpoint has a throughput point below the independent drafter. Additional epochs do not establish a decoding improvement in this fixed-pool setting. The highest-throughput fitted endpoint remains one epoch. We freeze this checkpoint before longer quality evaluation. This result concerns the tested frozen-drafter configuration, supervision and pool, and does not establish a general failure of SD² or its other training options.

Table 57: PARD capped answer quality on 128 exposed MATH development requests, after an eight-request pilot passed. Both methods use the same 2048-token cap and public BF16 eager runtime. Cap hits include all outputs reaching the cap. None ends in EOS exactly at the cap.

Setting	Tok/s	AR ratio [95% CI]	Correct	Cap hits
Matched eager AR	31.67	1.000 [1.000, 1.000]	103/128	11
Released PARD	105.63	3.335 [3.222, 3.451]	105/128	8

PARD has five beneficial and three adverse correctness discordances against AR, a difference of +1.56 percentage points. A conservative paired 95% interval is $[-6.34, 9.30]$ points: form exact 97.5% Clopper-Pearson intervals for each discordance probability and combine their opposite endpoints using a Bonferroni bound. This construction requires IID request pairs and a fixed candidate. It does not establish a one-percentage-point noninferiority margin. The descriptive paired bootstrap interval is $[-2.34, 5.47]$ points. Sparse or absent discordances can make such intervals misleadingly narrow. These external-baseline measurements use development questions.

PARD matches AR’s full capped token sequence on 28 of 128 requests. Its mean output length is 664.5 tokens versus AR’s 690.4, so the 3.34-fold token-throughput ratio differs from the 3.47-fold request-time ratio. Every measured PARD request retains a separate, exactly matching verifier-observed replica. These are development outcomes under the specified runtime and scorer, not untouched confirmation or an isolated algorithm ranking across runtimes.

Table 58: Selected one-epoch SD² on the same 128 exposed MATH development requests, with a 2048-token cap. The public mixed-precision runtime and matched AR control are retained. Both have 11 cap-length outputs, none with EOS exactly at the cap.

Setting	Tok/s	AR ratio [95% CI]	Correct	Cap hits
Matched AR	12.48	1.000 [1.000, 1.000]	103/128	11
Selected SD ²	19.92	1.596 [1.560, 1.632]	103/128	11

SD² and matched AR both score 103 of 128, with three beneficial and three adverse correctness discordances. The conservative paired 95% interval is $[-7.04, 7.04]$ percentage points, which does not establish the one-point noninferiority margin. The descriptive paired bootstrap interval is $[-3.91, 3.91]$ points. Full capped sequences agree on 42 requests. Mean output lengths are 679.1 versus 680.9 tokens, so the 1.60-fold token-throughput and request-time ratios are close. Both shards retain the same selected training and inference fingerprints, and saved answers reproduce under the pinned scorer.

The public physical-buffer limit can end an untimed warmup before its token cap. We record this termination only during warmup. Measured requests must still reach EOS or the cap and pass all actual-verifier checks. A replacement eight-request pilot passed before the complete quality run. Failed preliminary attempts remain excluded from the reported timings. These results concern the tested frozen-drafter configuration and its public runtime, not all SD² training options.

E ACCEPTANCE AND NUMERICAL AGREEMENT

A retention value of 90% means the reused drafter produces 90% as many tokens per second as the target’s own drafter. It is not the fraction of speedup above AR. We also report the ratio of summed AR request time to summed relay request time. This differs from a throughput ratio when output lengths differ.

Acceptance explains how speed is obtained. We report the pooled number of tokens advanced per proposal-verification cycle, including the implementation’s target-supplied progress. This is distinct from the fraction of proposed draft tokens accepted. Appendix E gives the position-wise curves and precise counting convention.

Let c_R be the average relay cycle time and a_R its average progress. Its decode time per output token is approximately c_R/a_R . If c_A is AR time per token, then the decode-only approximation is

$$\text{AR-relative decode speed} \approx \frac{a_R c_A}{c_R}. \quad (5)$$

Removing source conditioning reduces cycle cost. Imperfect feature matching may reduce progress. The map is useful when the saved computation outweighs the extra cycles. We measure end-to-end speed directly and use this approximation to interpret the component measurements.

We pool all recorded proposal cycles rather than averaging each request’s mean. This weights cycles equally. DFlash’s recorded length is one already target-selected position plus the accepted draft prefix. EAGLE-3’s length follows its shared decoder’s progress count. These quantities measure work advanced per cycle, including target-supplied progress. A proposed-token acceptance fraction requires separating those positions and accounting for the number proposed.

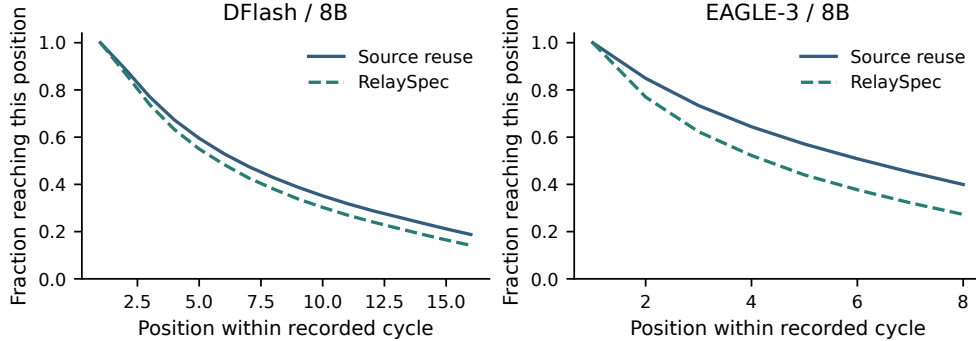


Figure 21: Historical 8B source-reuse comparisons. Each curve gives the fraction of recorded cycles reaching each position. Summing the curve values gives mean recorded progress per cycle. Solid and dashed lines show source reuse and RelaySpec, respectively.

The main MATH-500 table reports measured full-sequence agreement with AR. A fresh diagnostic selects eight prompts with early differences and saves target inputs, cache lengths, positions and the two leading logits at every call. The DFlash paths share the same first differing choice across native, source and relay drafting. The observed BF16 logits include ties or changed ordering with gaps up to 0.25 at those positions. Such traces identify numerical score differences in the selected cases. They do not establish that all mismatches have the same cause. The task-quality intervals in Table 1 remain the empirical quality evidence.

E.1 SAMPLED DECODING WITH FROZEN INTERFACES

We test whether the frozen 512-record retargeting maps also support nonzero-temperature generation. A seeded operational pilot checks byte-identical checkpoint controls in both families at temperatures 0.6 and 1.0: tokens, acceptance and verifier-call trajectories reproduce. An earlier unseeded pilot is excluded from reproducible sampling evidence. The measurement cohort is fixed before decoding: 32 previously exposed GSM8K questions, two sampling seed bases, both families and both temperatures. These questions are disjoint from the entire 256-question reserve and a separate initial eight-question sampling study. The latter is not pooled into this result. Each family/temperature cell contains 64 outputs per method, from 32 question clusters. All 768 declared outputs are retained and scored, with no 2,048-token cap hits.

RelaySpec achieves 3.90 times AR throughput at temperature 0.6 and 3.79 at 1.0 with DFlash, and 2.64 and 2.47 times AR with EAGLE-3 (Table 59). It retains 92.6% and 94.4% of native DFlash

throughput, and 97.7% and 95.5% of native EAGLE-3 throughput. Thus applicability extends beyond greedy decoding in this setting. We report request-time ratios separately because sampled methods can generate different output lengths. These comparisons use the original retargeting maps, not the compact native students.

A runtime control matters for this comparison. Released DFlash proposes greedily even when its target samples. An earlier RelaySpec run instead sampled draft proposals and retained 87.7% and 80.2% of native throughput. The reported DFlash repeat matches the released proposal policy on the same 32 questions and both seeds. A separate eight-question full-projection control then reproduces all native outputs and acceptance trajectories at both temperatures, with approximately 99% throughput retention. The policy correction is not a learned-interface or algorithmic contribution.

The bootstrap resamples questions jointly across both seeds and all paired methods, using 10,000 draws at seed 1729. These intervals condition on two sampled seeds and do not measure fitting-seed or development selection uncertainty. Sparse correctness discordances can produce zero-width empirical bootstrap intervals, which cannot establish tight quality noninferiority. Observed scores and the operational reproduction check also do not prove distributional exactness. This remains a small, exposed development study with one request per worker.

Family / T	Method	Tokens/s	AR ratio (95% CI)	Time ratio	Tokens	Correct
DFlash / 0.6	AR	37.96	1.00 [1.00,1.00]	1.00	274.9	60/64
DFlash / 0.6	Native	160.16	4.22 [3.95,4.51]	4.20	276.5	61/64
DFlash / 0.6	RelaySpec	148.23	3.90 [3.65,4.17]	3.95	271.6	60/64
DFlash / 1.0	AR	37.26	1.00 [1.00,1.00]	1.00	281.9	59/64
DFlash / 1.0	Native	149.65	4.02 [3.75,4.31]	3.96	286.2	62/64
DFlash / 1.0	RelaySpec	141.29	3.79 [3.56,4.05]	3.79	282.0	61/64
EAGLE-3 / 0.6	AR	36.80	1.00 [1.00,1.00]	1.00	274.9	60/64
EAGLE-3 / 0.6	Native	99.44	2.70 [2.60,2.82]	2.69	276.1	61/64
EAGLE-3 / 0.6	RelaySpec	97.12	2.64 [2.51,2.77]	2.58	281.5	61/64
EAGLE-3 / 1.0	AR	37.39	1.00 [1.00,1.00]	1.00	281.9	59/64
EAGLE-3 / 1.0	Native	96.89	2.59 [2.47,2.72]	2.64	276.6	61/64
EAGLE-3 / 1.0	RelaySpec	92.49	2.47 [2.36,2.60]	2.46	283.9	60/64

Table 59: Sampled GSM8K decoding, 32 fixed questions and two sampling seeds. Each row has 64 scored outputs. Time ratio is summed AR request time relative to method time, while Tokens is mean output length. Native is target-specific and RelaySpec retargets the released 4B drafter. Paired 95% intervals cluster both seeds by question. All methods avoid the cap.

F MATCHED-COUNT TRAINING COMPOSITION

We compare two 2,048-record DFlash-8B fitting pools. The math-only arm uses NuminaMath records. The mixed arm replaces half with 1,024 Dolly general-instruction records.² Exact and lexical filters exclude matches to fitting validation, recorded evaluation prompts and the confirmation and reserve sets. The common validation set contains 1,024 math and 1,024 general-instruction records. Its cached feature files are identical across arms.

Each arm fits dense, linear-512 and MLP-512 maps at seed 1729, plus a second dense seed, 1730. Each fit uses 8,192 batch-four updates, learning rate 0.0006, no regularization and the same relative interface objective. We retain the 2,048-update checkpoint as a fitting diagnostic, but preselect the 8,192-update endpoint for decoding rather than selecting by validation. Four independent fits run simultaneously on four L40S GPUs. Feature extraction took 111.4 seconds for the math cache and 107.1 for the mixed cache. The full fitting allocations took 4m36s and 4m21s, respectively. Extraction and fitting are separate costs.

Equal record counts and updates do not imply equal token exposure. With a 192-token record cap, the math pool contains 385,492 non-padding tokens and the mixed pool 316,518. Sixteen passes expose

²[databricks/databricks-dolly-15k](https://databricks.com/databricks-dolly-15k), revision bdd27f4d94b9, CC-BY-SA 3.0. General instructions can include mathematical questions.

6,167,872 and 5,064,288 tokens, respectively. The experiment therefore measures composition at matched records and updates, not matched tokens.

Table 60: Composition fitting at the preselected 8,192-update endpoint. Validation columns use the same held-out features in both arms. Reported seconds cover per-worker data access, updates and training-log writes, excluding validation, export and setup. Training losses concern different input distributions and are not directly comparable across arms.

Mapper	Training	Train loss	Math val.	General val.	Fit seconds
Dense	math	0.1695	0.1805	0.3329	177.0
Linear-512	math	0.2629	0.2685	0.4719	114.5
MLP-512	math	0.2812	0.2878	0.4928	110.3
Dense (seed 2)	math	0.1695	0.1805	0.3328	173.6
Dense	mixed	0.1854	0.1990	0.2234	167.0
Linear-512	mixed	0.3027	0.3015	0.3293	97.1
MLP-512	mixed	0.3221	0.3233	0.3479	110.2
Dense (seed 2)	mixed	0.1852	0.1990	0.2231	152.7

Mixed training lowers dense general-instruction validation error by 32.9%, from 0.3329 to 0.2234, while raising math error by 10.2%, from 0.1805 to 0.1990. The second dense seed reproduces this tradeoff. At width 512, the linear and MLP maps show general-error reductions of 30.2% and 29.4%, with math-error increases of 12.3% in both cases. The matched linear map retains lower validation error than the MLP in both domains and both training arms. All three architectures improve in both validation domains between 2,048 and 8,192 updates.

Fixed cross-domain decoding protocol. Before observing endpoint results, we declare 32 development inputs from each of MATH-500, GSM8K, HumanEval and MT-Bench. The 128 initial inputs expand to 160 timed requests because MT-Bench includes two turns. All eight fixed endpoints run alongside AR, native DFlash and source reuse, producing 1,760 measured outputs capped at 2,048 tokens. A separate eight-input operational pilot completes in 2m58s before this full run. Methods rotate within requests in the same runtime. Ratios compare mixed and math-only fits of the same architecture and seed. Paired bootstrap intervals resample whole conversations for MT-Bench and individual requests elsewhere. They describe this development sample, without multiplicity adjustment or fitting-seed uncertainty.

The feature-domain tradeoff carries through to decoding, but depends on mapper capacity and task. Mixed dense training changes throughput by -1.2% on MATH, $+1.6\%$ on GSM8K, $+2.9\%$ on HumanEval and $+4.9\%$ on MT-Bench. The second dense seed gives -1.7% , $+1.2\%$, $+2.4\%$ and $+4.7\%$, respectively. The first seed’s MATH interval includes no change. Linear-512 and MLP-512 gain 11.6% and 10.8% on chat, but lose 6.2% and 7.8% on MATH and 2.6% and 4.1% on GSM8K. Dense remains faster than either smaller map on every task in both arms. General-domain feature improvement therefore supports useful workload-specific gains, not a uniform advantage from mixed training.

Every matched math/mixed pair produces identical generated token sequences on all 160 requests, including conversation turns. Thus these measured throughput gains do not arise from shorter outputs. All eight maps score 22/32 on MATH and 30/32 on GSM8K. AR scores 22/32 and 31/32. Four MATH outputs per method reach the cap, and none elsewhere does. The conservative paired accuracy-difference interval between matched arms is $[-12.80, 12.80]$ percentage points on each math benchmark. Equal observed outcomes on this small sample do not establish tight population quality noninferiority. Code and conversation outputs remain unscored for task quality. The full decoding allocation took 52m30s.

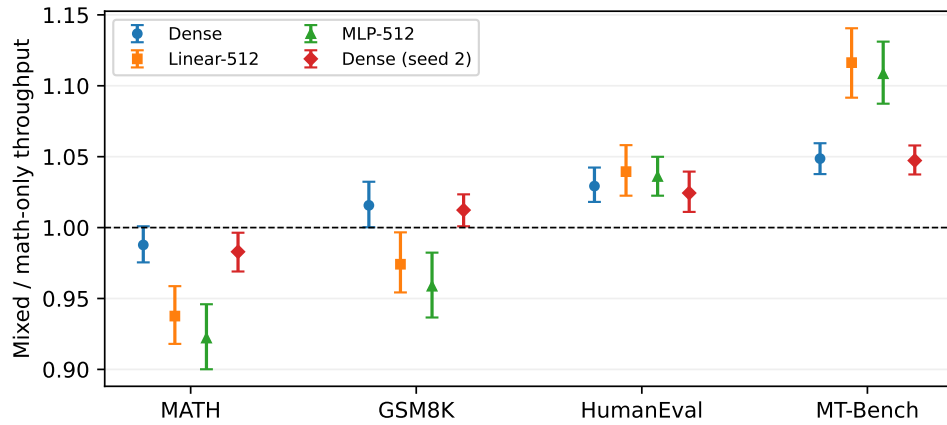


Figure 22: Effect of training composition on measured DFlash-8B throughput. Values above one favor mixed training. Marker shapes distinguish the four matched comparisons. Bars show paired 95% intervals. Code and conversation throughput does not establish output quality.

Table 61: All predeclared composition decoding comparisons. Throughput includes prompt processing. Ratios are mixed divided by math-only throughput, with paired 95% intervals. MT-Bench has 64 timed turns from 32 conversations. Each other task has 32 requests.

Task	Mapper	Math tokens/s	Mixed tokens/s	Ratio (95% CI)
MATH	Dense	201.1	198.6	0.988 [0.975,1.001]
GSM8K	Dense	156.5	159.0	1.016 [1.000,1.032]
HumanEval	Dense	138.1	142.2	1.029 [1.018,1.042]
MT-Bench	Dense	71.1	74.6	1.049 [1.038,1.059]
MATH	Linear-512	165.4	155.1	0.938 [0.918,0.959]
GSM8K	Linear-512	120.3	117.2	0.974 [0.954,0.997]
HumanEval	Linear-512	69.9	72.7	1.039 [1.022,1.058]
MT-Bench	Linear-512	49.1	54.8	1.116 [1.092,1.141]
MATH	MLP-512	155.6	143.5	0.922 [0.900,0.946]
GSM8K	MLP-512	112.7	108.0	0.959 [0.937,0.982]
HumanEval	MLP-512	64.4	66.7	1.036 [1.022,1.050]
MT-Bench	MLP-512	46.5	51.5	1.108 [1.087,1.131]
MATH	Dense (seed 2)	201.6	198.2	0.983 [0.969,0.996]
GSM8K	Dense (seed 2)	156.7	158.6	1.012 [1.001,1.023]
HumanEval	Dense (seed 2)	138.0	141.4	1.024 [1.011,1.039]
MT-Bench	Dense (seed 2)	71.1	74.5	1.047 [1.037,1.058]

Table 62: Composition evaluation: correct-answer and output-cap counts. MATH and GSM8K columns show correct answers followed by capped outputs, each out of 32. Code cap counts are out of 32 and chat cap counts out of 64 turns. HumanEval execution and MT-Bench answer quality are not scored in this experiment. Capped outputs remain in every aggregate.

Method	MATH correct / cap	GSM8K correct / cap	Code cap	Chat cap
AR	22 / 4	31 / 0	0	0
Native DFlash	22 / 4	30 / 0	0	0
Source reuse	22 / 4	30 / 0	0	0
Dense, math	22 / 4	30 / 0	0	0
Dense, mixed	22 / 4	30 / 0	0	0
Linear-512, math	22 / 4	30 / 0	0	0
Linear-512, mixed	22 / 4	30 / 0	0	0
MLP-512, math	22 / 4	30 / 0	0	0
MLP-512, mixed	22 / 4	30 / 0	0	0
Dense (seed 2), math	22 / 4	30 / 0	0	0
Dense (seed 2), mixed	22 / 4	30 / 0	0	0

F.1 COMPOSITION WITH COMPACT INTERFACES

We separately screen two-layer interfaces using 512 math records or 256 math plus 256 general-instruction records, at 8,192 updates and seed 1729. Both use target layers [25,33]. Retargeted maps fit the source-drafter interface. Native students fit the released native projection and normalization. Thus this comparison changes calibration composition at fixed interface capacity and record count, but not necessarily equal token exposure.

The largest observed difference is on retargeted dialogue: mixed calibration retains 110.8% [107.8,112.5]% of math-only throughput. All seven other intervals include parity. Every paired output matches within these capped screens. This is a workload-specific exploratory finding from eight conversations and one fitting seed, not evidence that mixed calibration improves decoding across workloads or answer quality.

Table 63: Compact-interface calibration composition. Ratios compare both checkpoints within the same worker. Paired 95% bootstrap intervals use requests, or whole conversations for dialogue, and exclude fitting-seed and selection uncertainty. Code and math screens use 128-token caps. dialogue uses 256 tokens per turn.

Interface	Workload	Units	Mixed / math throughput (95% CI)
Native	Code	32	100.6% [98.6, 102.7]
Native	Dialogue	8	101.1% [98.8, 102.7]
Native	GSM8K	32	101.2% [99.6, 103.0]
Native	MATH	32	101.5% [99.4, 103.4]
Retargeted	Code	32	102.1% [98.8, 105.6]
Retargeted	Dialogue	8	110.8% [107.8, 112.5]
Retargeted	GSM8K	32	99.3% [96.8, 101.9]
Retargeted	MATH	32	101.1% [96.7, 105.5]

We then repeat fitting at seed 1730 and compare both seeds on sixteen additional development conversations, disjoint from the initial eight. Within each worker, all four compact maps are paired. Mixed/math throughput ratios are 107.50% [104.34,110.44]% at seed 1729 and 107.52% [104.44,110.21]% at seed 1730 for retargeting. Native students give 101.09% [100.12,102.08]% and 100.28% [99.47,101.14]%, respectively. All 32 paired turn outputs match at each seed. Both seeds share the same conversations, so they are fitting replications rather than independent benchmark samples. This extension remains development evidence, with the same 256-token cap and no dialogue-quality scoring.

The fitting diagnostics help delimit the result. At seed 1729, mixed calibration lowers general-instruction validation error from 0.445 to 0.289 for retargeting and from 0.0468 to 0.0298 for native students. Both improve substantially in relative terms, but only retargeting shows a sizeable dialogue

throughput gain. Math errors increase from 0.227 to 0.251 and from 0.0241 to 0.0267, respectively. These diagnostics use the same 64 validation records per domain. The pattern suggests that domain coverage matters more for the harder retargeted interface in these tests. It also shows why proportional feature-error improvements alone do not predict proportional decoding gains.

G FROZEN-CHECKPOINT GSM8K CONFIRMATION

We freeze the dense 512- and 2,048-example checkpoints at 8,192 updates and seed 1729 before generating confirmation outputs. The fixed test contains 256 GSM8K questions, sampled after exact and lexical exclusion against fitting data and collected evaluation history. A separate 256-question reserve was left unused by this experiment. Subsequently, all 256 reserve questions were consumed by four separately frozen studies: 64 for the interface/SVD comparison and 64 different questions for native student confirmation, followed by 64 further questions for short-fit native confirmation and the final 64 for native EAGLE-3 confirmation (Appendix C). No reserve questions remain unused. Operational pilots use eight previously exposed GSM8K questions, not confirmation requests. The same confirmation questions are evaluated in both drafter families, so the second family is not an independent benchmark sample.

The primary speed criterion requires the lower endpoint of the paired 95% throughput-ratio interval to exceed 0.95. DFlash retains 97.48% [96.90,98.06]% and EAGLE-3 retains 98.90% [98.14,99.66]%. Throughput includes prompt processing. Resampling uses requests as paired units, 10,000 draws and seed 1729. These intervals condition on the frozen fitting checkpoints rather than estimating fitting-seed variation.

Table 64: All frozen-confirmation methods on 256 GSM8K requests per family. At cap counts outputs of exactly 2,048 tokens. Comparisons are within each family’s runtime.

Family	Method	Tokens/s	Correct	At cap
DFlash	AR	38.0	236/256	0
DFlash	Native drafter	170.3	236/256	0
DFlash	Source reuse	99.0	236/256	0
DFlash	Dense 512	157.8	236/256	0
DFlash	Dense 2,048	161.9	236/256	0
EAGLE-3	AR	36.9	236/256	0
EAGLE-3	Native drafter	100.9	238/256	0
EAGLE-3	Source reuse	70.7	238/256	0
EAGLE-3	Dense 512	100.0	238/256	0
EAGLE-3	Dense 2,048	101.1	238/256	0

The predeclared quality criterion requires the conservative paired 95% accuracy-difference lower bound relative to AR to exceed -1 percentage point. DFlash has two beneficial and two adverse discordances, giving $[-3.07, 3.07]$ points. EAGLE-3 has two beneficial and no adverse discordances, giving $[-1.63, 3.14]$ points. Both quality outcomes are inconclusive under this criterion. Therefore the joint speed-and-quality criterion is not established, despite both speed criteria passing. Within each family, the two mapper sizes have identical per-question correctness, with a conservative interval of ± 1.70 points. DFlash also has identical token outputs between mapper sizes on all 256 requests, while EAGLE-3 has identical outputs on 254. Comparisons with AR retain the finite-precision distinctions discussed elsewhere. No reserve extension or outcome-dependent tuning is performed. Exclusion checks concern the recorded local history and lexical matches, not unknown pretraining exposure or semantic independence.

H DATA CHECKS AND DEVELOPMENT EXPOSURE

The 4,096-record fitting manifest contains distinct problems with solutions. The loader renders both fields before applying the 192-token cap. Repeating the manifest changes presentations, not distinct records. The 32-question development set informed normalization and objective choices.

The historical 468-question complement is computed from saved full-run outputs and retains that exposure history.

Normalized exact matching finds zero overlaps between the 4,096 fitting and 1,250 evaluation problem texts. For a second check, each text is split into unique normalized tokens. Token-set Jaccard overlap is the number of shared tokens divided by the number of tokens in either set. Each evaluation question is compared with its closest fitting record. Table 65 reports the resulting similarities.

Table 65: Closest fitting-to-evaluation token-set overlap. Threshold columns count evaluation records at or above the stated overlap.

Task	Maximum overlap	≥ 0.80	≥ 0.90	≥ 0.95
MATH-500	0.980	47	13	4
GSM8K	0.378	0	0	0
HumanEval	0.290	0	0	0
MBPP	0.333	0	0	0
MT-Bench	0.571	0	0	0

MATH-500 includes 47 questions above 0.80 overlap and four above 0.95. These shared templates qualify the interpretation of math generalization. The fixed-map code and conversation evaluations give additional task coverage without changing the fitted map.

I FITTING TIME AND ISOLATED MEMORY

Table 66: Selected map size and warm fitting-loop duration on four RTX 6000 Ada GPUs. Models are loaded before the recorded loop begins.

Drafter	Target	Trainable weights (millions)	Fitting loop (seconds)
DFlash	8B	52.43	85.6
DFlash	14B	65.54	118.0
EAGLE-3	8B	52.43	86.7
EAGLE-3	14B	65.54	118.5

The fitting-loop timer covers source and target feature computation and matrix optimization. A complete setup also loads model weights, writes the map, reloads it and initializes generation. Published original training counts in Table 70 provide context for the additional-data requirement. They are not a wall-time or token-budget comparison with fitting a new native drafter.

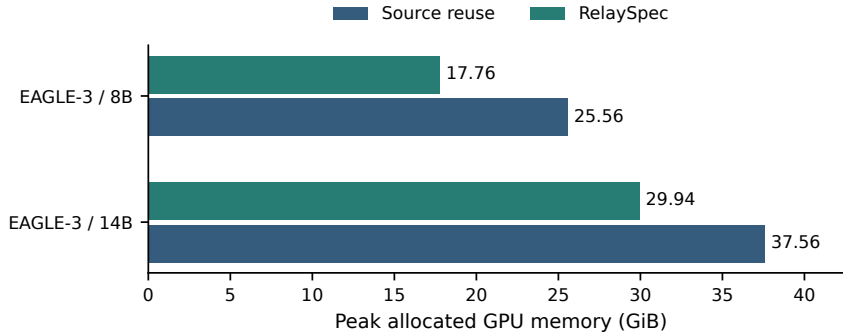


Figure 23: Isolated EAGLE-3 peak allocated memory on RTX 6000 Ada. Each method runs in a fresh process on eight paired prompts per target. Source removal saves 7.80 GiB at 8B and 7.63 GiB at 14B.

GPU peak counters are reset before the measured request. Allocated memory counts live tensor allocations and differs from reserved allocator memory. The mixed-arm processes used for timing are not the basis of this source-removal measurement.

I.1 SELECTIVE FEATURE CAPTURE DURING GENERATION

A small conditioning interface also makes it unnecessary to retain all verifier-layer outputs. We implement selective forward hooks for the requested target taps, preserving the final normalization for the last layer. The transformer, mapper weights, verifier decisions and KV-cache policy are unchanged. This is an implementation consequence of the interface, not a new compression algorithm. Even the five-tap mapper benefits: reducing the tap count is not required for the basic capture optimization.

On Qwen3-8B/DFlash, we compare the same frozen checkpoint with full hidden-state return and selective capture. Each method has two repetitions in reversed order on one constructed archive prompt per length, with a 64-token output cap. Every pair matches output token hashes, output lengths, acceptance trajectories and target/draft calls. At 31,526 input tokens, the two-tap mapper reduces peak allocated memory from 32.73 to 25.03 GiB (23.5%), and the five-tap mapper from 32.98 to 25.87 GiB. A stricter control immediately releases prefill outputs and concatenated features after conditioning in both methods: savings remain 7.70 and 7.11 GiB, respectively. In that control, paired request time ratios range from 0.994 to 1.002 across the two lengths and maps, so these runs show memory savings without a material runtime change.

Input tokens	Taps	Release prefill	All states (GiB)	Selected (GiB)	Saved (GiB)
2,273	5	No	18.35	17.83	0.52
2,273	2	No	18.22	17.66	0.56
8,993	5	No	21.70	19.67	2.03
8,993	2	No	21.54	19.35	2.20
16,826	5	No	25.62	21.83	3.80
16,826	2	No	25.43	21.33	4.11
31,526	5	No	32.98	25.87	7.11
31,526	2	No	32.73	25.03	7.70
16,826	5	Yes	25.62	21.83	3.80
16,826	2	Yes	25.43	21.32	4.11
31,526	5	Yes	32.98	25.87	7.11
31,526	2	Yes	32.73	25.03	7.70

Table 67: Selective-capture memory diagnostic on L40S. Values are mean peak allocated GiB over two order-reversed repetitions. These repeats are not independent prompts. Both frozen mapper copies are resident for both methods. Release-prefill rows control buffer lifetime after conditioning. Allocator-reserved memory is not the reported quantity.

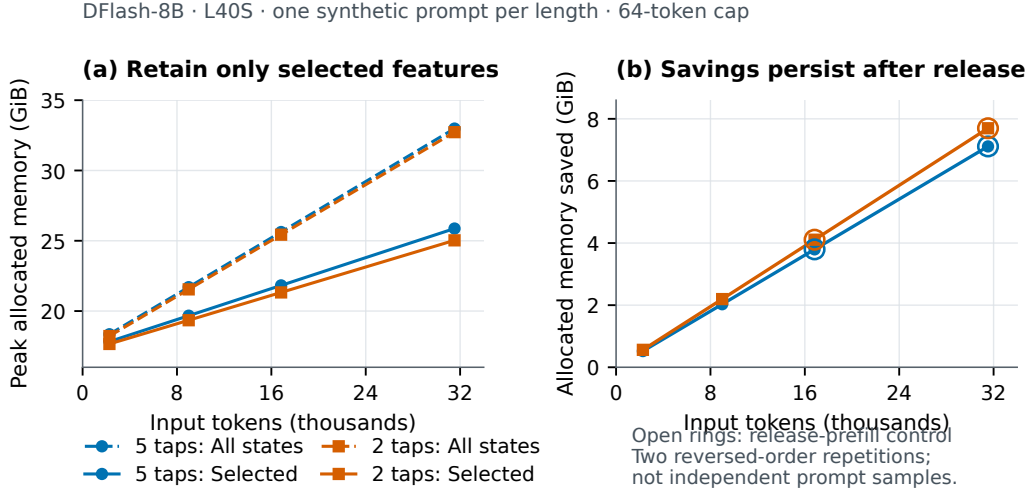


Figure 24: Selective feature capture on DFlash-8B/L40S. Left: peak allocated memory when returning all states or retaining two/five taps. Right: savings, with open rings for the release-prefill lifetime control. Each length uses one constructed archive prompt and two order-reversed executions with a 64-token cap. All paired outputs and acceptance trajectories match. These allocation measurements do not establish long-context quality or a material speed improvement.

A separate correctness check on eight code requests and eight two-turn dialogue conversations also preserves every output and acceptance trajectory for both the five-tap and two-tap mappers, with two order-reversed repetitions. These repeated executions do not add independent quality observations.

The synthetic inputs isolate sequence-length-dependent allocation and do not establish document understanding, long-context quality or concurrent-request capacity. Absolute values include model weights and the resident comparison maps. These results are distinct from the training-record scaling experiment and from the isolated source-removal experiment above.

J GENERATED-ROLLOUT CALIBRATION AND NATIVE-DRAFTER COMPARISON

Table 68: Separate NuminaMath evaluation, 128 requests, 2,048-token cap. End-to-end tokens/s in a batch-invariant BF16 runtime. The same AR measurement is used for both speculative runs. Intervals are paired request bootstraps, conditional on this fitted checkpoint.

Run	AR tok/s	Native tok/s	Relay tok/s	Relay/native [95% CI]
Original assignment	26.05	163.57	167.83	1.026 [1.018, 1.034]
Reversed GPUs	26.05	163.67	167.97	1.026 [1.018, 1.034]

This study instantiates Equation 4 with a larger calibration corpus and a batch-invariant inference runtime. It is a separate operating point from the original short-conversation fitting recipe. The source is Qwen3-4B and the target is Qwen3-8B. The inherited 4B DFlash drafter, target, source fusion, normalization, embedding and vocabulary head remain frozen. The five bias-free maps have 2560×4096 weights each, totaling 52,428,800 trainable parameters. Their block-output taps are [1,9,17,25,33], using zero-based indices.

Data and supervision. We use the pinned NuminaMath-CoT source shard and a custom split, not MATH-500. Normalized question text is grouped after replacing numbers and punctuation, retaining one representative per group. Deterministic source-proportional sampling yields 16,384 training, 256 development and 4,096 evaluation records. Evaluation uses the first 128 records of that evaluation

split. These checks prevent exact and defined-template group overlap within this corpus, but do not establish semantic independence from pretraining or other project data. The frozen target generates training solutions with a 4,096-output-token cap. Both frozen models then process the same complete token sequences. Segment-stratified sampling retains 25% of positions, giving 5,398,695 positions per epoch. Each record has equal total loss weight.

Objective and optimization. For a selected position, let $z = \text{concat}_i W_i h_i$. The layer term averages five relative squared errors, and the context term compares $N_{\text{out}}(Fz)$ with $N_{\text{out}}(Fs)$, both with the additive denominator constant 10^{-6} . The source normalization is

$$N_{\text{out}}(u) = \gamma \odot u \left(\frac{1}{2560} \sum_j u_j^2 + 10^{-6} \right)^{-1/2}, \quad (6)$$

with frozen γ . Unlike Equation 2’s direct-context DFlash map, the per-layer maps receive raw inputs. Folding produces $W_{\text{fold}} = [F_1 W_1, \dots, F_5 W_5]$, so inference still uses one linear projection followed by the inherited output normalization. The layer and context terms have equal coefficients. No cross-entropy, KL divergence or acceptance-rate objective is used.

Xavier initialization uses seed 42. AdamW has peak learning rate 10^{-3} , zero weight decay, 5% warmup and cosine decay. Batches contain 2,048 positions and gradient norm is clipped at one. Three fixed epochs yield 7,911 updates, with no validation-based checkpoint selection. Epoch losses are 0.738, 0.330 and 0.240. Their measured computation times are 281.1, 293.1 and 290.7 seconds. BF16 folded-export relative context error is 2.15×10^{-5} under the recorded export check. Rollout generation, feature capture and validation, and fitting are separate stages. Their times are 96.4 minutes on two GPUs, 52.0 minutes on two GPUs and 14.4 minutes on one GPU, respectively.

Inference and repeat. All arms use vLLM 0.28, BF16 and batch-invariant execution on L40S GPUs, greedy decoding with thinking disabled, no prefix caching and one measured request per worker. The 128 outputs contain 124,899 tokens per arm, with 104 ordinary stops and 24 length stops at 2,048 tokens. Timers include prefill and generation, excluding model setup and four warmups. Requests that trigger new compilation have one warmed repeat, with cold timings retained in the record. The second speculative run reverses GPU assignments and reuses the original AR measurement. All prompt IDs, generated token arrays and finish reasons agree across AR, native and relay, and across both speculative runs. Table 68 reports both runs rather than selecting the faster one.

This experiment follows the supplied archived training, mapper, export and benchmark implementation on newly generated rollouts. All historical rollout shard hashes differ, and the fitted weights differ as well. The execution therefore reproduces the specified recipe and measured behavior on a new data realization, not the original training tensors. Capture uses eager batch-invariant execution after compiled capture failed the recorded sampled/dense feature check. The corrected capture passes that check without relaxing its tolerance. These details and the complete source/configuration fingerprints accompany the result artifacts.

The 2.6% native-throughput advantage has a paired request interval above parity in both runs, conditional on one fitted checkpoint. The comparison changes inherited drafter release, and does not equate native drafter training costs. There is no matched normalized-dense fit on these same generated rollouts. Consequently it does not isolate the benefit of layer supervision, more data, longer answers or the runtime change.

K FROZEN LLAMA AND CROSS-TOKENIZER EXTENSION

Table 69: Frozen 128-question MATH evaluation. End-to-end tokens/s, 2,048-token cap, FP32 target and BF16 drafter. Every mapper arm matches the full AR token sequence on all 128 requests. Original and selected recipes differ in data and optimization.

Drafter source → target	Fit records	AR tok/s	Relay tok/s	T_R/T_A	Exact AR
Llama-8B → Llama-3B	4,096	45.73	114.93	2.51	128/128
Llama-8B → Llama-3B	16,384	45.73	115.44	2.52	128/128
Qwen-4B → Llama-8B	4,096	21.61	51.97	2.41	128/128
Qwen-4B → Llama-8B	8,192	21.61	59.41	2.75	128/128

Model and feature pairs. The within-family pair reuses the Llama-3.1-8B DFlash UltraChat drafter with a Llama-3.2-3B target. Source taps [1,8,15,22,29] of width 4096 map from target taps [1,7,13,19,25] of width 3072. The cross-family pair reuses the Qwen3-4B DFlash block-16 drafter with Llama-3.1-8B. Source taps [1,9,17,25,33] have width 2560 and target taps [1,8,15,22,29] width 4096. All indices identify zero-based block outputs. Pinned model revisions, source code and checkpoint fingerprints are included in the configurations. These two pairs extend architectural and tokenizer coverage, without constituting a broad multi-family survey.

Text alignment and verification. Source and target tokenize identical plain problem/solution text separately, without their different chat wrappers. Features are paired only at shared token-end character offsets, after the same text prefix. Input text is capped at 1,024 tokens per tokenizer, with at most 32 common positions retained per record. At generation time, each source draft chunk is decoded and re-encoded in the target vocabulary. The target verifies this candidate sequence using its own logits and commits its matching prefix plus its own next token. The committed text supplies the next source-vocabulary anchor. Vocabulary-intersection tables use direct ID translation when every token in a chunk is covered. Partial coverage falls back to full decode/re-encode. Tokenwise text equivalence does not guarantee canonical chunk tokenization, so the fast path can change proposed token boundaries. The target still verifies the resulting target-token sequence. This bridge is tested for greedy decoding, not heterogeneous vocabulary probability correction under sampling.

Search and frozen selection. The expanded campaign uses nested 4,096, 8,192 and 16,384 NuminaMath records, with 256 shared fitting-validation records. Dense context regression and the raw-input layer-plus-context recipe use the same feature caches, but differ in normalization and supervision. Twelve initial fits run for 12 epochs, saving epochs 1, 3, 6 and 12. Four follow-ups vary learning rate or extend to 24 epochs. Each update averages 64 records, with equal record weights. AdamW, seed 42, zero decay, clipping one, and 5% warmup followed by cosine decay are fixed. This is supplied solution text, not the generated-rollout corpus in Appendix J.

Development uses 74 short checkpoint/control arms, 24 coarse block arms and 16 longer arms. These reuse questions and are not independent datasets. Coarse block lengths span 6–32. Finer checks use 9/10/11 for Llama and 15/16/17 for cross-family transfer. A separate earlier 16-question check has a 1,024-output-token cap. It is not pooled with Table 69. The selected Llama candidate is the 16,384-record dense map at epoch 6 of a 12-epoch schedule, with 1,536 updates and block length 10. The selected cross-family candidate is the 8,192-record dense map at epoch 3 of a 24-epoch schedule, with 384 updates and block length 16. Its peak learning rate is 10^{-3} and it has 786,414 sampled-position exposures. An epoch revisits cached positions, rather than generating new data. The historical 4,096-record controls use a different fitting recipe, so the old-to-selected gain is a recipe comparison, not isolated data scaling.

Loss and normalization details. The dense extension uses the normalized-input context map, with an additive 10^{-6} context-loss denominator rather than the original smallest-normal FP32 floor. Its fitting input RMS epsilon is 10^{-6} , while deployment preserves the existing target input epsilon 10^{-5} . The output epsilon is 10^{-6} . A fixed-feature comparison over 256 validation records and 8,192 positions per pair measures mean context relative MSE of 5.14×10^{-7} and 5.49×10^{-7} from this epsilon difference. The runtime and weights were not changed after selection. This diagnostic quantifies a recipe discrepancy and does not remove it.

The layer-plus-context alternatives do not beat the selected dense recipe in development. This does not conflict with the native-parity rollout result, which changes the training text, position sampling and runtime. The two studies do not establish a globally optimal loss. Full fitting trajectories and all measured candidates remain in the accompanying campaign record.

Final 128-question protocol. We shuffle the pinned MATH-500 records with seed 1729 and take indices 40 through 167, excluding the 40 used in recent family development and checks. Both pairs receive the same 128 questions, with a 2,048-output-token cap. These are additional questions for this campaign, not globally unseen records in the broader project. Four GPUs execute independent requests in sixteen waves, two workers per pair. Every worker shares the frozen models and one AR reference between original and selected maps, rotating method order. Each mapper has exactly 128 unique requests. Token arrays are saved outside the request timer, checked against their hashes and compared directly. All four mapper arms agree with AR on 128/128 sequences. The within-family outputs total 62,608 tokens, with eight cap stops. Cross-family outputs total 92,610 tokens, with 24 cap stops.

Target and mapper weights, target head and target computation are FP32, with TF32 disabled. The frozen drafter, source embedding/head and mapped context are BF16. The source transformer is unloaded after retaining required vocabulary tensors. The matched AR reference uses the same FP32 target. End-to-end throughput includes prompt processing and generation, while excluding model loading, tokenization and final text decoding. Decode-only rates are 116.00/116.52 tokens/s for original/selected Llama maps and 52.29/59.83 for the cross-family maps. Main-paper values include prefill. Paired 95% intervals use 10,000 request resamples with seed 1729, conditioning on the selected checkpoint and runtime.

The original-to-selected request-throughput ratios are 1.0045 [0.9937,1.0154] for Llama and 1.1432 [1.1155,1.1723] for cross-family. The Llama interval includes parity, favoring retention of the cheaper original mapper. Cross-family tuning yields a measured improvement, without isolating data count from optimization. No independent fitting-seed replication is included in this final comparison.

Verification work and request heterogeneity. The selected cross-family map is faster on 122/128 requests. Total verifier calls fall from 28,000 to 24,487, increasing final target tokens per call from 3.31 to 3.78. For Llama, only 58/128 requests improve and calls change from 16,066 to 16,094, leaving progress at about 3.90. This accounts for the direction of the aggregate contrast without isolating data count as its cause (Figure 25). The cross-family gain also appears in all four fixed output-length bins (Figure 26). Figure 27 shows the observed prompt and capped completion lengths.

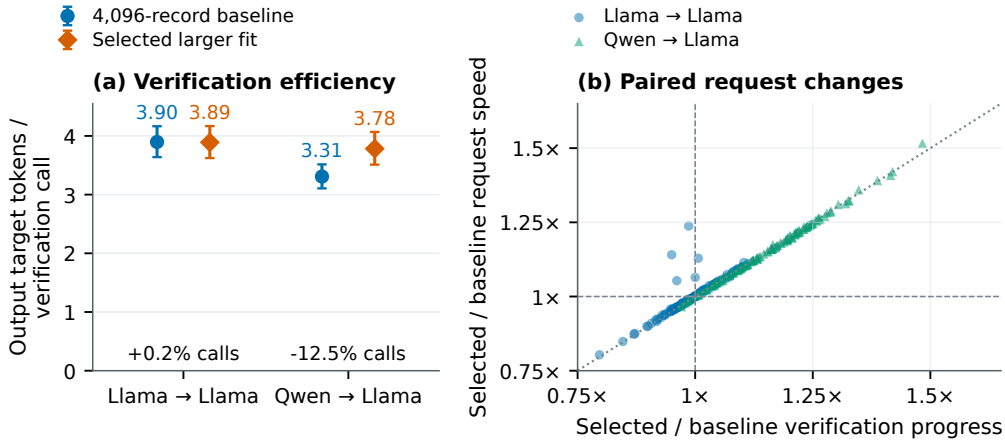


Figure 25: Verification efficiency on the frozen 128-question family evaluation. (a) Final target tokens divided by verification-loop calls, with paired 95% request intervals. Prefill calls are excluded, but output counts include the first token and EOS/cap trimming. This is effective progress, not proposal acceptance fraction. (b) Per-request changes in progress and throughput for identical outputs. The diagonal denotes equal ratios, not a fitted model.

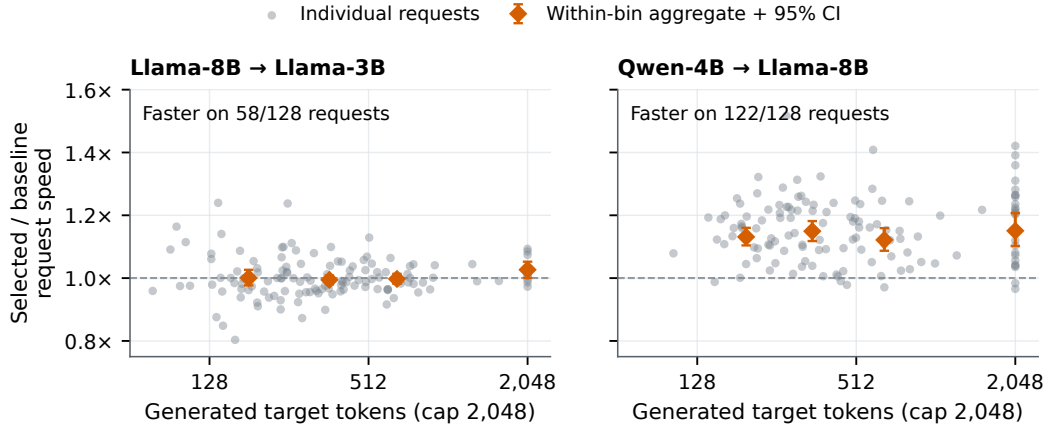


Figure 26: Paired request-speed gains versus generated target-token length. Each point is old/selected request latency for identical outputs. Diamonds aggregate four fixed length bins with paired 95% intervals. Bins contain 49/42/26/11 Llama and 31/45/25/27 Qwen-to-Llama requests. The 2,048-token cap censors longer completions. This is descriptive analysis of the frozen evaluation.

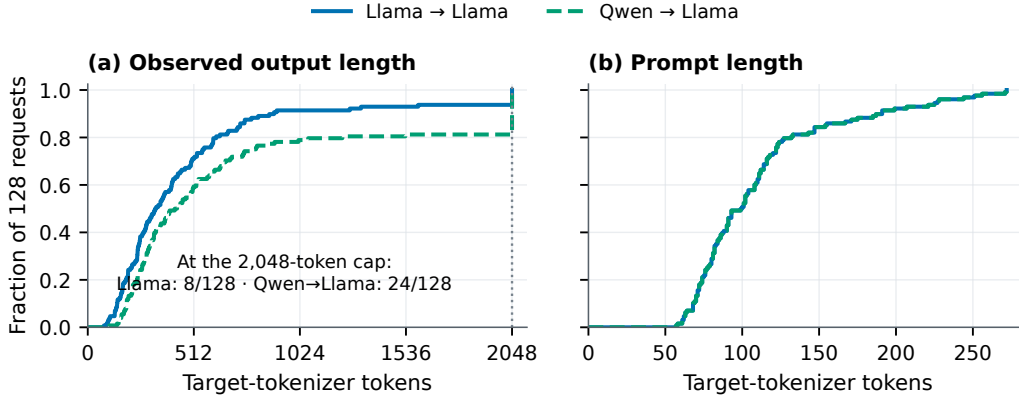


Figure 27: Observed lengths on the same 128 MATH questions under each target tokenizer. The capped output distributions include 8/128 Llama and 24/128 cross-family cap stops. Empirical distributions describe these recorded prompts and completions, not unrestricted generation lengths or answer quality.

Numerical agreement and its limits. Earlier BF16 target runs matched AR on only 4/8 within-family and 0/8 cross-family pilot requests for both old and layer-plus-context maps. Replaying identical prefixes exposed target-score differences across execution shapes and cache histories. FP32 target execution improved agreement in the recorded diagnostics without changing the acceptance rule, and the expanded mixed-precision configuration matches on all 128 requests here. These results do not validate earlier BF16 output streams or prove agreement on untested inputs. Exact sequence equality is an implementation check, not mathematical answer correctness. In the new comparisons, each mapper necessarily inherits the matched AR answers and its cap-related truncation on the tested prompts.

L ADDITIONAL RELATED WORK

Improving drafting. EAGLE predicts future features with shifted tokens (Li et al., 2024b). EAGLE-3 combines feature layers and simulated drafting steps (Li et al., 2025). DFlash predicts a block in

parallel (Chen et al., 2026), while Medusa uses several prediction heads (Cai et al., 2024). GRIFFIN addresses training-to-decoding token alignment (Hu et al., 2025). RepSpec changes the training structure (Huo et al., 2026), while VSD and Draft-OPD optimize accepted proposals (Zou et al., 2026; Lei et al., 2026). RelaySpec inherits the trained drafter and changes only its conditioning input.

Proposal structure and execution. EAGLE-2 and OPT-Tree adapt proposal trees (Li et al., 2024c; Wang et al., 2025a). SpecInfer organizes tree verification (Miao et al., 2024). Distributed speculative inference and speculative speculative decoding overlap computation (Timor et al., 2025b; Kumar et al., 2026). Draft & Verify skips target layers (Zhang et al., 2024), LayerSkip trains early exits (Elhoushi et al., 2024), and Lookahead checks candidates without an auxiliary drafter (Fu et al., 2024). RelaySpec retains an external drafter and holds its released proposal structure fixed. We evaluate the cost and accepted progress of changing its feature source.

M INHERITED DRAFTER TRAINING CONTEXT

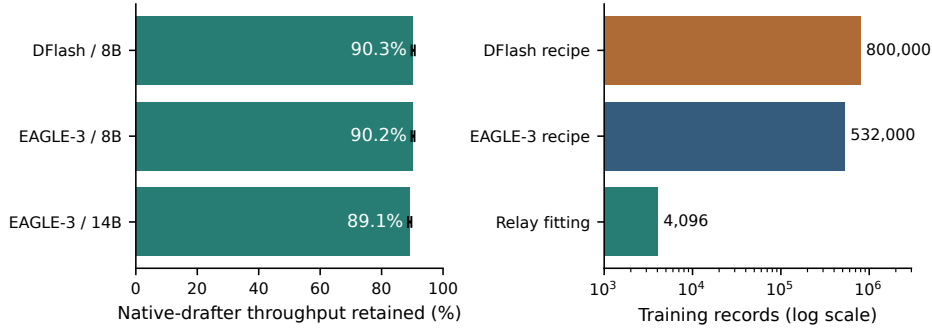


Figure 28: Left: fraction of native target-specific drafter throughput retained by RelaySpec in paired MATH-500 runs. Right: published original DFlash and EAGLE-3 recipe sizes versus the additional RelaySpec fitting set. Recipe counts provide training-scale context, not a matched training experiment or an audit of each released checkpoint.

Table 70: Original drafter training and additional target adaptation. Record counts describe different training stages and objectives.

Stage	Records	Relative to relay fit	Weights learned
Published DFlash recipe (Chen et al., 2026)	~800,000	~195×	Drafter
Published EAGLE-3 recipe (Li et al., 2025)	~532,000	~130×	Drafter
RelaySpec target adaptation	4,096	1×	Feature map

The DFlash recipe reports about 800K target-generated records. The EAGLE-3 recipe uses about 68K ShareGPT and 464K UltraChat entries. Relay fitting uses 0.51% and 0.77% of these respective record counts. This comparison describes marginal adaptation: the original drafter’s training remains inherited. Record lengths, objectives and hardware differ, so these ratios do not represent equivalent reductions in training compute. The evaluated EAGLE-3 checkpoints come from DeepSpec, while the table’s EAGLE-3 count refers specifically to the published training recipe.